
Significance of Microstructural Reinforcement for Concrete in Tension

Barzin Mobasher

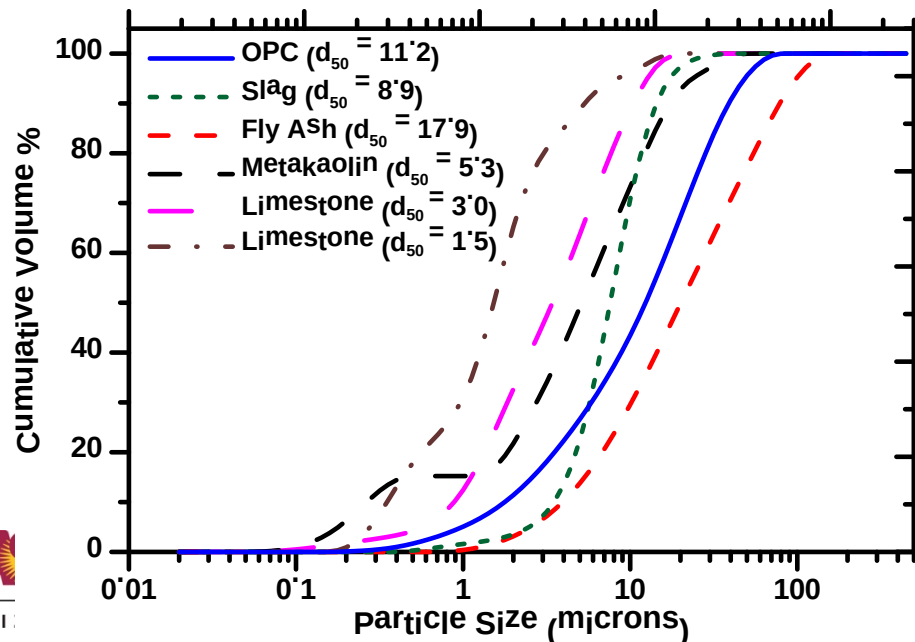
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The Sustainability Imperative- I

Concrete Volumetric demand and its Carbon footprint (SCM)

- Use of new cement formulations and Supplementary Cementitious Materials (SCM) is recognized as one of the most important sustainability aspects of concrete industry's effort to address its carbon footprint. (Damtoft et al. 2008a)
- Categories of inorganic powders with similar and compatible chemical and mineral characteristics to conventional portland cement and are referred to as mineral admixtures.
- ASTM designations: Type I/II ordinary portland cement (OPC) conforming to ASTM C 150, Class F, and C fly ash and metakaolin, or natural pozzolans conforming to ASTM C618, slag conforming to ASTM C 989, limestone powder conforming to ASTM C 568, and microsilica (silica fume) conforming to ASTM C 1240.



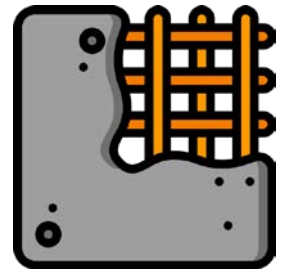
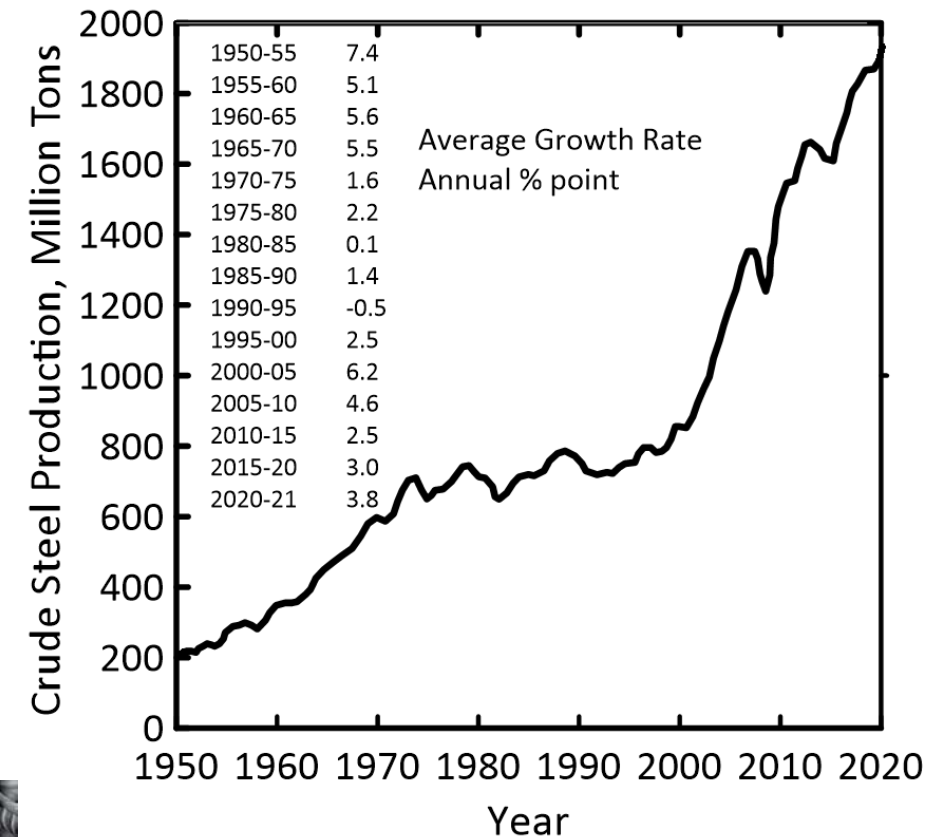
Portland Cement Particles



Silica Fume Particles

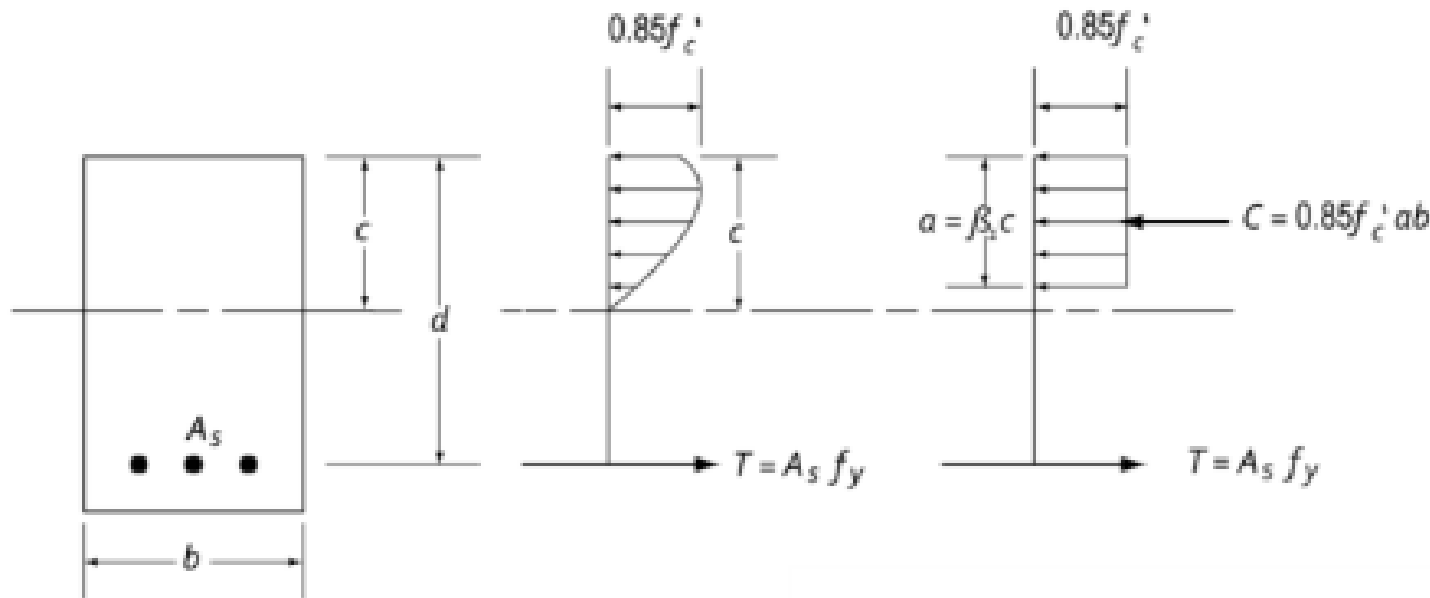
Sustainability Imperative II- Reinforcement

- World Production of Steel in the past 70 years has increased 10 folds
- Cost of Concrete Structures, 30% concrete, 20% steel, 50% forming
- Crude steel production increased from 800 to 2000 MT/Y since 1990
- 50% of production, 1000 MT/Y is by China, 10x of India, 2nd producer.
- For concrete consumption of 10Bt, rebar will be 522 MT
- On Average 120 kg of steel rebar per cubic meter of concrete
- 100% recyclable, raw feed supplies of ore will exhaust due to demand.



Concepts for the structural design are more than 120 years old

- Structural Design of reinforced concrete has been practically unchanged in 120+ years
- Guidelines developed in 1907, modified in 1930s, and codified in 1940s.
- Approximately 60-70% of concrete's volume that is subjected to tensile stress is still ignored and thus a waste of resource, providing no contribution other than providing a cover for the steel rebar.



Rectangular Stress Block
Charles S. Whitney, 1930s

$$M_u = \phi A_s f_y d \left(1 - 0.59 \frac{\rho f_y}{f'_c} \right)$$

Formulas.

Position of neutral axis,

$$k = \sqrt{2n \left(p + p' \frac{d'}{d} \right) + [n(p + p')]^2} - n(p + p'). \quad (26)$$

Position of resultant of compressive stress, $C + C'$,

$$x = \frac{k + \frac{d'C'}{dC}}{1 + \frac{C'}{C}}; \text{ in which } \frac{C'}{C} = \frac{2p'n \left(k + \frac{d'}{d} \right)}{k^2}. \quad (27)$$

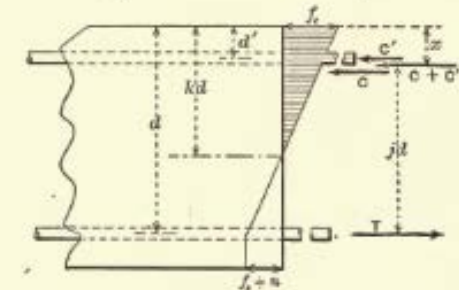


FIG. 67.

Arm of resisting couple,

$$j = \left(1 - \frac{x}{d} \right). \quad (28)$$

Moment of resistance,

$$M_s = f_s p j \cdot b d^2, \quad (29)$$

$$M_s = \frac{1}{2} f_c k (1 - \frac{1}{2} k) b d^2 + f'_s p' b d (d - d'). \quad (30)$$

Fibre stresses,

$$f_s = \frac{M + j d}{A}, \quad (31)$$

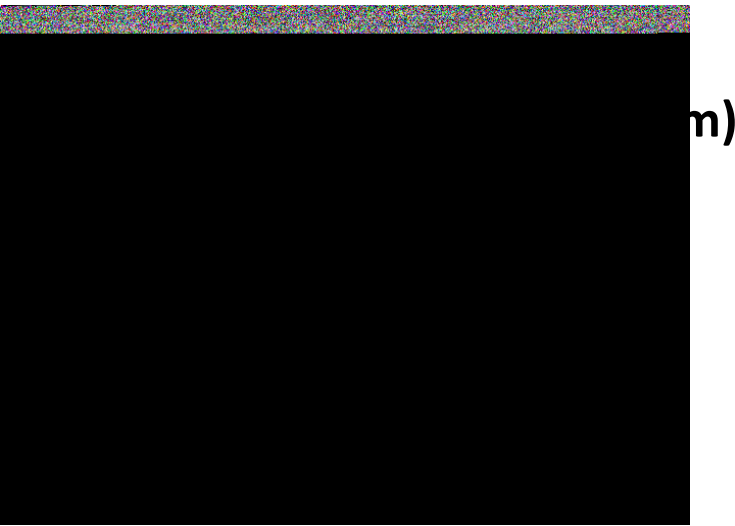
$$f_c = \frac{k}{n(1-k)} \cdot f_s, \quad (32)$$

$$f'_s = \frac{n \left(k - \frac{d'}{d} \right)}{k} \cdot f_c = \frac{k d - d'}{d - k d} \cdot f_s. \quad (33)$$

Ultimate Loads formula by Tuneaure 1907

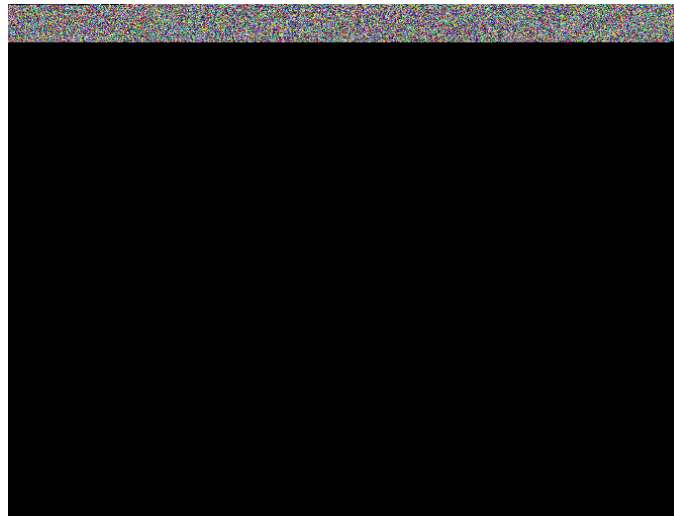
Supplementary vs. Primary Reinforcement

Manipulation of reinforcement at a microscale is a game changer in terms of Carbon footprint
Microstructure level toughening will have a profound impact at the macro-structural level



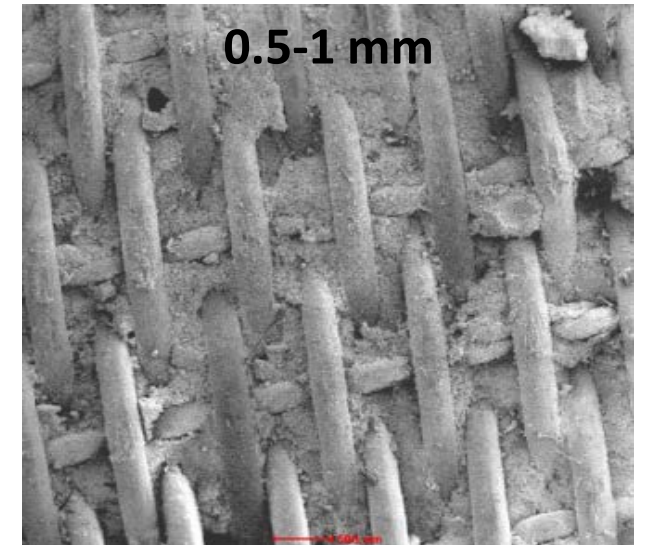
Ordinary reinforced concrete slab, 5-10" thick,
standard rebars, 12"-18" apart

Reinforcing Bars: Macro-structure
reinforcement



Fiber reinforced Concrete (FRC), at the meso-
structure with $\frac{1}{2}$ " long fibers, $\frac{1}{2}$ " (10mm)
millimeters apart

Fiber reinforced Concrete: Meso-
structure reinforcement



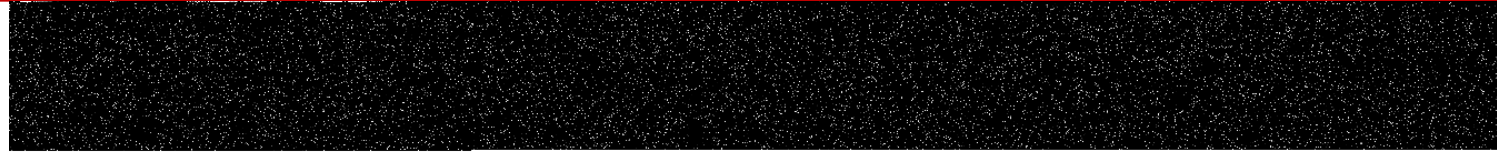
Textile reinforced Concrete (TRC), at the
micro-structure with long fibers, woven,
only millimeters apart

Textile Reinforced Concrete: Micro-
structure reinforcement

Delayed Activation of Reinforcement

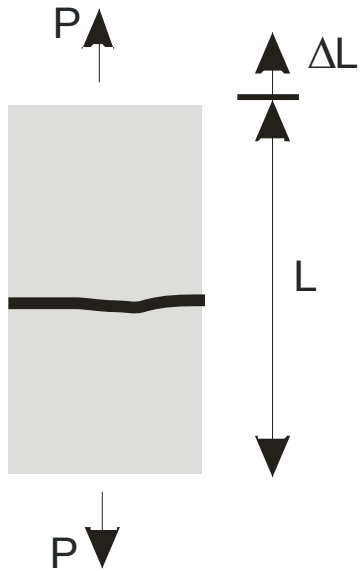
Inhibited Localization

Quantified Nature of Microcracking



- H. Stang, B. Mobasher, S.P. Shah, Quantitative damage characterization in polypropylene fiber reinforced concrete, Cement and Concrete Research, Volume 20, Issue 4, 1990, Pp. 540-558, [https://doi.org/10.1016/0008-8846\(90\)90098-I](https://doi.org/10.1016/0008-8846(90)90098-I)
- B. Mobasher, H. Stang, S.P. Shah, Microcracking in fiber reinforced concrete, Cement and Concrete Research, Vol 20, Issue 5, 1990, pp. 665-676, [https://doi.org/10.1016/0008-8846\(90\)90098-I](https://doi.org/10.1016/0008-8846(90)90098-I)

Strain Softening

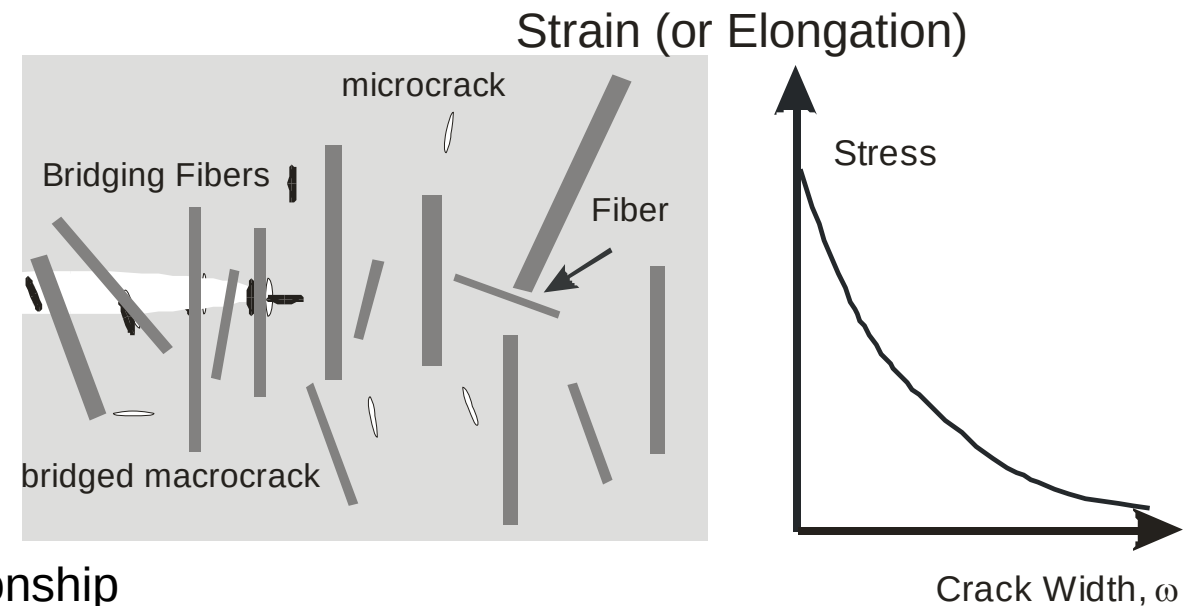
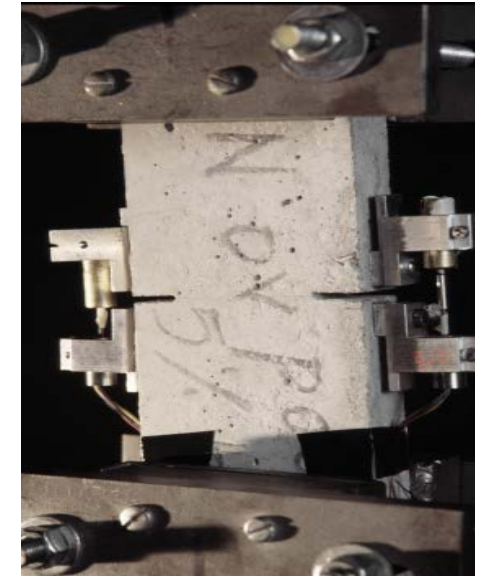
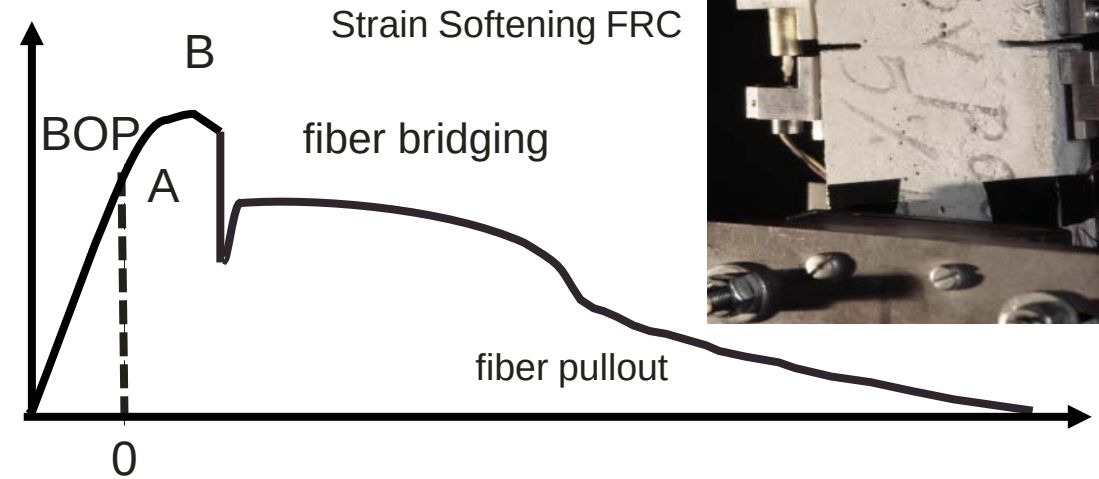


$$\omega = \frac{\Delta L}{L}$$

$$\sigma = P/A$$

Stress-Crack
width relationship

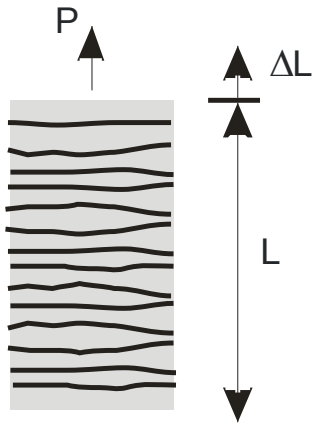
Strain Softening Behavior



- Single bridged crack
- Fiber pullout
- Stress-crack width relationship

Strain hardening

- Multiple Parallel cracks
- Fiber pullout
- Stiffness reduction crack spacing
- Strain capacity and ductility

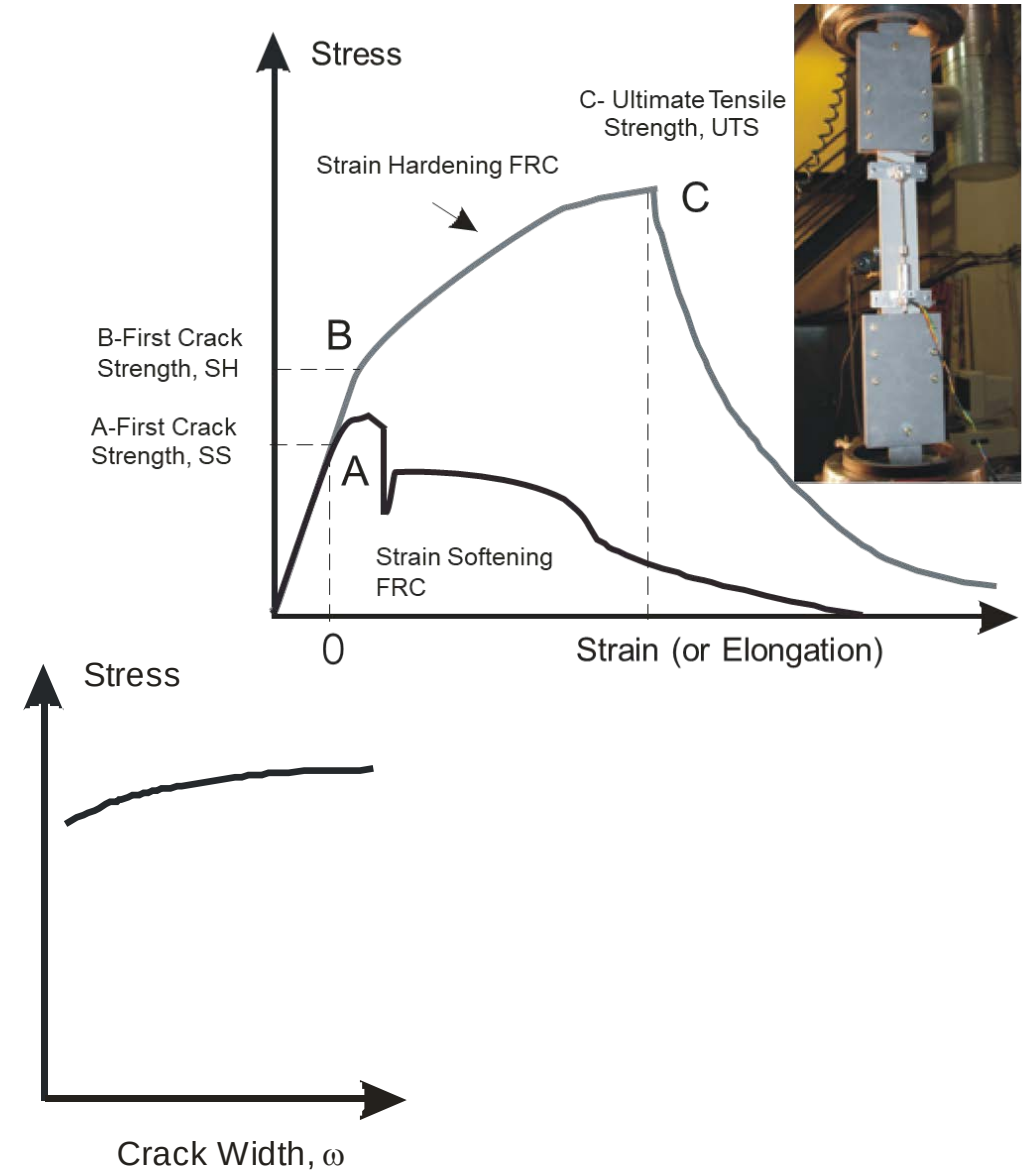
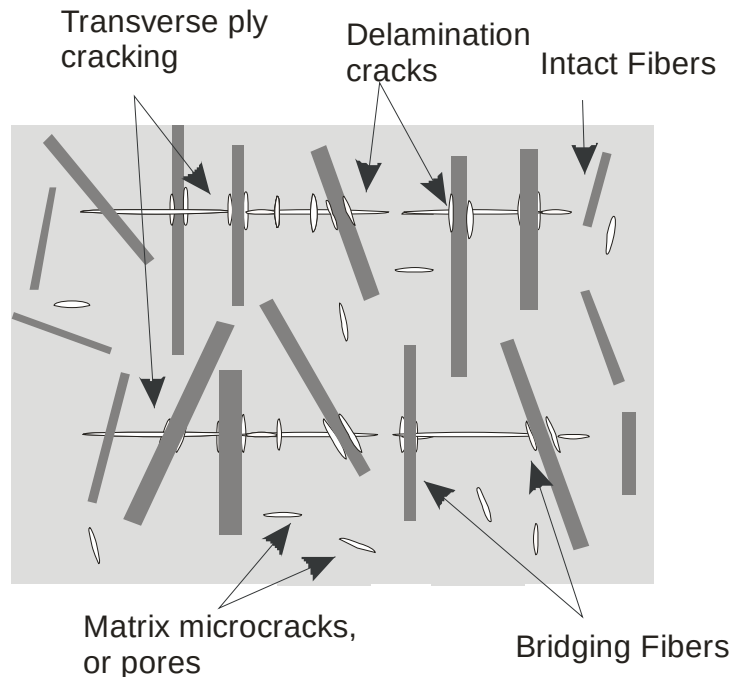


$$\varepsilon = \Delta L / L$$

$$\sigma = P / A$$

Avg Stress- Smeared
Strain Relationship

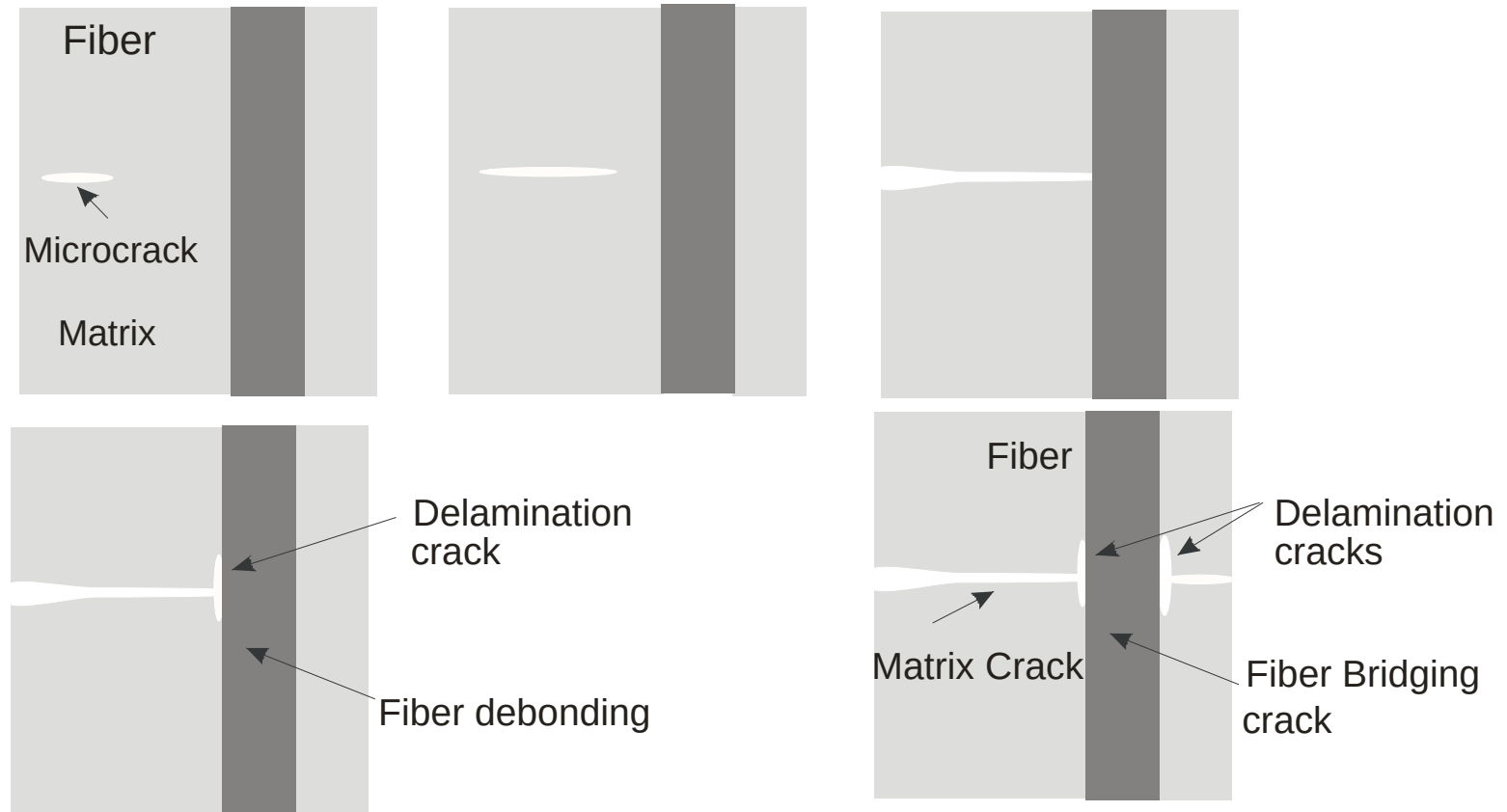
Strain hardening Behavior



Cracking Sequences: initiation and growth, deflection, & Bridging perspective

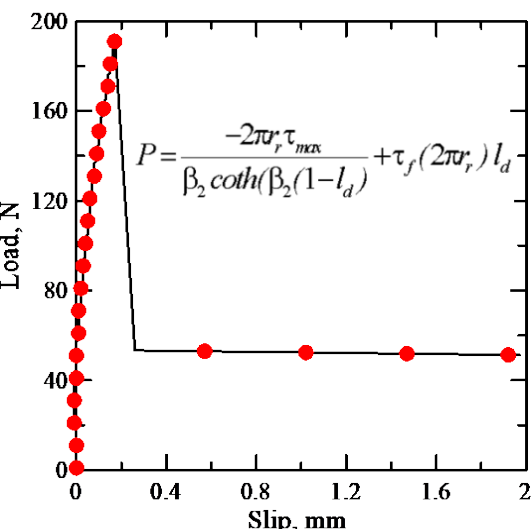
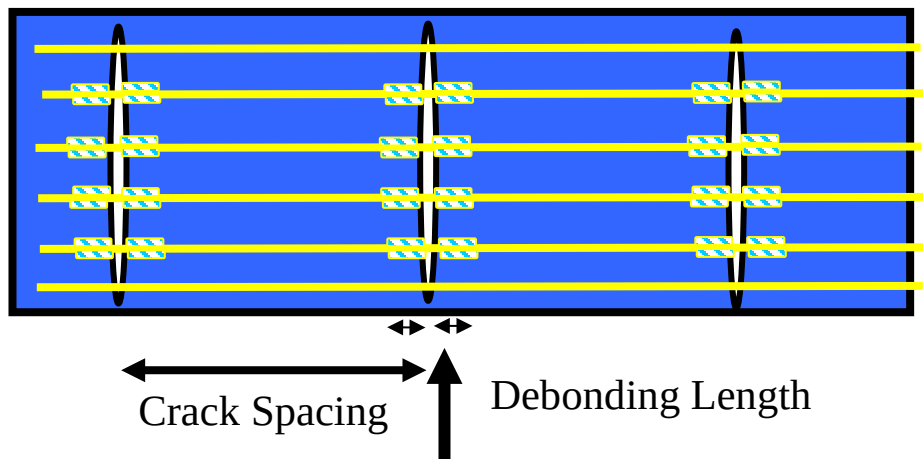
The cracking process starts from slow crack growth of a microcrack from a pore or aggregate interface, crack propagation, hindered by fibers, resulting in crack deflection, delamination, fiber bridging, pullout

1. Stable microcrack to macro crack growth
2. Crack coalescence and growth
3. Crack arrest by fibers
4. Fiber debonding and crack deflection
5. Interface debonding
6. Full fiber debonding growth
7. Fiber bridging

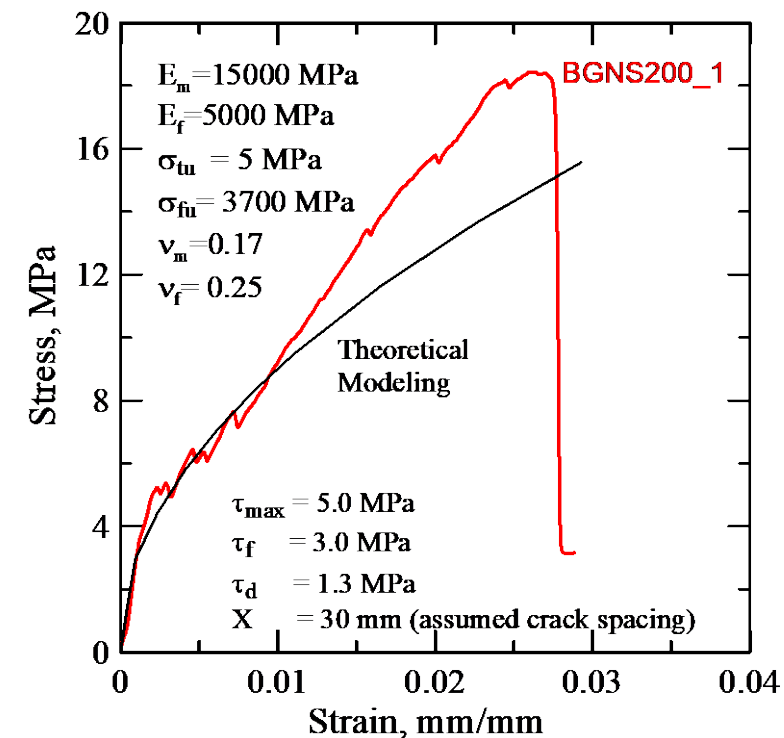
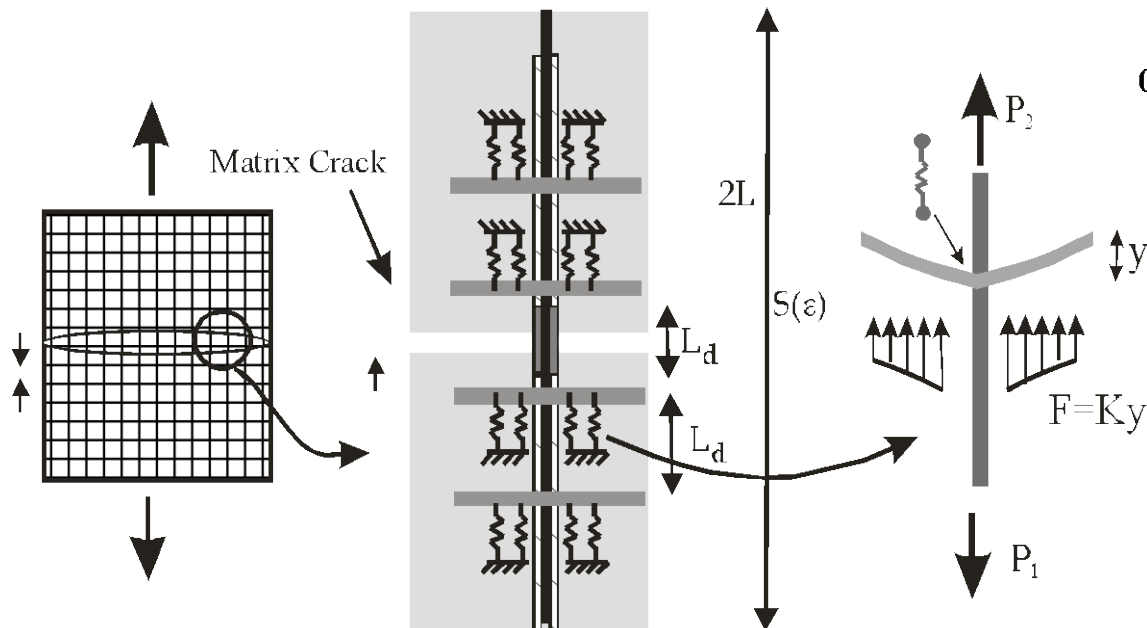


None of these mechanisms occur with concrete unreinforced at the microstructural level

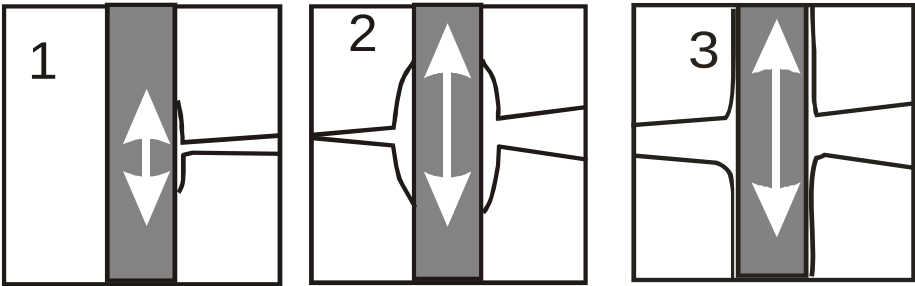
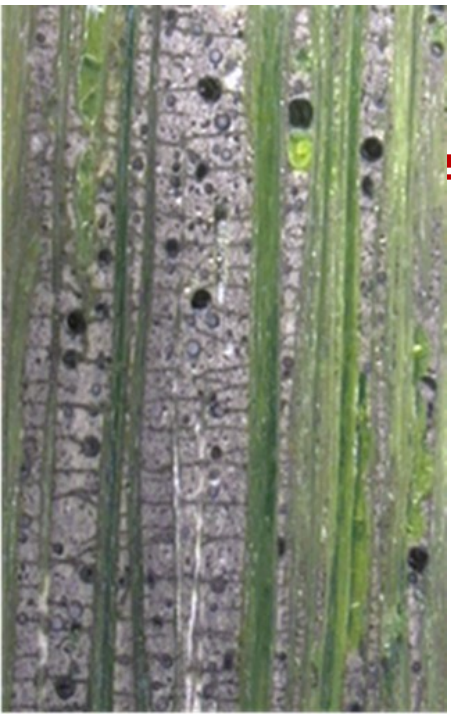
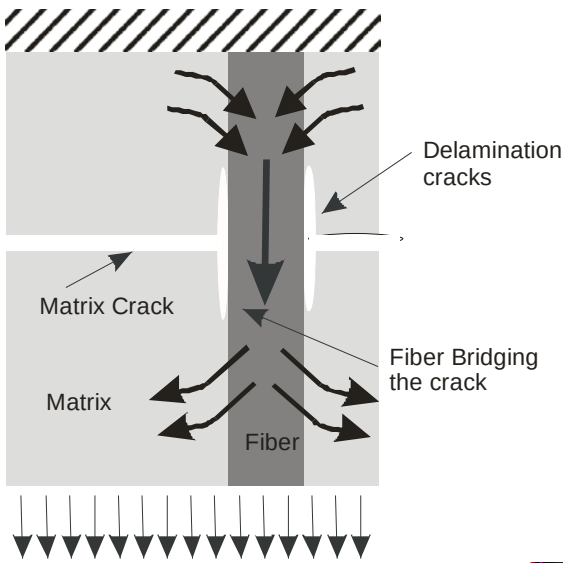
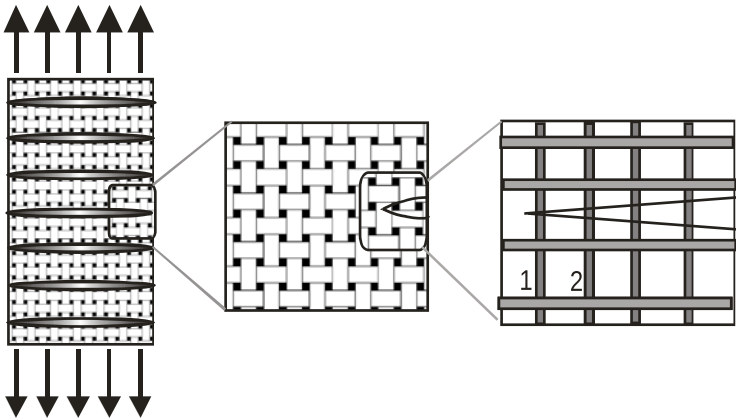
Modeling of delamination in textiles- Continuous Fibers



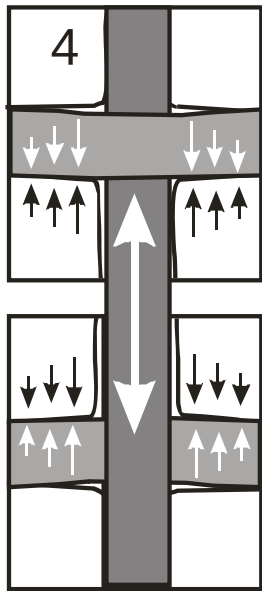
A | Pullout Slip response



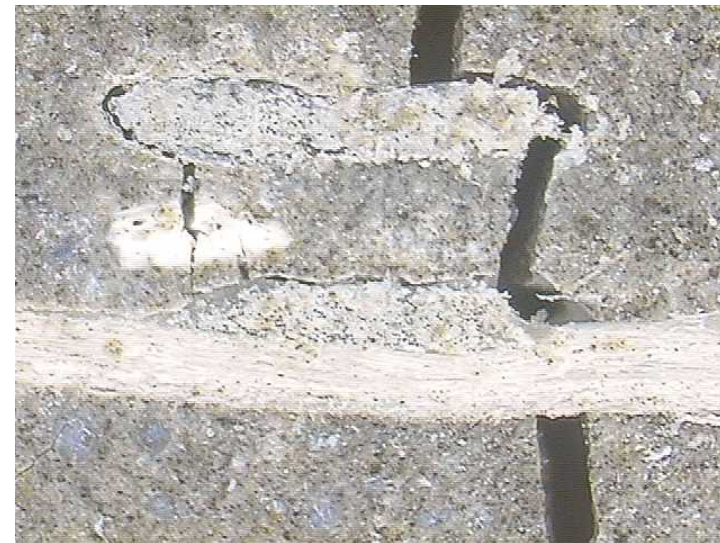
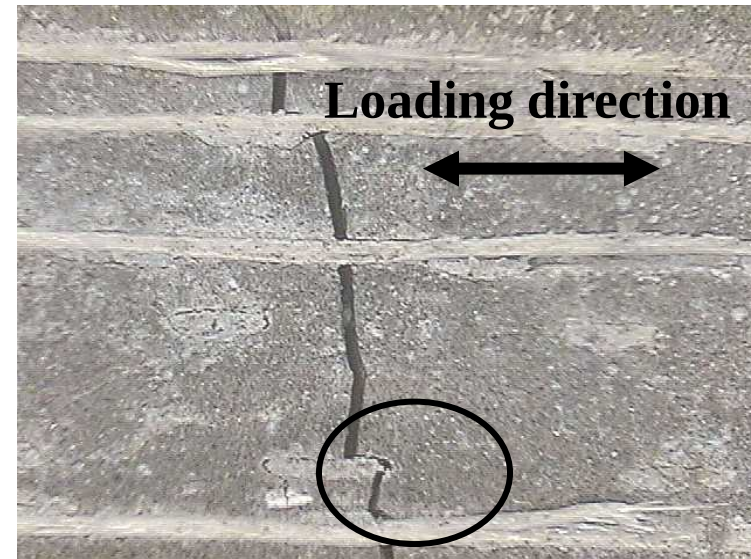
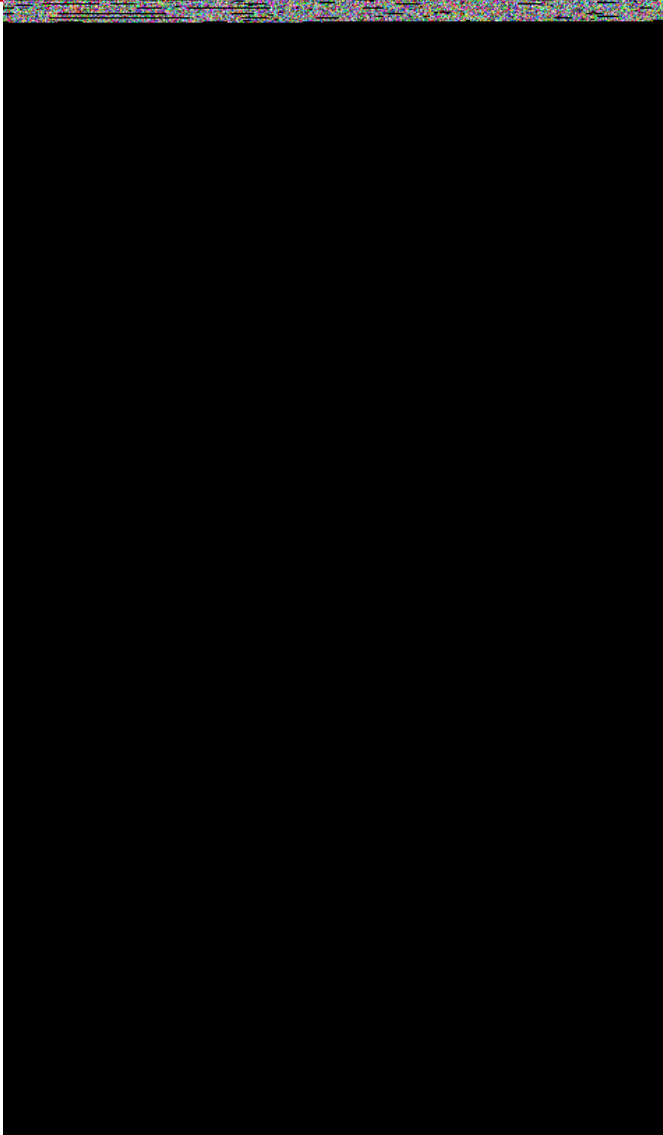
Tension Stiffening: major mode of toughening at the structural level



Crack Deflection Yam Debonding



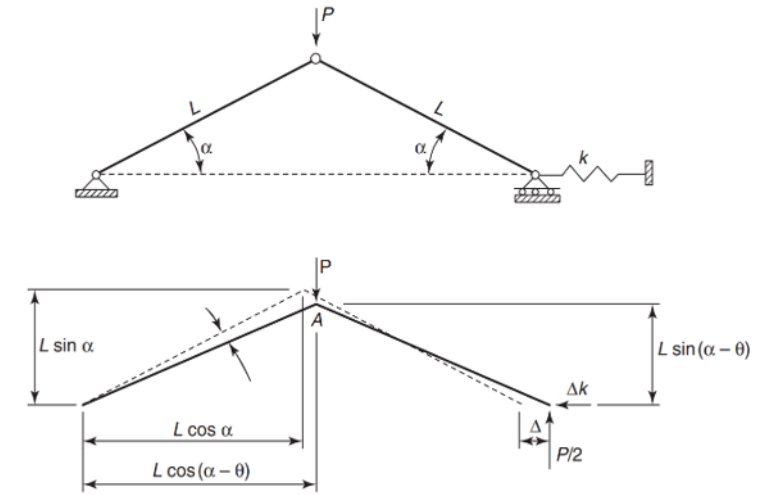
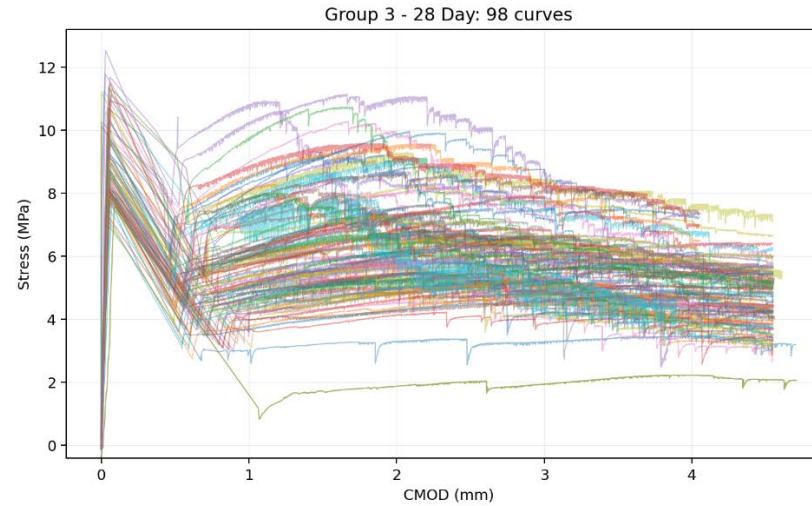
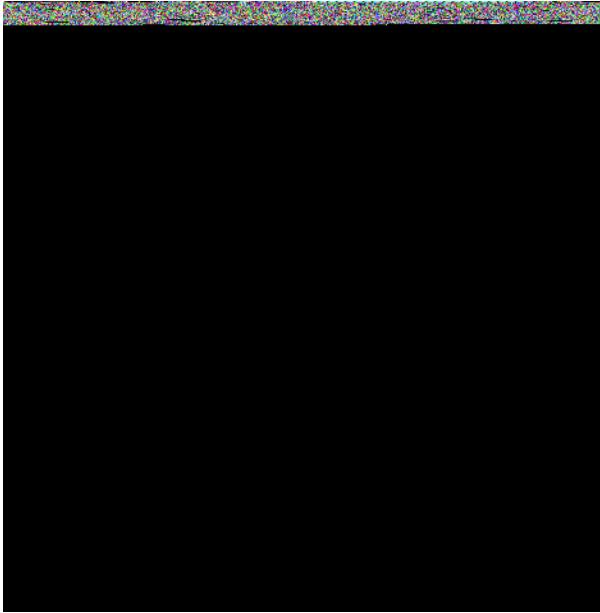
Optical Microscopy of Crack Deflection Mechanisms



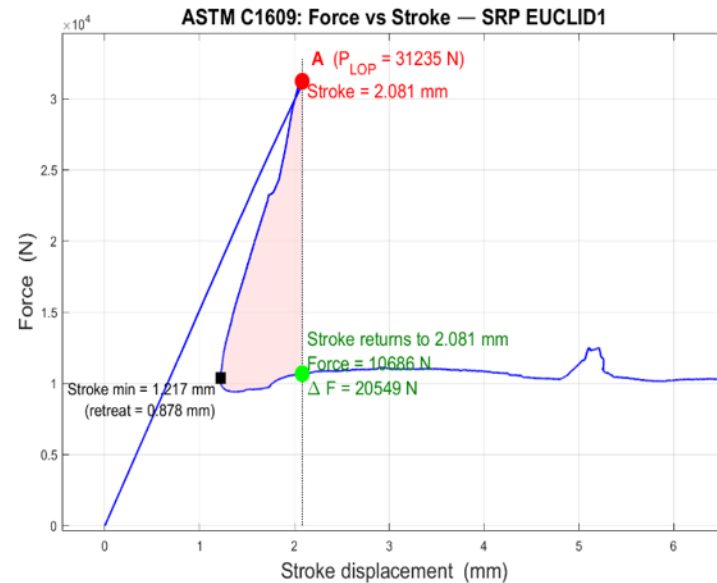
Potential mechanisms for toughening Concrete

- **The challenge is to reach the Theoretical Concept of tensile strength**
 - Reduce the pore size significantly to eliminate the internal microcracking, UHPC, HPC, ACK model
 - Eliminate small pores, flaws, connectivity, Packing, MDF, DSP, Bache, H.H. 1981
 - Three times the Tensile strength is attainable. We only measure the lower bound values. Hutchinson, Marshal, Cox, Evans, Brittle Matrix whisker reinforced Ceramics.
- **What are the toughening Mechanisms that can be used:**
 - Crack path tortuosity, Crack deflection, bridging, fiber pullout, yielding, deflection
 - Delamination cracks
 - Multiple cracking, Tension stiffening
- **Delay localization to increase the strength of concrete in tension**
 - Structural Redundancy in the Limit States
 - Correlation of cracking with creation of new surfaces and energy dissipation
 - Delays in crack propagation, reducing w/c , porosity, and improving interfaces
 - Transverse Yarn anchorage by textiles

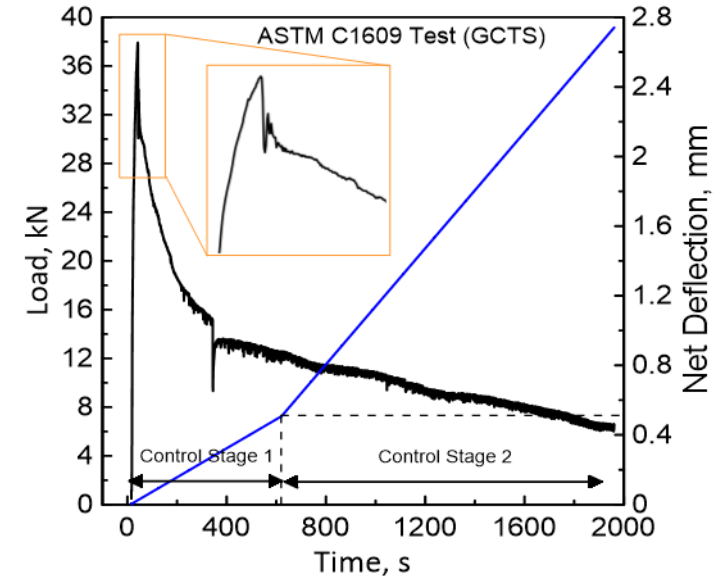
Localization Tension Snap-Back or Snap-Through Instability



Snap Through Simulation, Galambos



Snap back
Event



Controlled
test

Formulation of Theoretical R-Curves

- Stable and unstable crack growth

$$\alpha = \frac{a_0 + \Delta a}{b} \quad \begin{array}{ll} \alpha = 1 & \text{LEFM} \\ \alpha > 1 & \text{Quasi Brittle Materials} \end{array}$$

- Failure Conditions

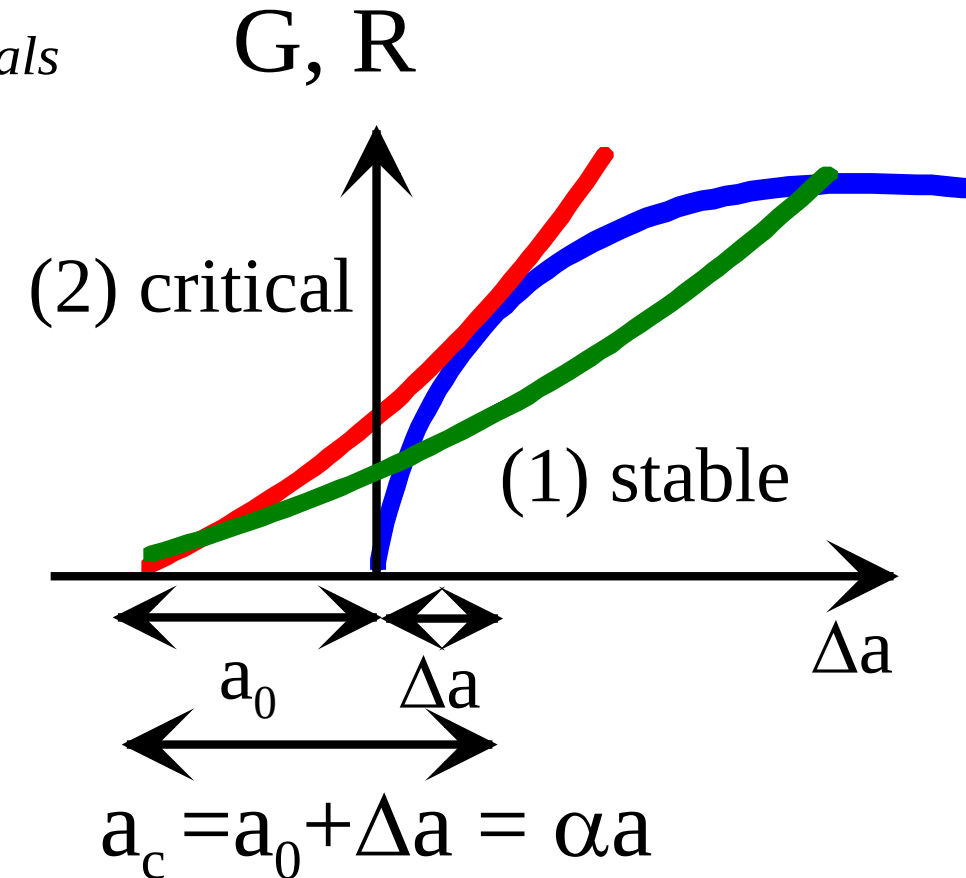
$$R = G, \quad \frac{\partial R}{\partial a} > \frac{\partial G}{\partial a} > 0 \quad (1) \text{ Stable}$$

$$R = G = \gamma, \quad \frac{\partial R}{\partial a} = \frac{\partial G}{\partial a} \quad (2) \text{ instability}$$

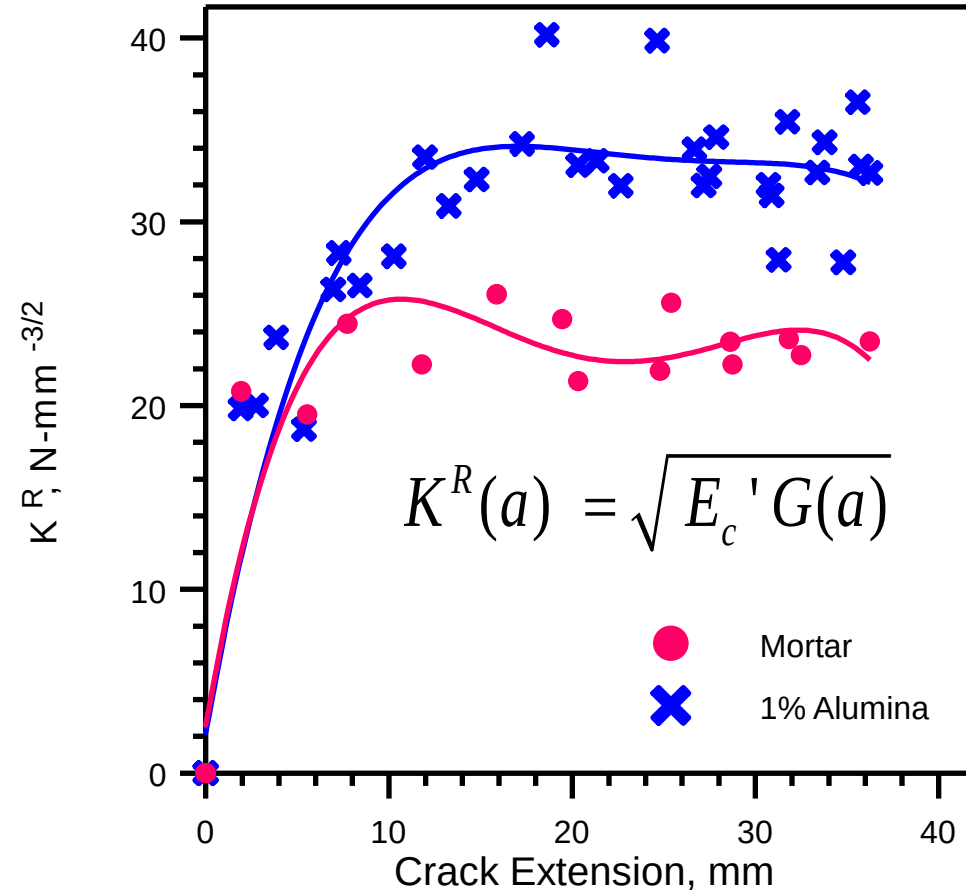
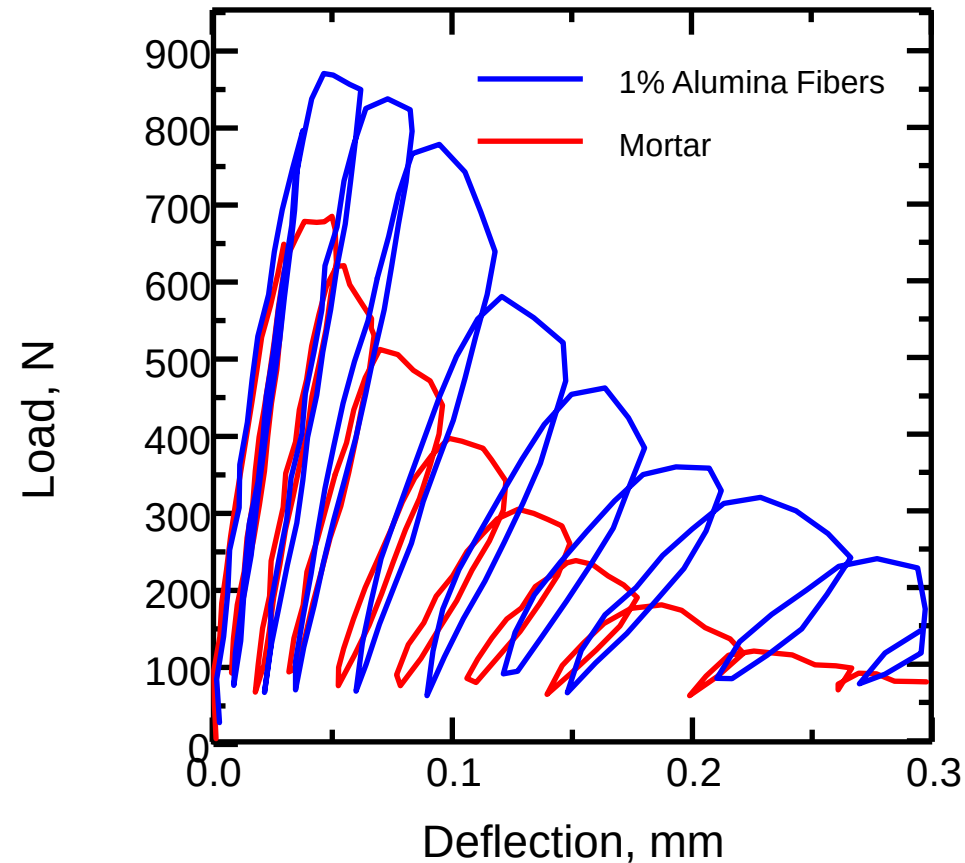
- Critical Condition

$$d_i = \frac{1}{2} + \frac{\alpha - 1}{\alpha} \pm \sqrt{\frac{1}{4} + \left(\frac{\alpha - 1}{\alpha}\right)^2} \quad i = 1, 2$$

$$R = \beta \left[1 - \frac{d_2}{d_1} \left[\frac{\alpha a_0 - a_0}{a - a_0} \right]^{(d_2 - d_1)} (a - a_0)^{d_2} \right]$$



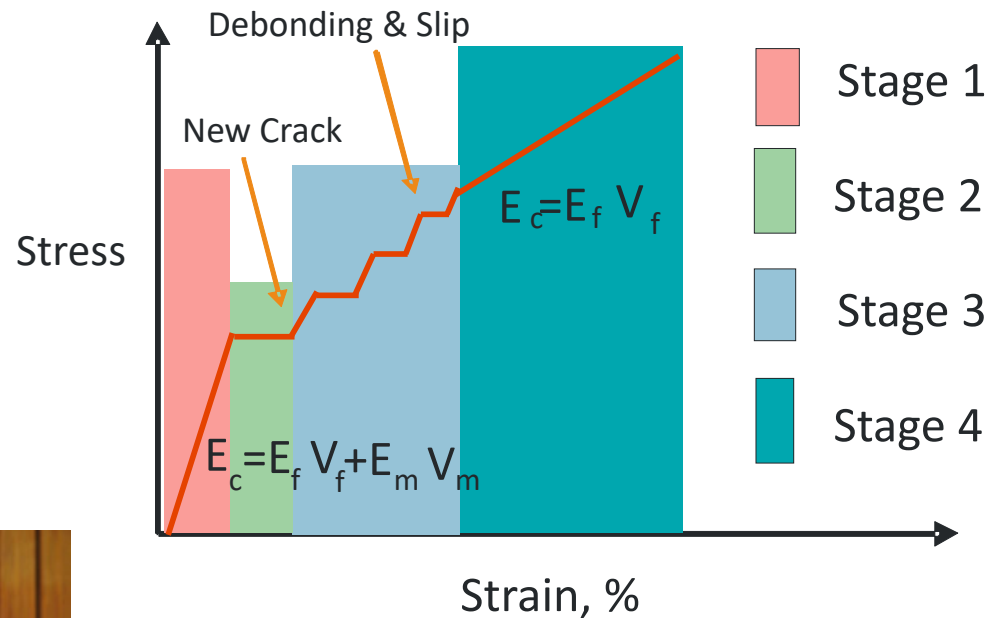
Cyclic Tests - Brittle Alumina Microfibers



$$G(a) = \frac{1}{2t} \frac{\partial C}{\partial a} P^2 + \frac{1}{2t} \frac{\partial \delta_r}{\partial a} P$$

Sequence of microcrack formation in continuous fiber systems- Strain hardening

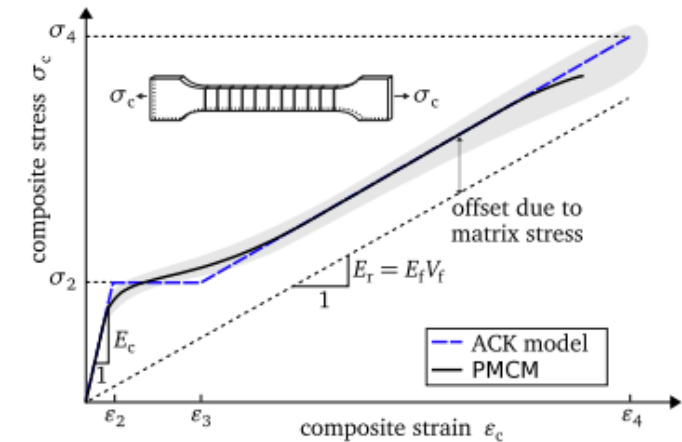
- Brittle Matrix Composites using ductile or brittle fibers
- The bond should not be as strong as possible as the failure of interface is beneficial to many degrees.
- Strong interfaces make brittle composites, whereas weaker interfaces make ductile composites



Two types of cracking:

- Transverse cracking in the paste. Inelastic Energy Sink
- Fiber slip after the matrix cracking. Dissipative frictional Energy
- Tension Stiffening Effect

A. Kelly, *Strong Solids* (1965)



Four input parameters:		stiffness	strength
	matrix	E_m	σ_{mu}
	reinforcement	E_c	σ_{fu}

Predicted stress-strain characteristics:

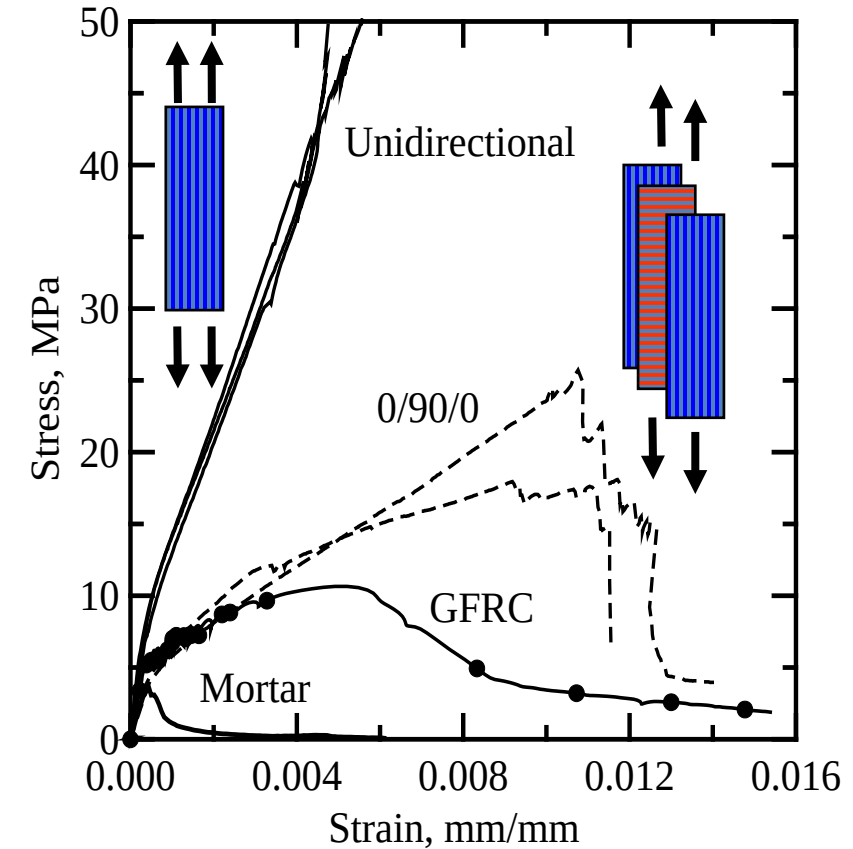
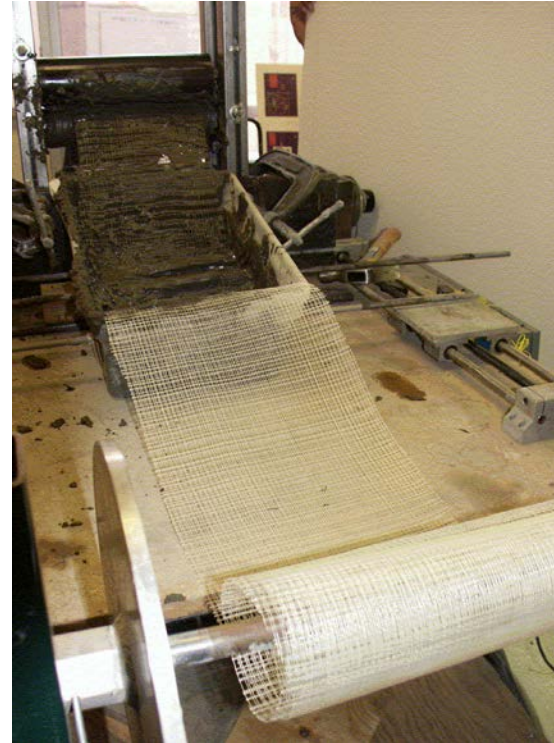
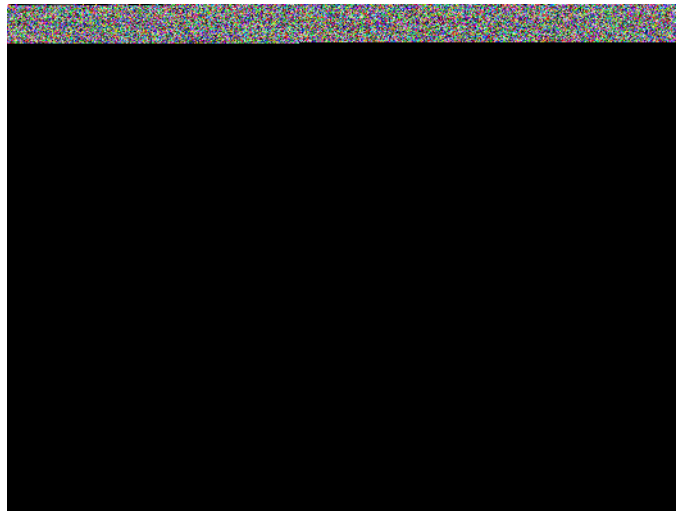
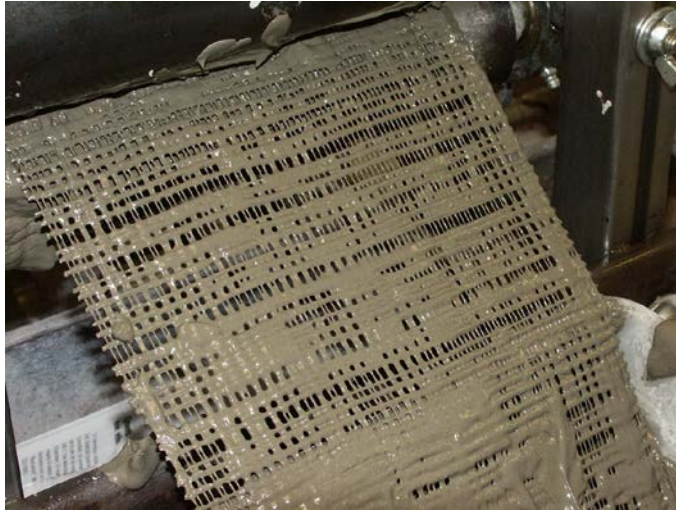
$$\epsilon_2 = \frac{\sigma_{mu}}{E_m}, \quad \sigma_2 = \epsilon_2 E_c$$

$$\epsilon_3 = \frac{\sigma_{mu}}{E_m} (1 + 0.666 \alpha_e), \quad \alpha_e = \frac{E_m (1 - V_f)}{E_f V_f}$$

$$\epsilon_4 = \epsilon_3 + \frac{\sigma_4 - \sigma_2}{E_f}, \quad \sigma_4 = \sigma_{fu} V_f$$

Microcracking in Fiber Reinforced Concrete,
Mobasher, Stang, and Shah, 1990, CCR,
[https://doi.org/10.1016/0008-8846\(90\)90001-E](https://doi.org/10.1016/0008-8846(90)90001-E)

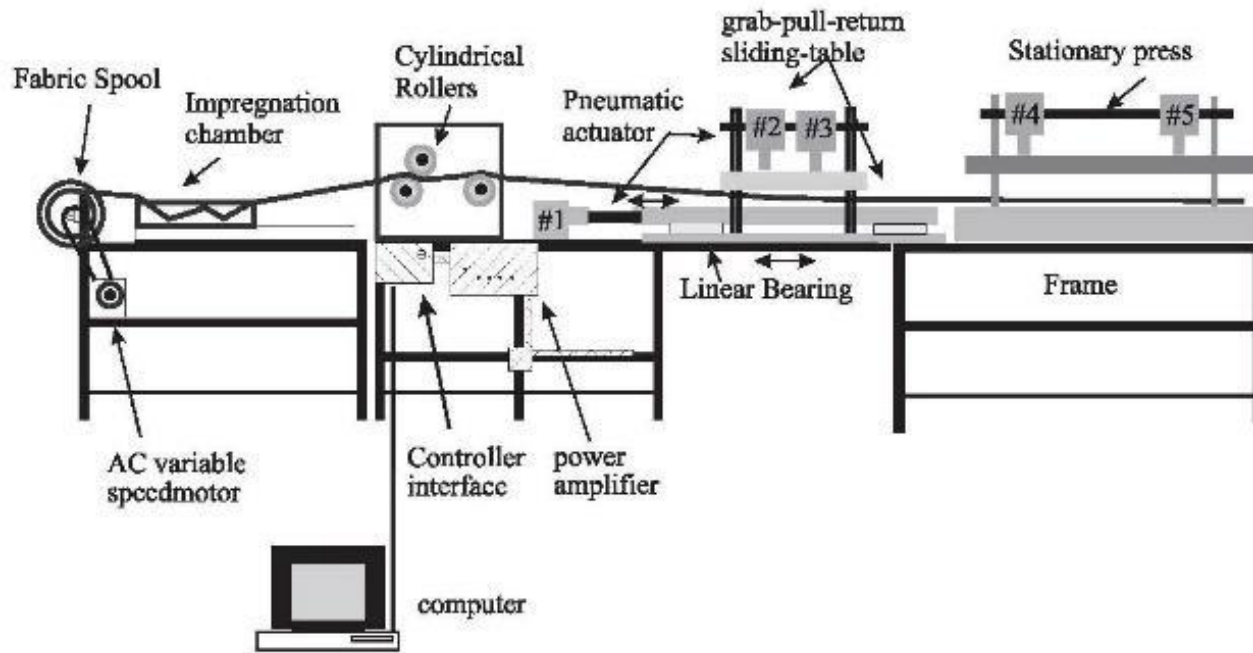
Textile Reinforced Concrete- Pultrusion based approaches



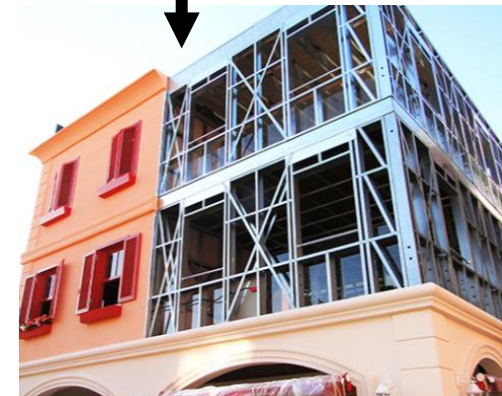
Tensile Strength = 2 MPa, strain capacity = 0.02%, 500 million tons of steel rebar
Pultruded Cement composites, Tensile Strength = 20 MPa, strain capacity = 1%

Structural Shapes using Textile Cement Composites

- Automated pultrusion system for full scale structural shapes with TRC laminates
- Competitive to wood and Light gage steel sections



Pultrusion Process Schematic Diagram



Peled, A., Mobasher, B., & Bentur, A. (2017). Introduction. In *Textile Reinforced Concrete* (pp. 1–12). <https://doi.org/10.1201/9781315119151-1>
Mobasher, B. (2011). Mechanics of fiber and textile reinforced cement composites.

Manufacturing techniques of TRC



a)



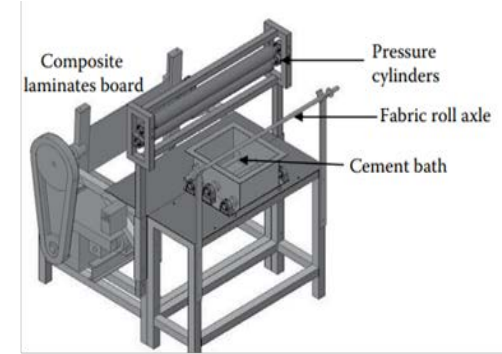
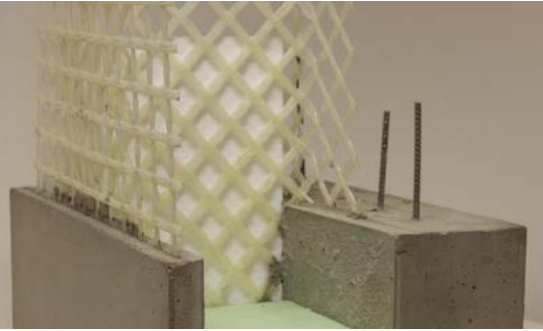
b)



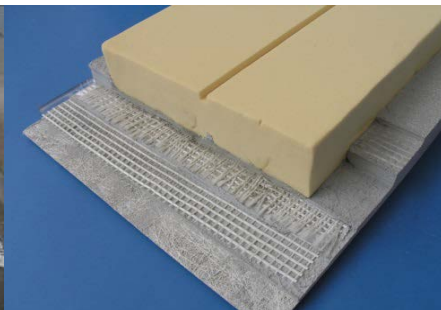
a)



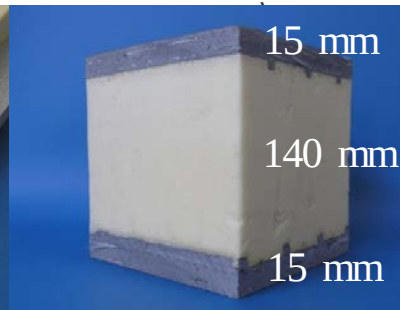
b)



(a)



(b)



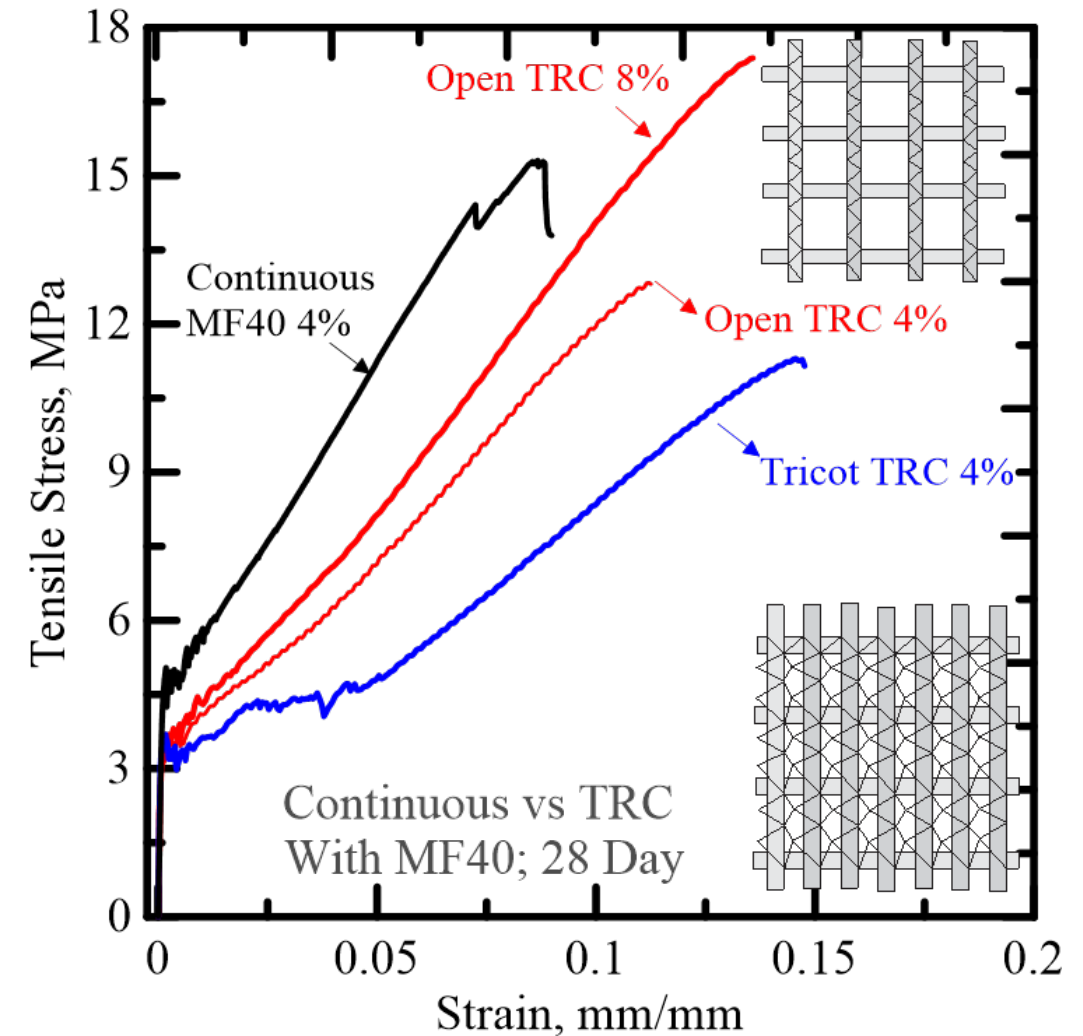
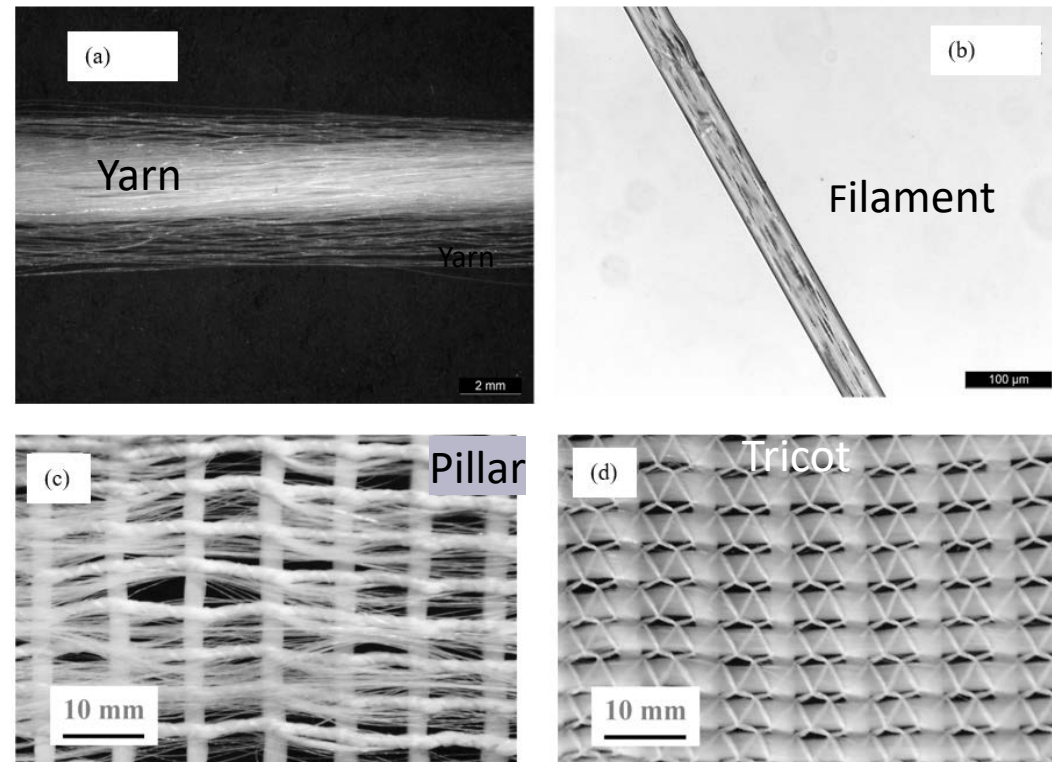
(c)



(d)

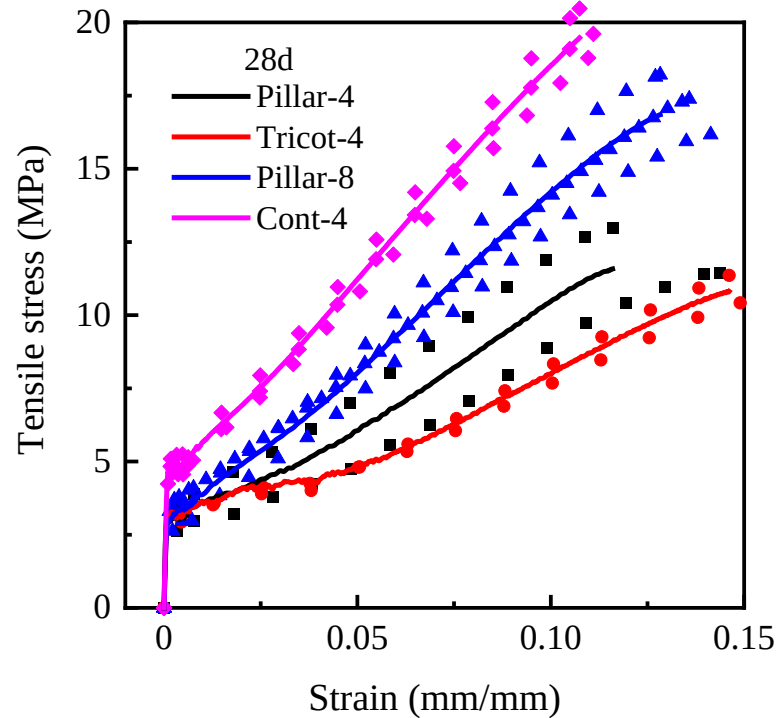
Distributed Cracking Mechanisms in Polymeric Textile TRC

- Warp knitting pattern (pillar vs. tricot) had no significant impact on tensile properties. Pillar knits are simpler, less labor-intensive stitching
- Doubling textile volume fraction (4% to 8%) increased tensile strength by 44%, post-crack modulus by 78%, and toughness by 47%
- DIC revealed random crack formation rather than sequential spacing reduction along the gage length
- Crack width limit-state of 0.5 mm is satisfied at 75% of ultimate tensile strength (8–12 MPa depending on textile content)

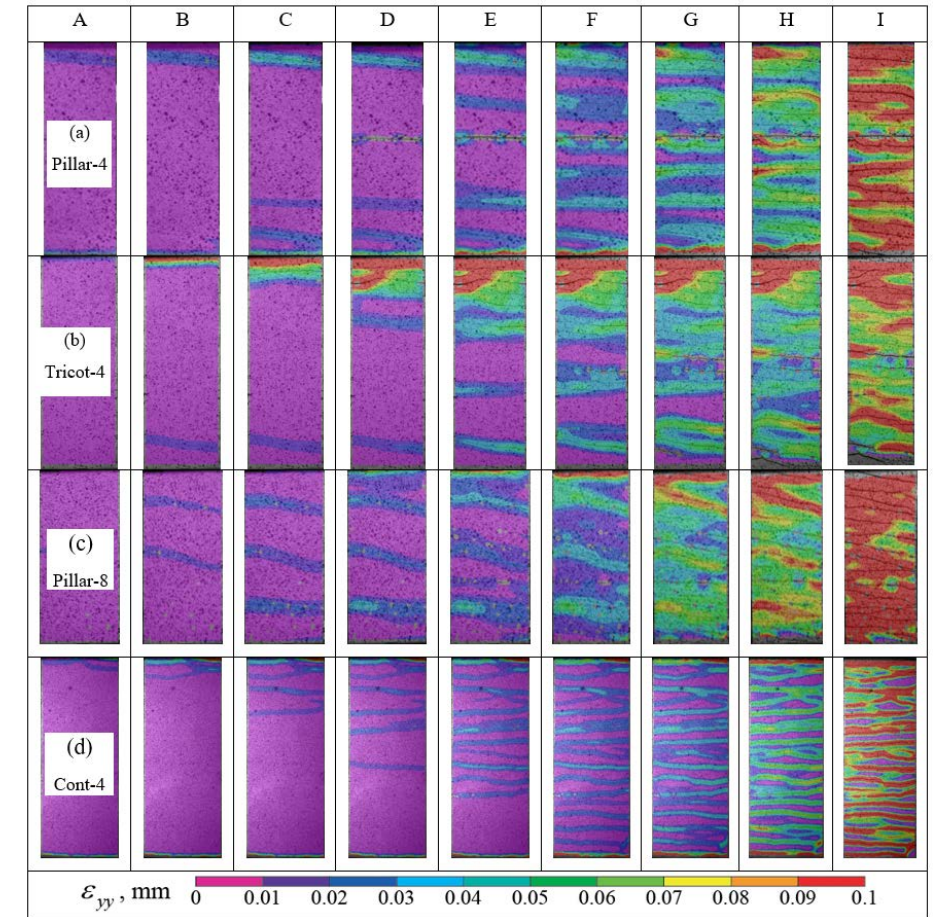
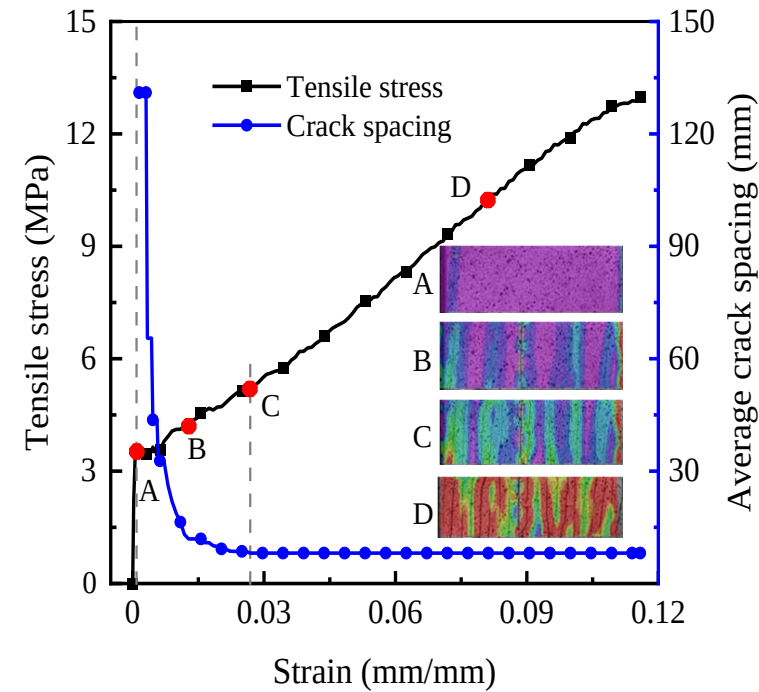


Dey, V., Li, A., Bauchmoyer, J., et al. "Distributed cracking mechanisms in micro-polypropylene based textile reinforced concrete," Materials and Structures, V. 56, No. 10, 2023, p. 175.

Distributed Cracking Mechanisms in TRC

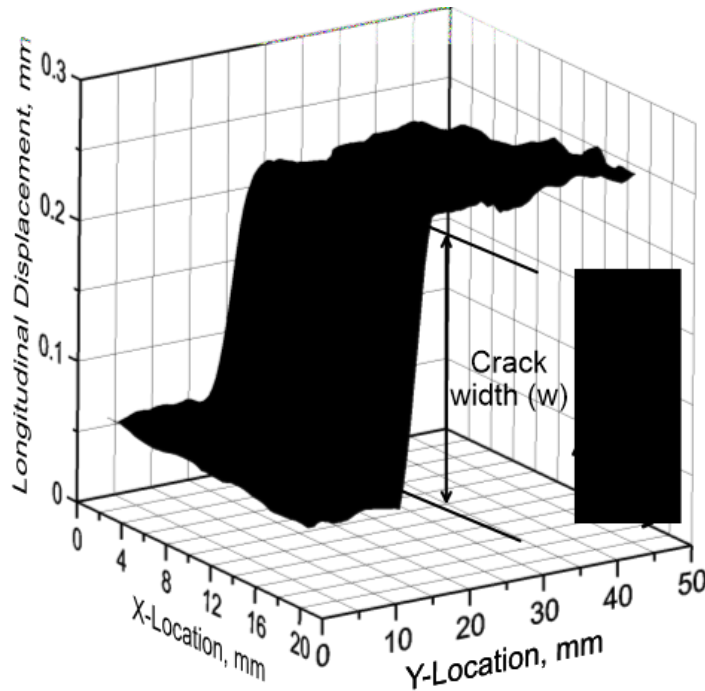


Crack spacing of pultruded pillar TRC specimen (Pillar-4): Stress-Strain vs. Crack location

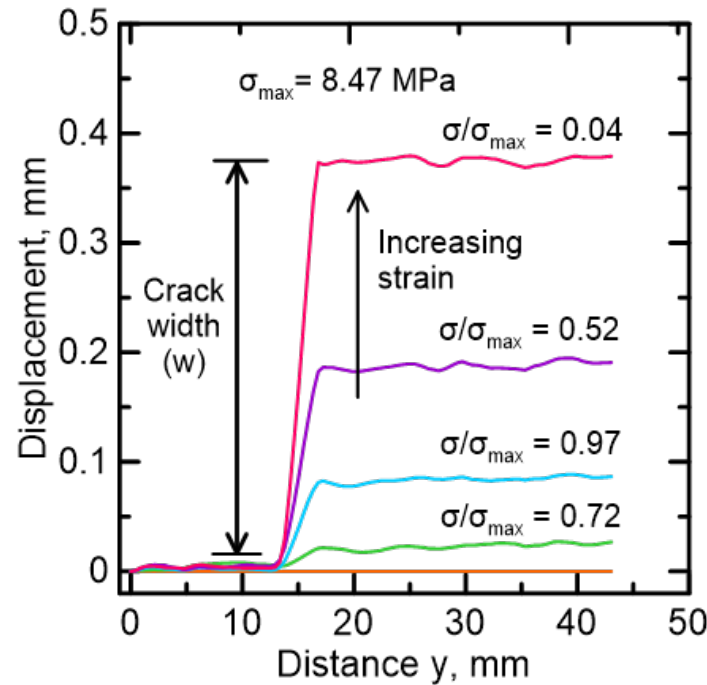


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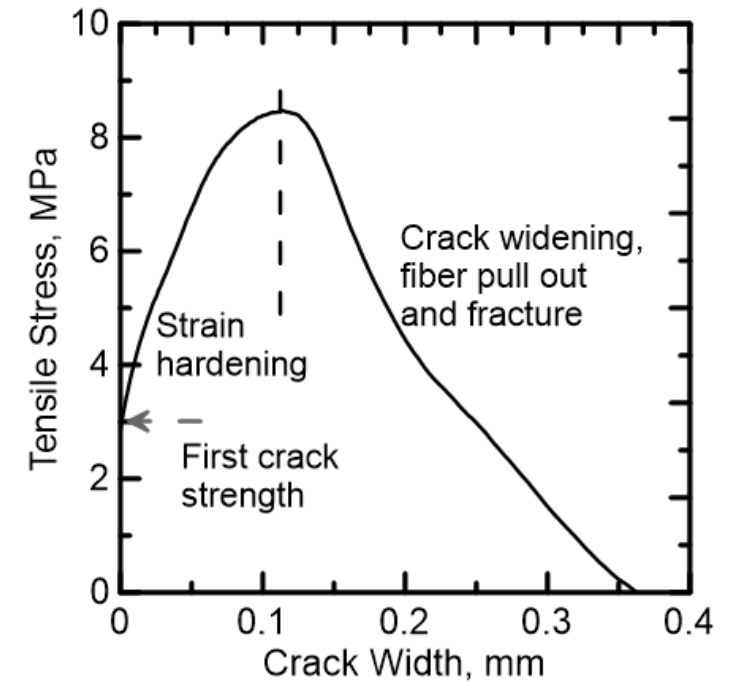
Crack Width Measurement for Strain Hardening in TRC



Displacement Field



Displacement Distribution Along Specimen

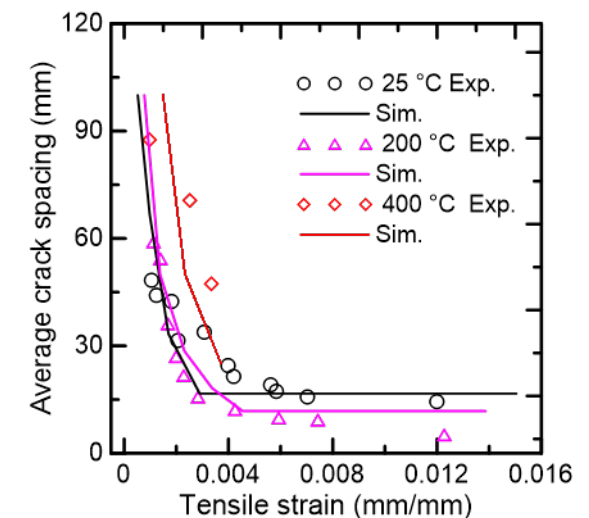
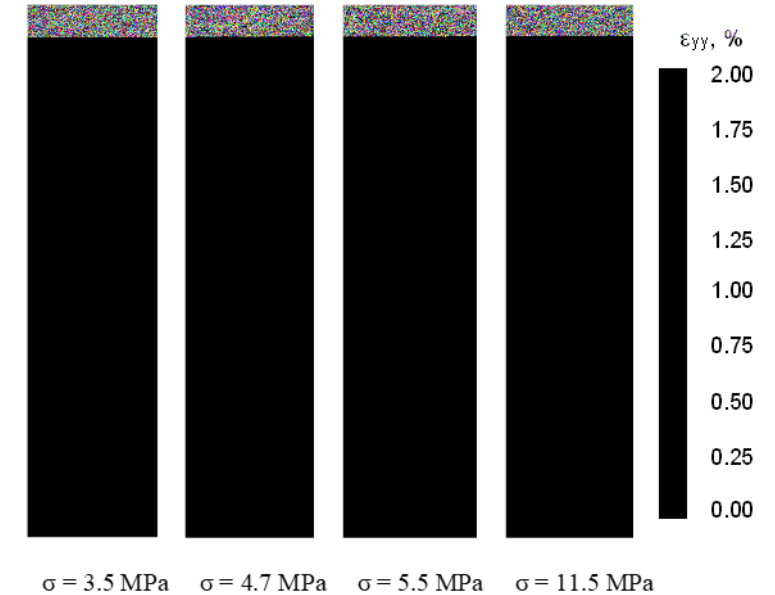
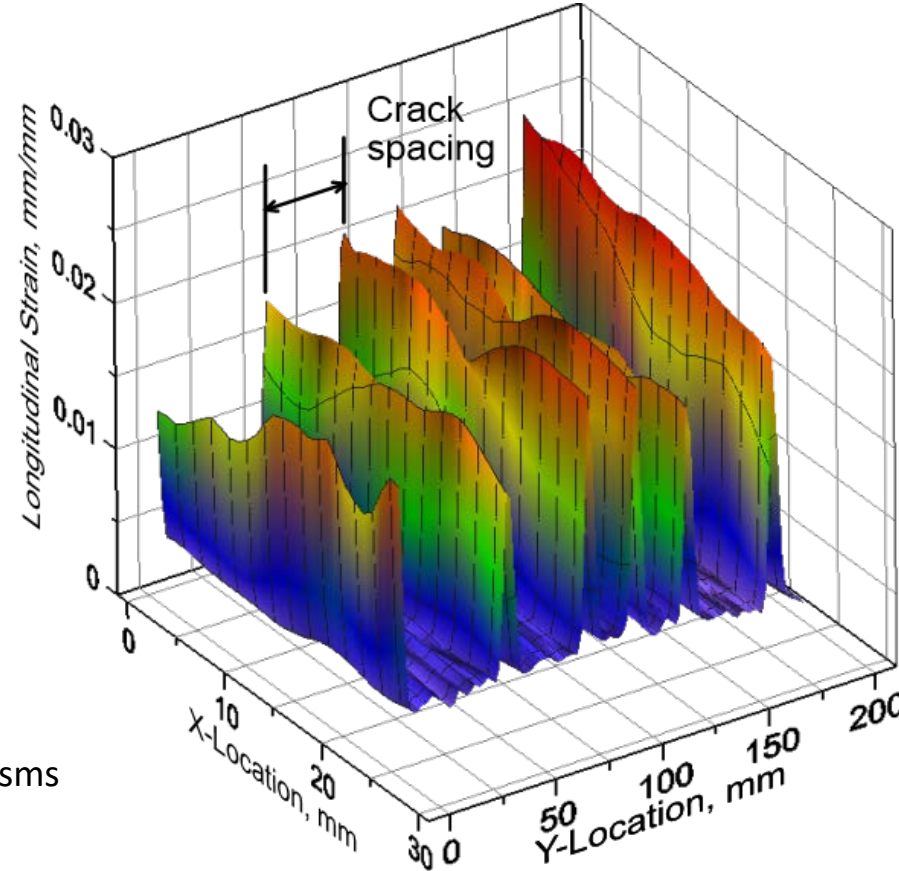


Stress-Crack Width Relationship

- Non-contact measurement
- Quasi-static to high speed
- Single crack and multiple cracks

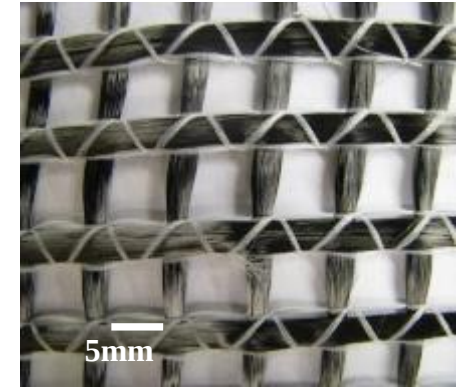
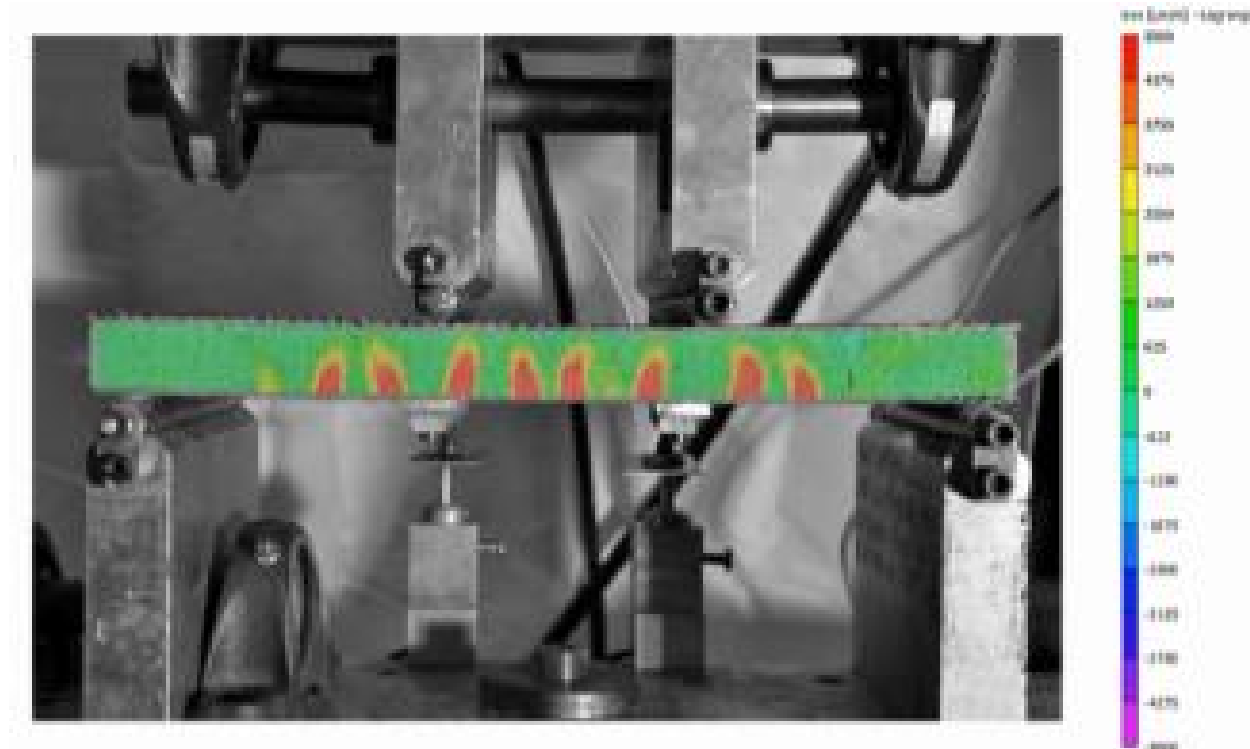
Ductility Induced enhanced behavior for structural composite applications

- Evolution of Crack Spacing in TRC
- Multiple cracking in tension
- Tension stiffening
- Development of parallel cracks
- Indication of toughening mechanisms
- Characteristic length in numerical modelling



Rambo, D. A. S., Yao, Y., et al. (2017). Experimental investigation and modelling of the temperature effects on the tensile behavior of textile reinforced refractory concretes. Cem. Concr. Compos. 75, 51-61.

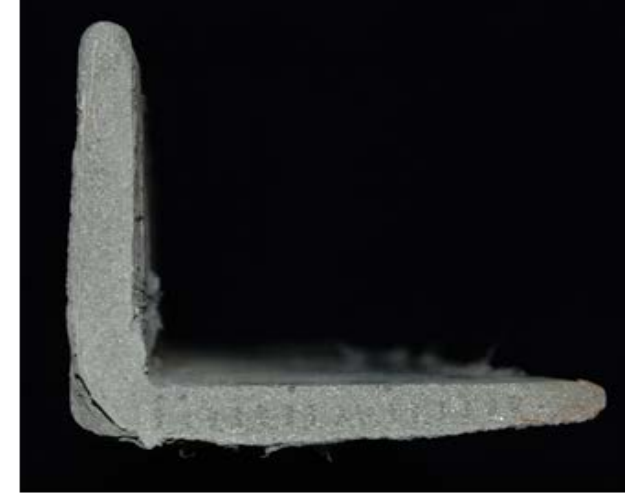
Case Studies addressing Textile Reinforced Concrete



First row from Left to right

1. Aerated concrete textile Sandwich composites
2. Ductile response of sandwich composite Panels
3. Carbon textile
4. Carbon textile beam-column repair

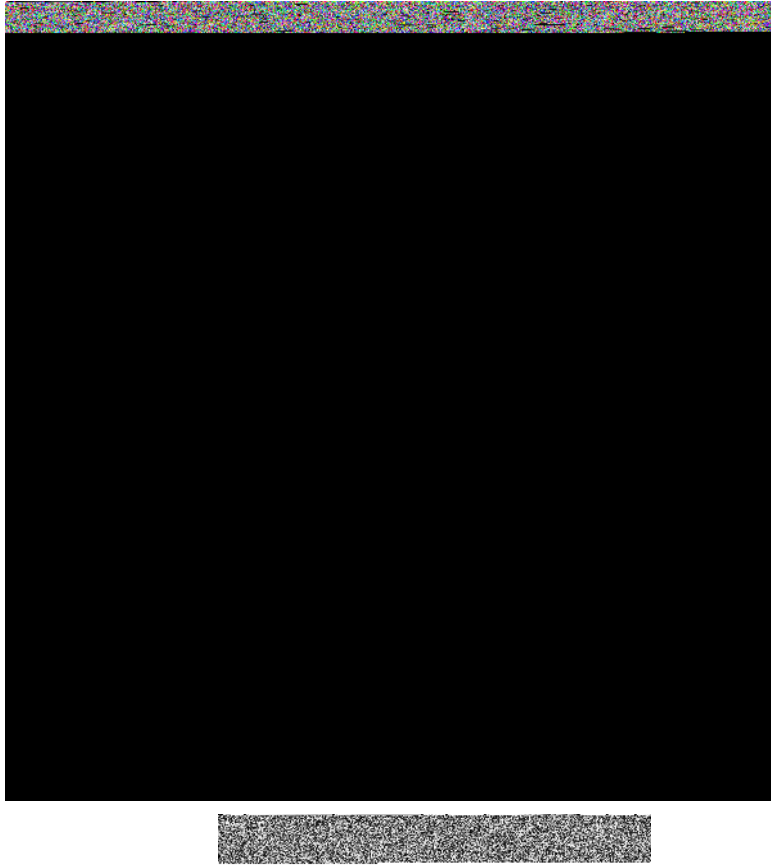
Pultruded Full Size TRC Structural Shapes



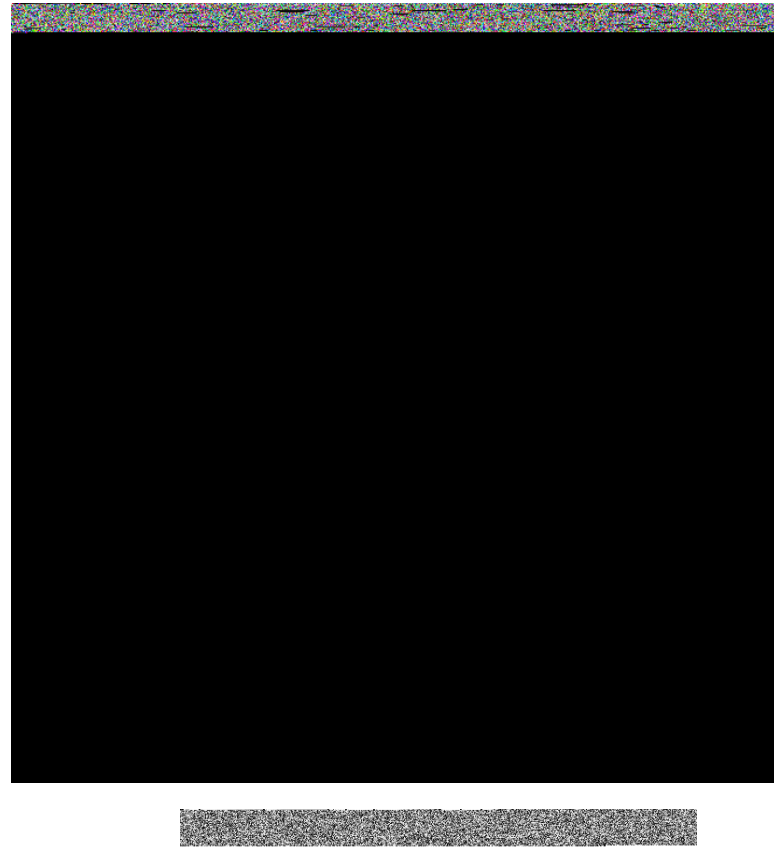
Cross section of pultruded shapes with TRC laminates

Tension and flexural response of PP Composite , $V_f=2.5\%$, with Analytical modeling

MAC, $V_f=2.5\%$, Tension Test

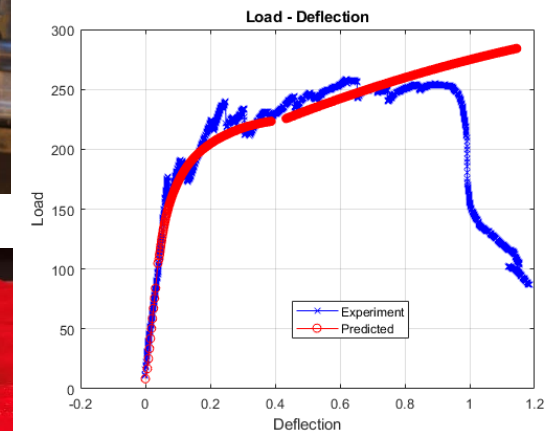
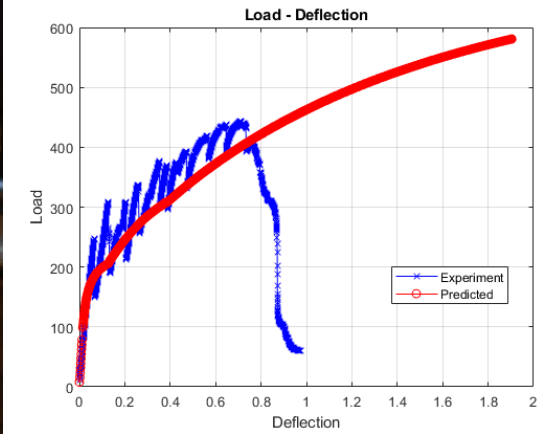
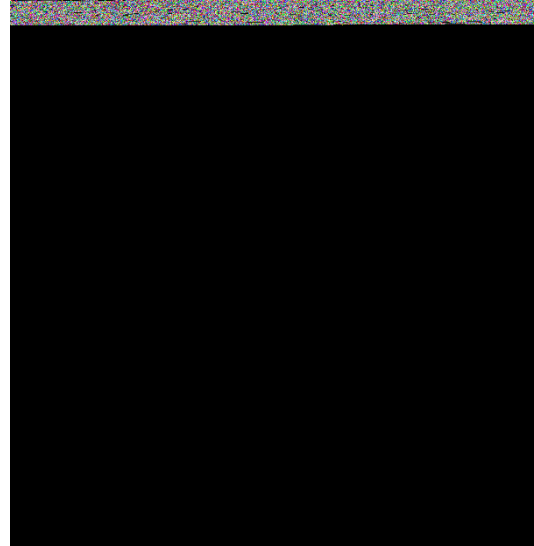
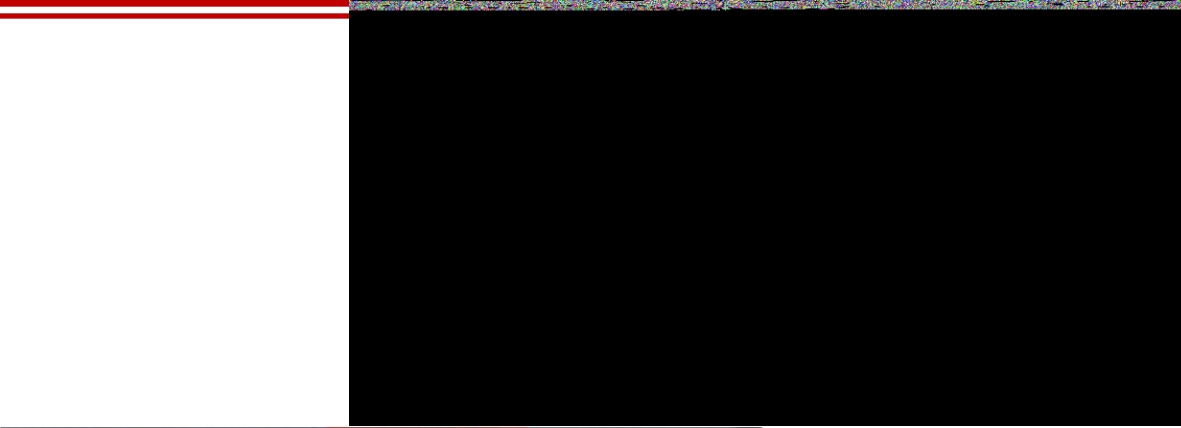


MAC, $V_f=2.5\%$, Flexural, simulation of test



Flexural Modeling for Structural Sections

28

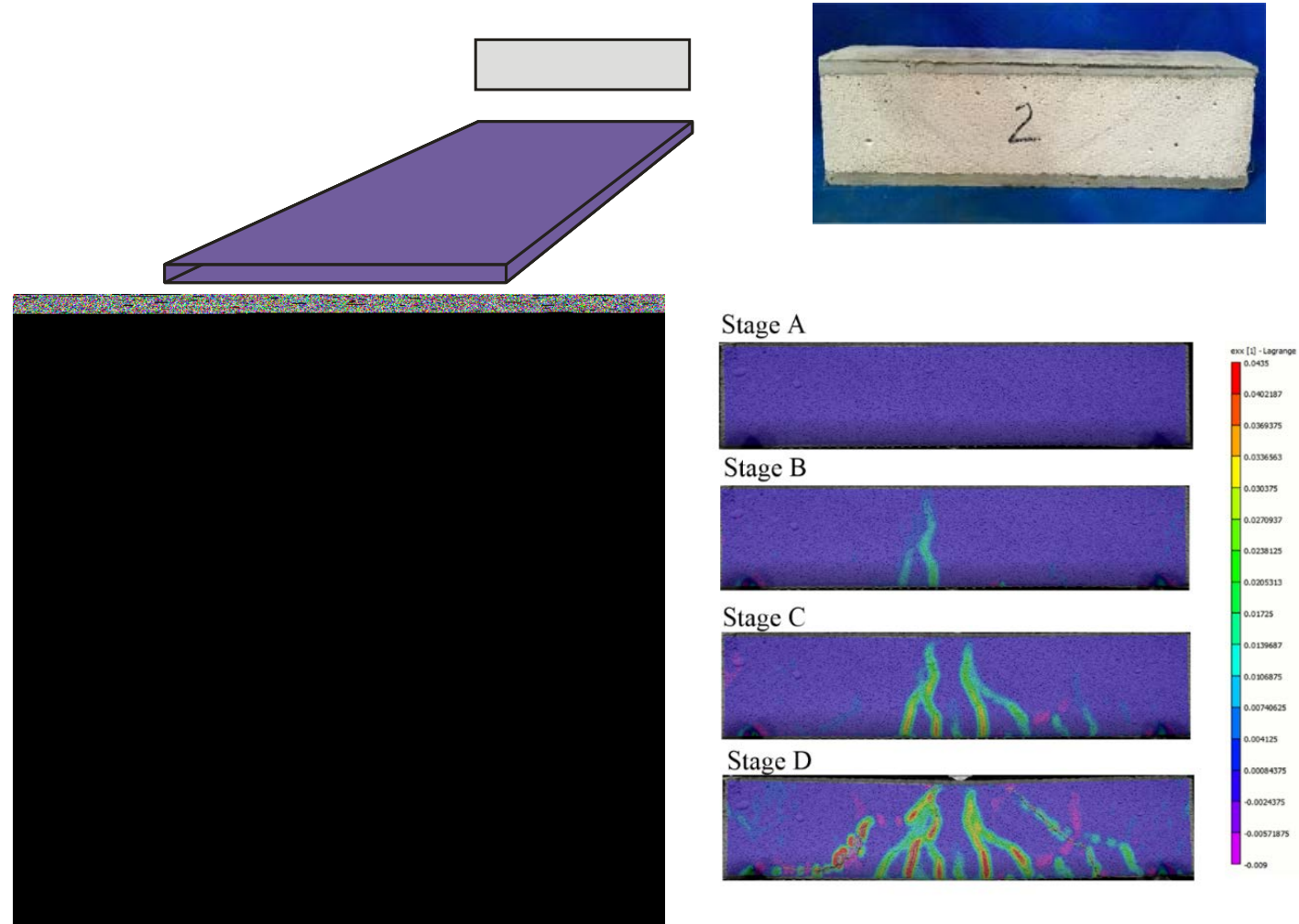
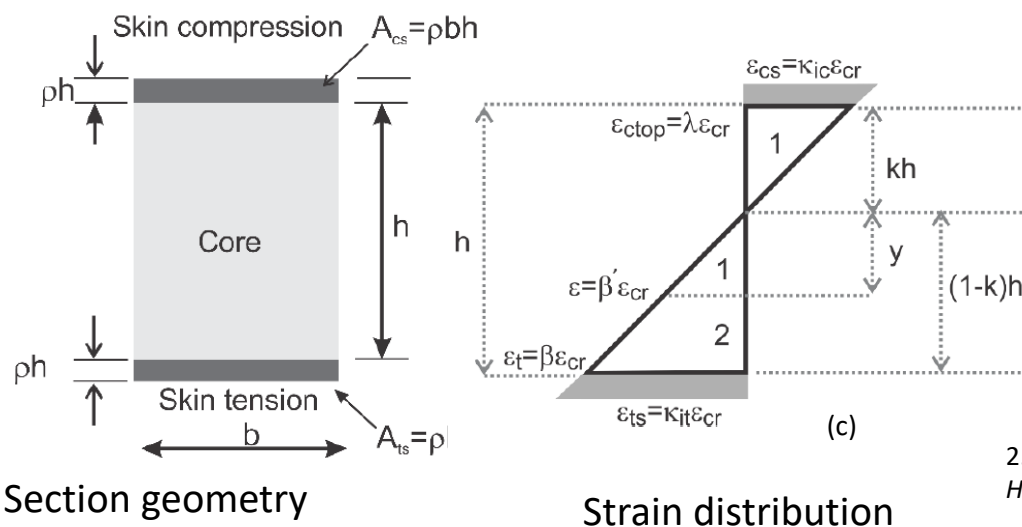


- I-section under Four-point bending test
- $80(b) \times 180(d) \times 1800(L)$ mm, t_w and $t_f = 12.5$ mm
- Textile fabrics: Carbon Textile Reinforcement

Pleesudjai, C., & Mobasher, B. (2023). Analytical moment-curvature solutions for generalized textile-reinforced concrete sections. *Engineering Structures*, 276(May 2022), 115317. <https://doi.org/10.1016/j.engstruct.2022.115317>

Flexural Modeling of Sandwich Composites

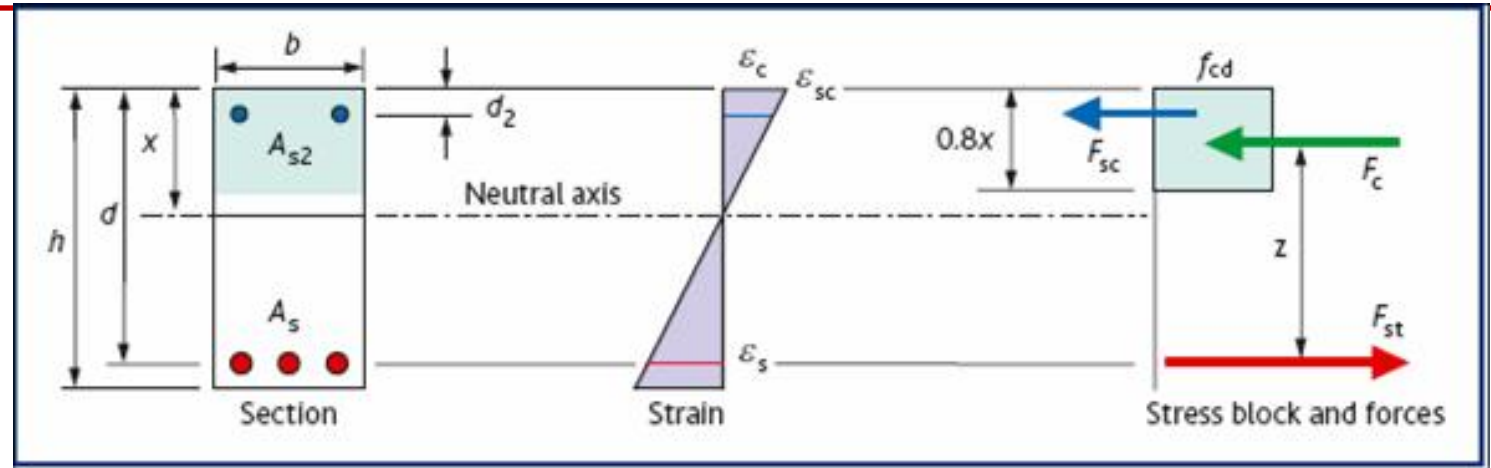
- “TRC-AAC”, brittle core, a shear dominated response, and deflection softening
- “TRC-FRAC”, Ductile core prevented premature shear failure, strain hardening in zone III
- Static test results of: Size A (50x50mm), Size B (50x100)mm, AAC vs FRAC cores
- Damage mechanisms in sandwich composites cannot be explained by a single mode of failure, but the interaction of different modes.



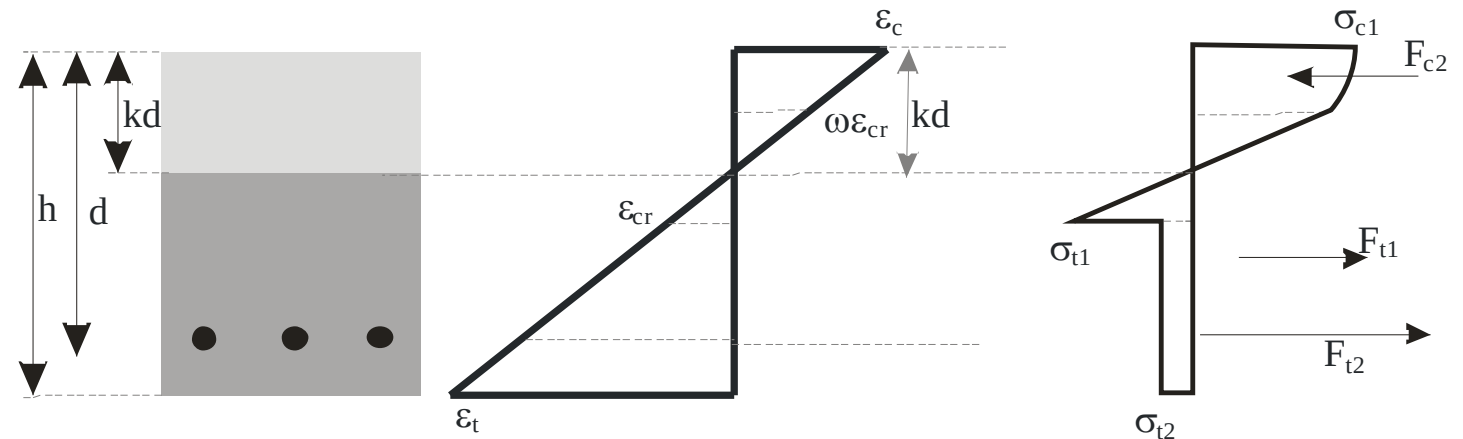
2. Pleesudjai, C., & Mobasher, B. (2023). Parametric Modeling of Flexural Response of Sandwich Composites. In *Strain Hardening Cementitious Composites. SHCC 2022* (pp. 199–208). Springer International Publishing.

Flexure design of RC

- Cross sectional analysis
- Plane section remains plane
- Compression: stress block method
- **Tension: ignored**
- Ultimate limit state capacity



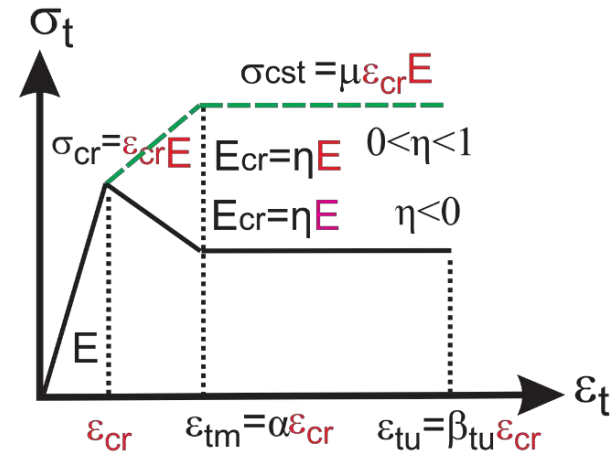
- Flexural design with rebar and fiber reinforcement.
- Stress distribution analogous to Whitney's stress block diagram
- Tensile stress-strain response applied over the tensile domain
- Integrate to compute the moment capacity



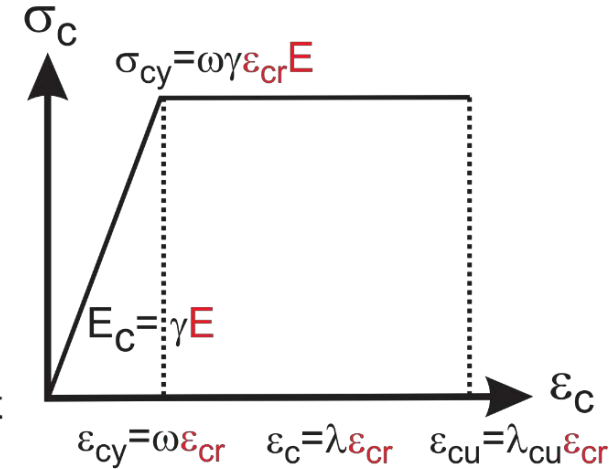
$$\mu = \frac{\sigma_{t2}}{\sigma_{t1}} \quad M_{ult} = \left[\frac{\mu \sqrt{f'_c}}{15.55\mu + 2\sqrt{f'_c}} \right] bd^2 E \epsilon_{cr}$$

Parameterized Material Models

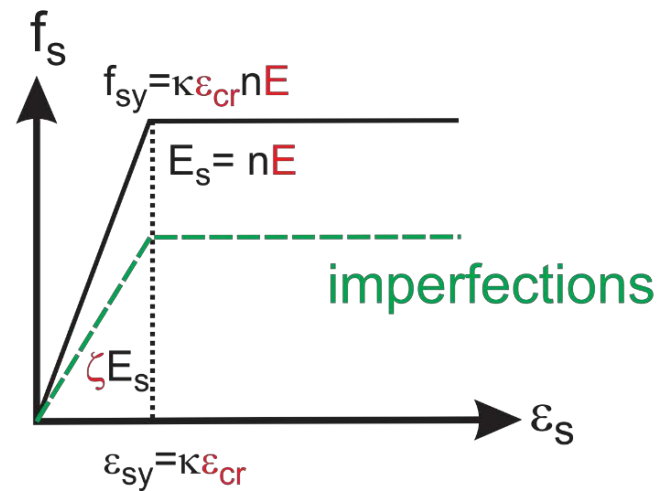
Tension



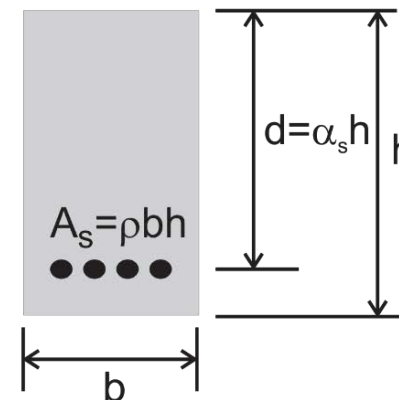
Compression



Steel rebar



Cross section



Derivation of Moment-Curvature Relationship

Impose $0 \leq \varepsilon_t \leq \varepsilon_{tu}$

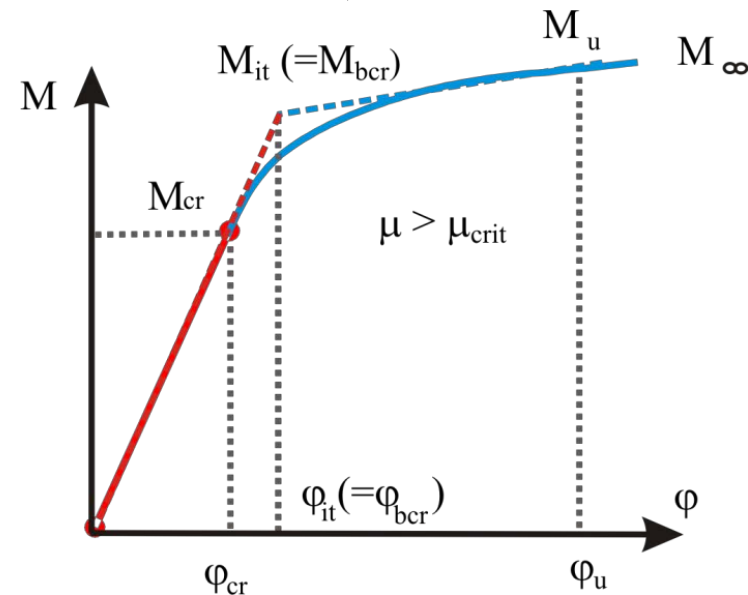
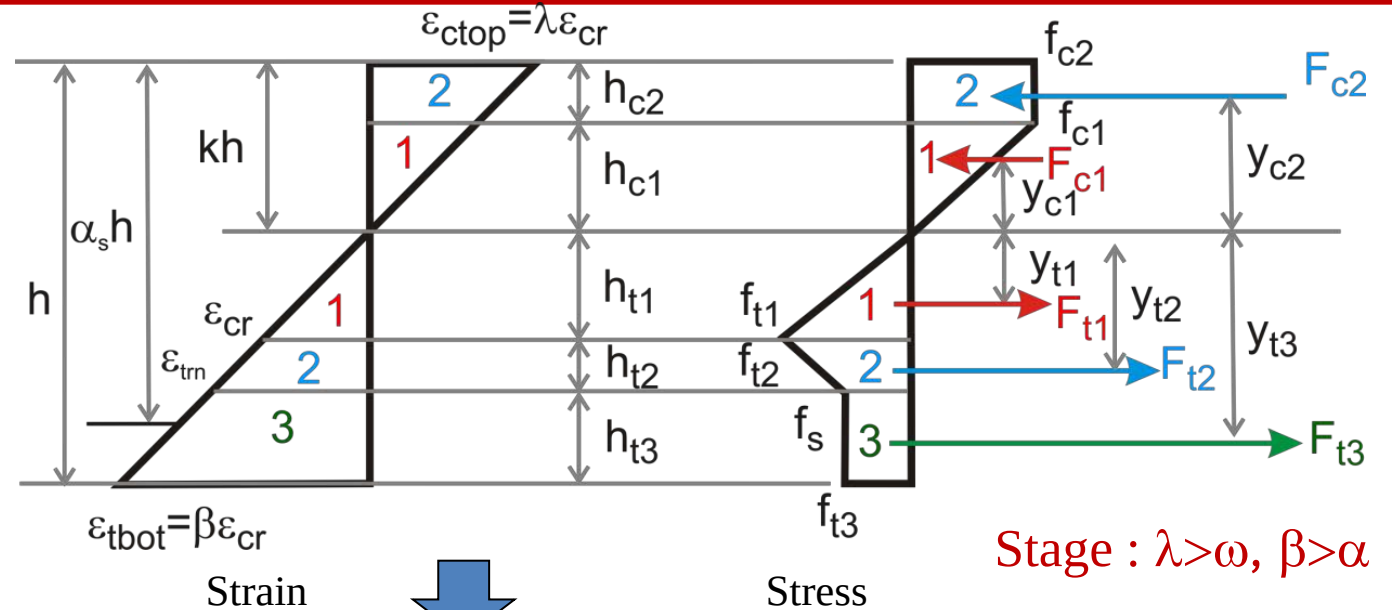
Strain Distribution
Stress Delivery

$\Sigma F = 0$, determine k

Moment:

$$M = \Sigma F_{ci} y_{ci} + \Sigma F_{ti} y_{ti}$$

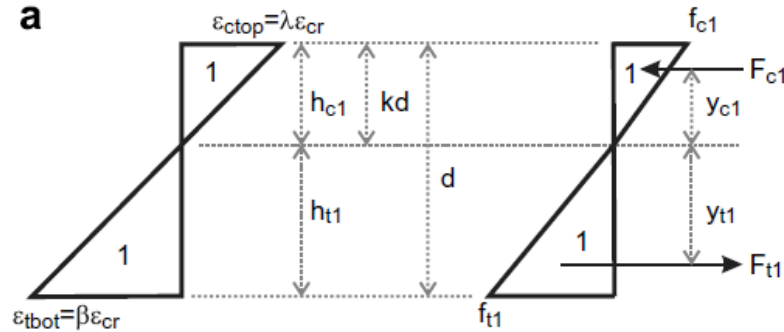
Curvature: $\phi = \varepsilon_c / kh$



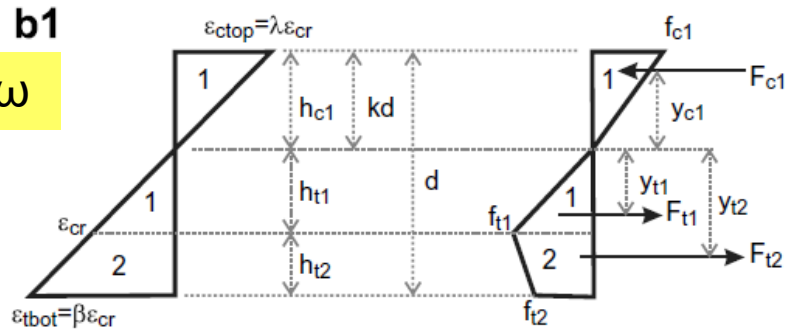
Simplified bilinear
moment-curvature

Stress and Strain Distributions in Different Stages

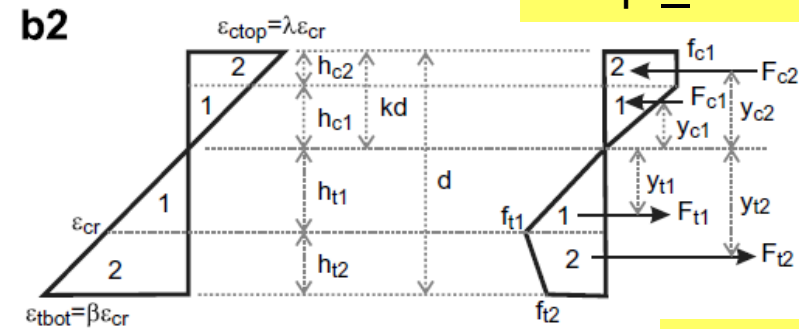
$$0 < \beta \leq 1 \text{ and } \lambda \leq \omega$$



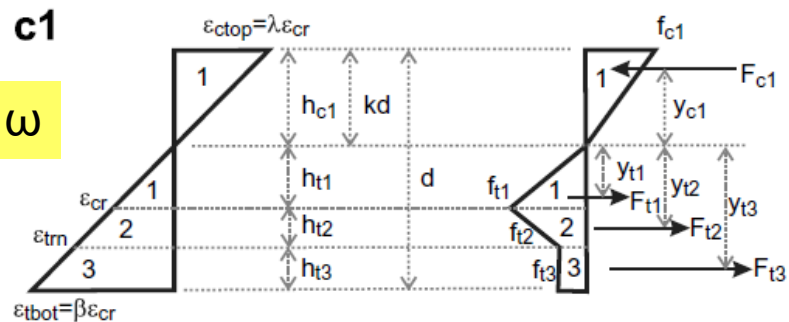
$$1 < \beta \leq \alpha \text{ and } \lambda \leq \omega$$



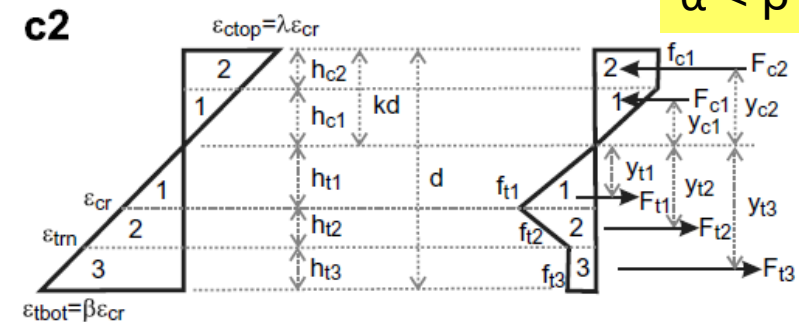
$$1 < \beta \leq \alpha \text{ and } \lambda > \omega$$



$$\alpha < \beta \leq \beta_{tu} \text{ and } \lambda \leq \omega$$



$$\alpha < \beta \leq \beta_{tu} \text{ and } \lambda > \omega$$



Closed Form Solutions for Strain Hardening/Softening materials from bilinear to quadrilinear, reinforced, sandwich

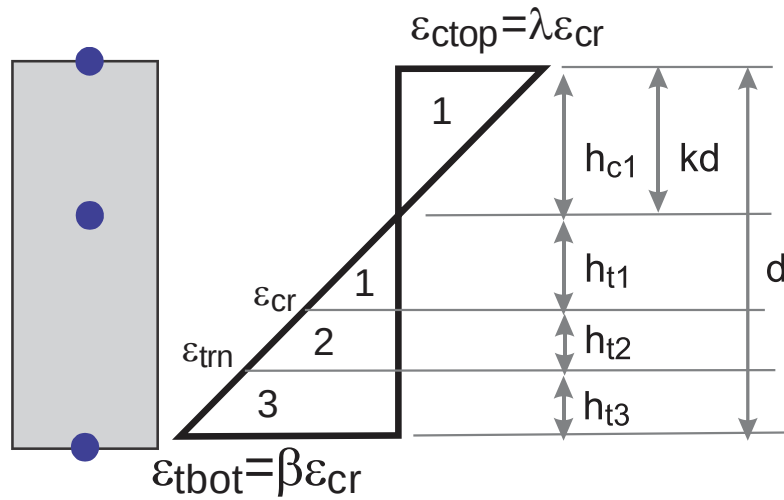
Table 6
Neutral axis depth ratio, normalized moment and curvature for each stage of normalized tensile strain at bottom fiber (β)

Stage	k	M'	ϕ'
1	$k_1 = \begin{cases} \frac{1}{2} & \text{for } \gamma = 1 \\ \frac{-1+\sqrt{\gamma}}{-1+\gamma} & \text{for } \gamma < 1 \text{ or } \gamma > 1 \end{cases}$	$M'_1 = \frac{2\beta[(\gamma-1)k_1^3 + 3k_1^2 - 3k_1 + 1]}{1-k_1}$	$\phi'_1 = \frac{\beta}{2(1-k_1)}$
2.1	$k_{21} = \frac{\beta^2\gamma + D_{21} - \sqrt{\gamma^2\beta^4 + D_{21}\gamma\beta^2}}{D_{21}}$ $D_{21} = \eta(\beta^2 - 2\beta + 1) + 2\beta - \beta^2\gamma - 1$	$M'_{21} = \frac{(2\beta\gamma + C_{21})k_{21}^3 - 3C_{21}k_{21}^2 + 3C_{21}k_{21} - C_{21}}{1-k_{21}}$ $C_{21} = \frac{-2\eta\beta^3 + 3\eta\beta^2 - 3\beta^2 - \eta + 1}{\beta^2}$	$\phi'_{21} = \frac{\beta}{2(1-k_{21})}$
2.2	$k_{22} = \frac{D_{22}}{D_{22} + 2\omega\gamma\beta}$ $D_{22} = \eta(\beta^2 - 2\beta + 1) + 2\beta + \omega^2\gamma - 1$	$M'_{22} = (3\omega\gamma + C_{22})k_{22}^2 - 2C_{22}k_{22} + C_{22}$ $C_{22} = \frac{2\eta\beta^3 - 3\eta\beta^2 + 3\beta^2 - \omega^3\gamma + \eta - 1}{\beta^2}$	$\phi'_{22} = \frac{\beta}{2(1-k_{22})}$
3.1	$k_{31} = \frac{D_{31} - \sqrt{\gamma\beta^2 D_{31}}}{D_{31} - \beta^2\gamma}$ $D_{31} = \eta(\alpha^2 - 2\alpha + 1) + 2\mu(\beta - \alpha) + 2\alpha - 1$	$M'_{31} = \frac{(C_{31} - 2\beta\gamma)k_{31}^3 - 3C_{31}k_{31}^2 + 3C_{31}k_{31} - C_{31}}{k_{31} - 1}$ $C_{31} = \frac{3(\mu\beta^2 - \mu\alpha^2 - \eta\alpha^2 + \alpha^2) + 2\eta\alpha^3 + \eta - 1}{\beta^2}$	$\phi'_{31} = \frac{\beta}{2(1-k_{31})}$
3.2	$k_{32} = \frac{D_{32}}{D_{32} + 2\omega\gamma\beta}$ $D_{32} = \omega^2\gamma + \eta\alpha^2 + 2(\mu\beta - \eta\alpha - \mu\alpha + \alpha) + \eta - 1$	$M'_{32} = (C_{32} + 3\omega\gamma)k_{32}^2 - 2C_{32}k_{32} + C_{32}$ $C_{32} = \frac{3(\mu\beta^2 - \mu\alpha^2 - \eta\alpha^2 + \alpha^2) + 2\eta\alpha^3 - \omega^3\gamma + \eta - 1}{\beta^2}$	$\phi'_{32} = \frac{\beta}{2(1-k_{32})}$

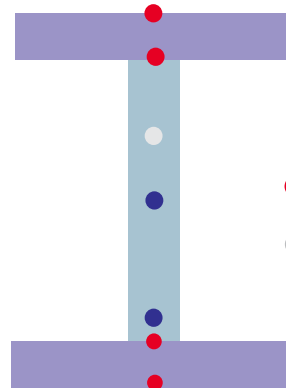
$$M = M' M_{cr} \quad \phi = \phi' \phi_{cr}$$

$$M_{cr} = \frac{1}{6} b d^2 E \varepsilon_{cr} \quad \phi_{cr} = \frac{2\varepsilon_{cr}}{d}$$

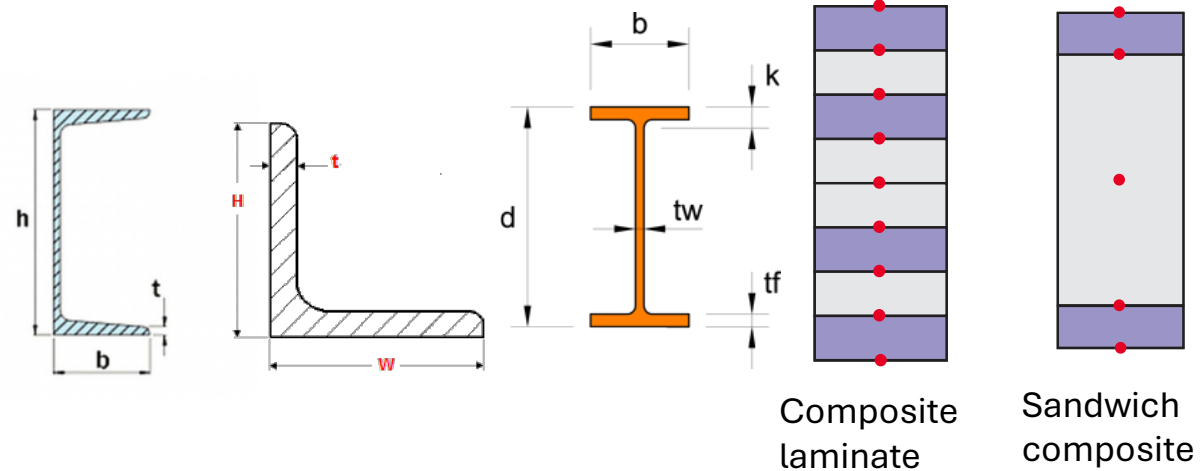
Geometrical and Material Control Points



Rectangular
section

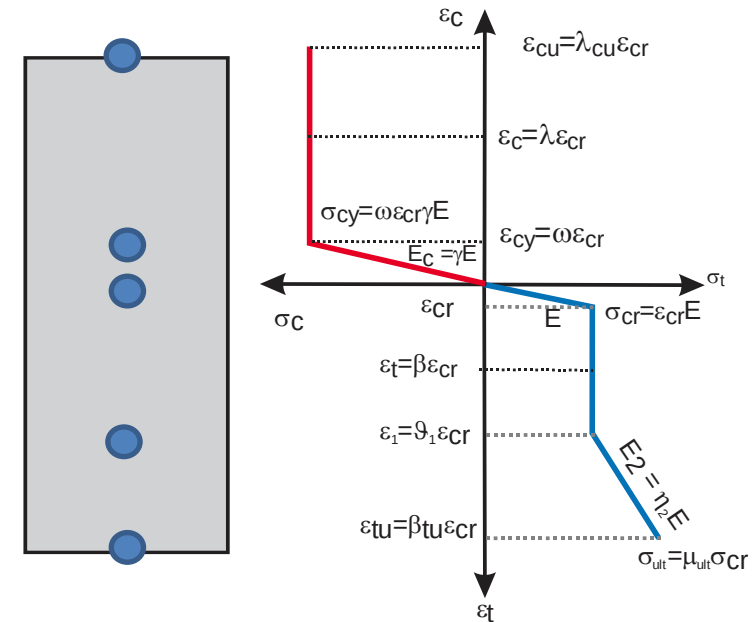


- stress-strain control point
- geometrical control point
- neutral axis control point

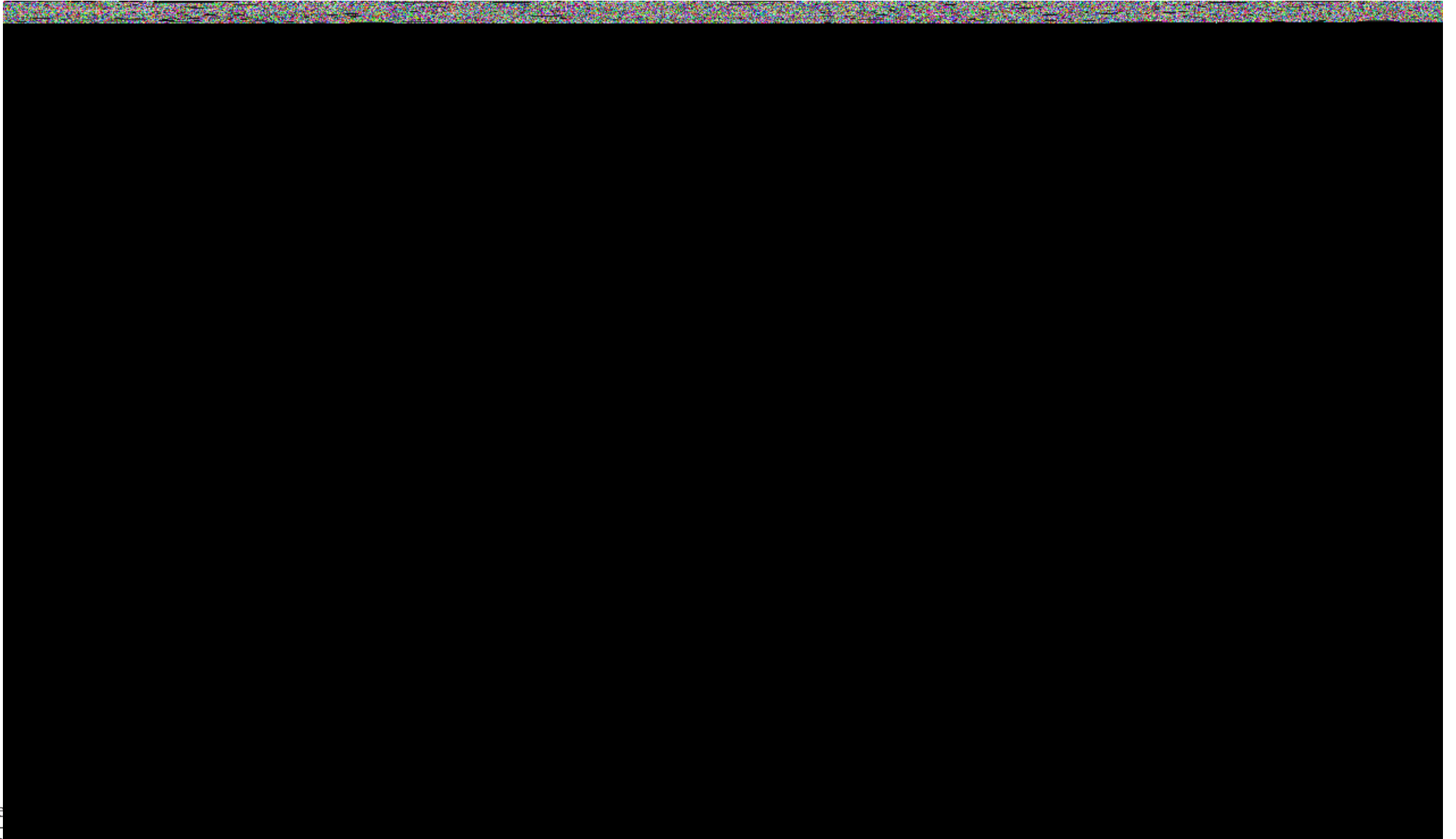


Composite
laminate

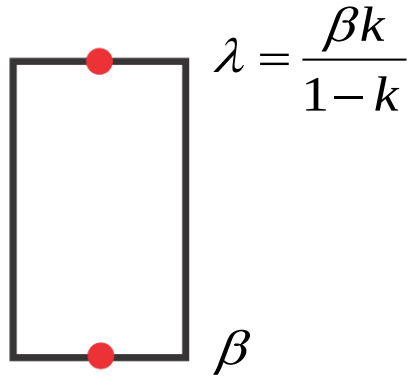
Sandwich
composite



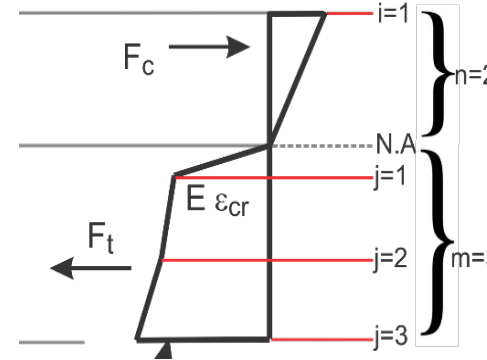
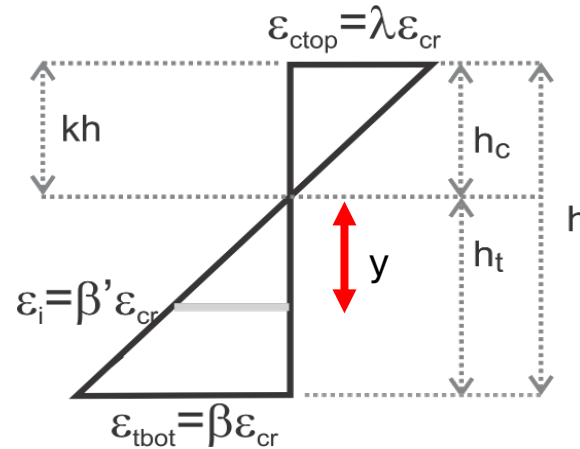
Using stress-strain, internal equilibrium, and section geometry, obtain moment-curvature to get load-deflection response



Solve for the Neutral axis, k , in closed-form for each applied strain, β



Start with $\beta, k, \sigma - \epsilon$,
geometry, parameters



$$y = \frac{\beta' \epsilon (1-k) h}{\beta \epsilon_{cr}}$$

$$dy = \frac{(1-k) h}{\beta} d\beta'$$

Change of variables to
unknowns β, k .
Express all equilibrium
in terms of knowns $\sigma - \epsilon$,
and geometry
parameters and
unknowns

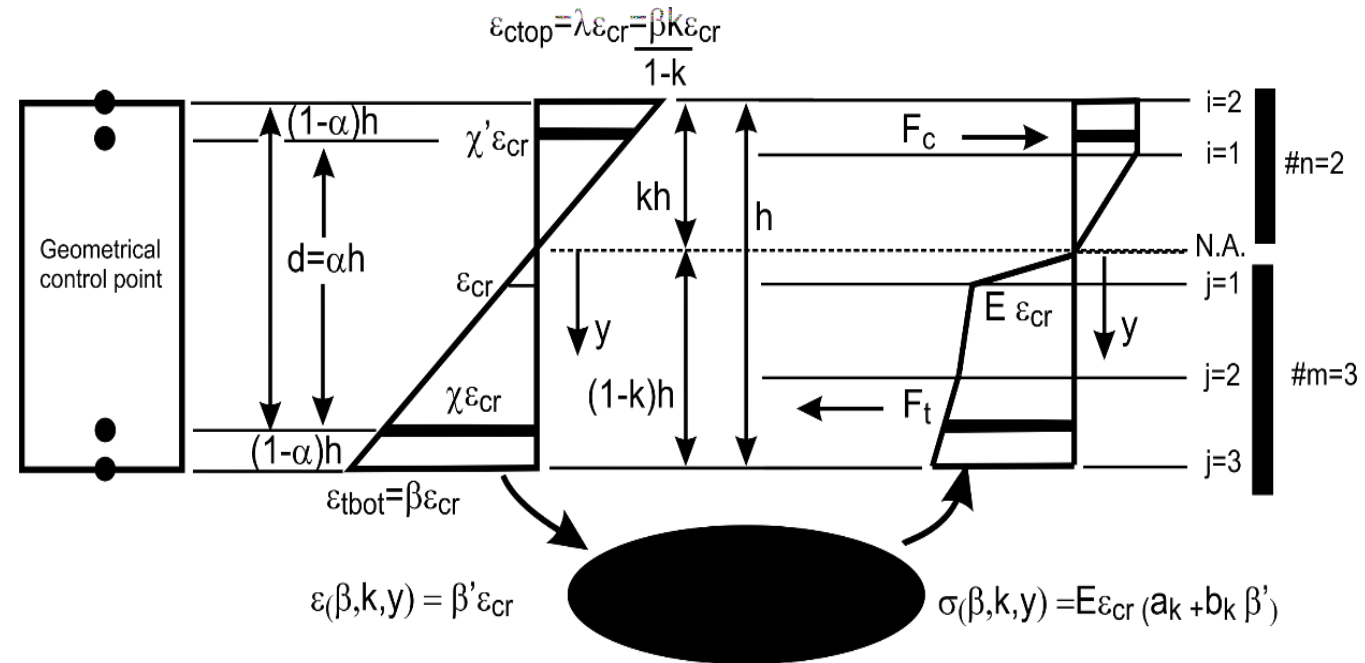
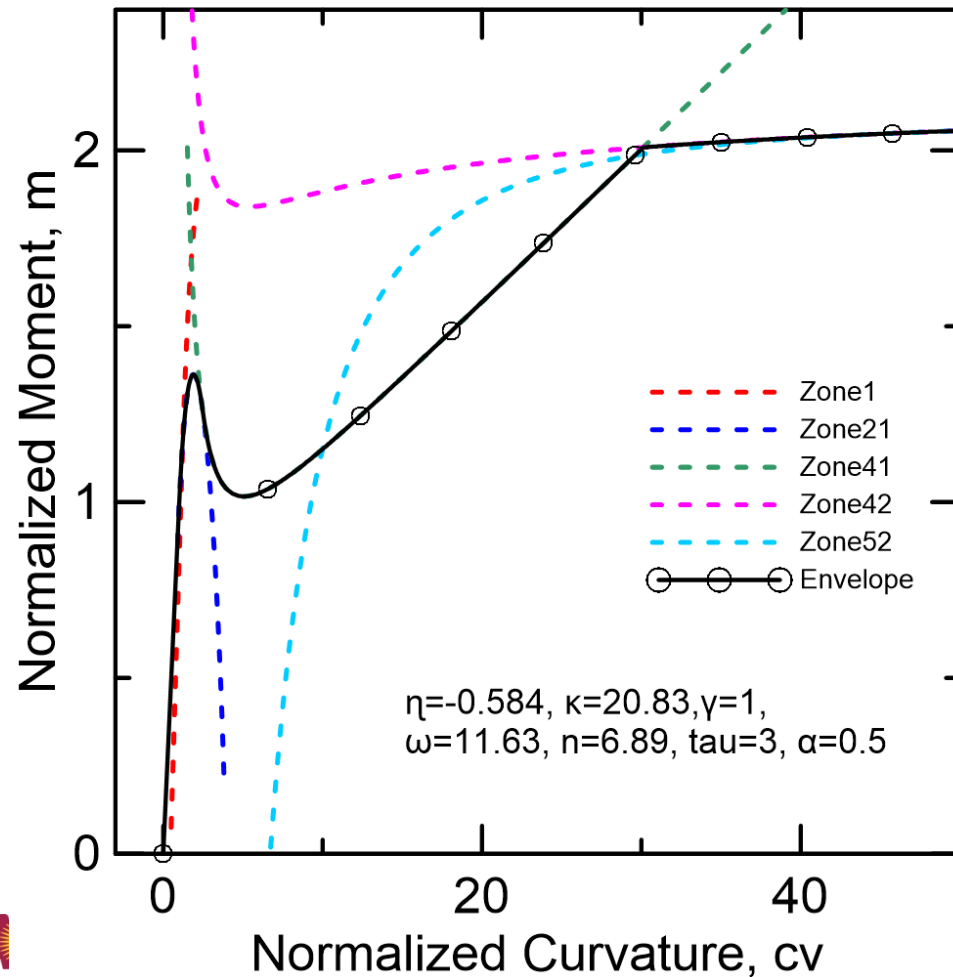
Solve for internal forces, and get the k , and moment as a function of imposed
strain β

$$\sum_0^h F = \int \sigma(\epsilon) dA = b \int_0^h \sigma(\epsilon) dy = 0$$

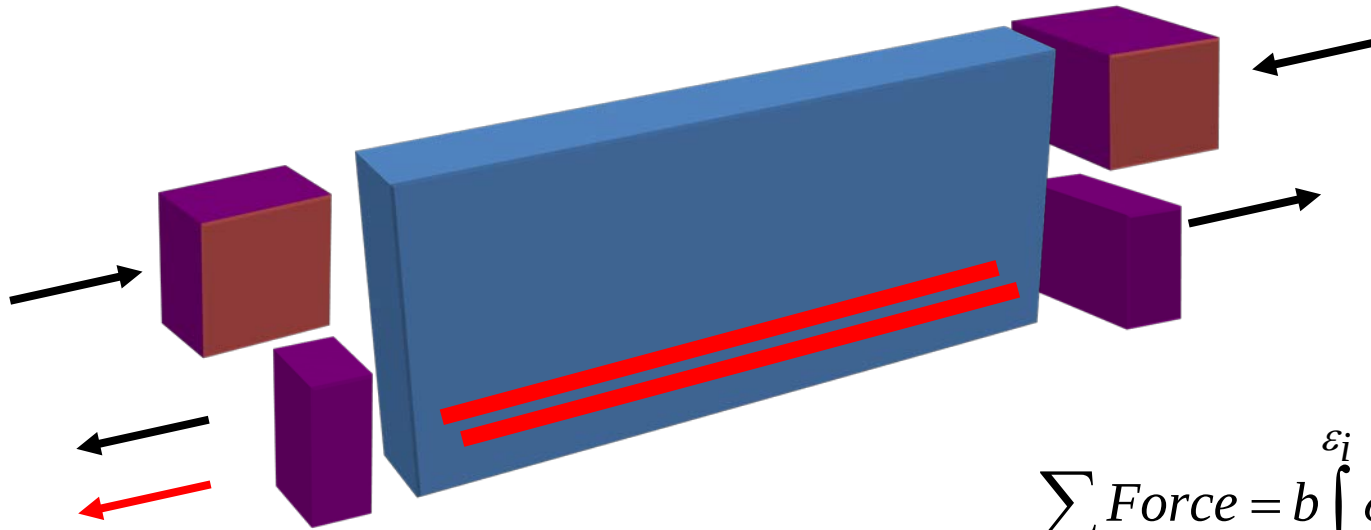
$$\sum_0^h F = \frac{bh(1-k)}{\beta} \int_{-\lambda}^{\beta} \sigma(\beta') \frac{(1-k)h}{\beta} d\beta'$$

$$\sum M = \int_0^h dM(y) = b \int_0^h \sigma(\beta') \bar{y} dy = \epsilon_{cr} E b \int_{-\lambda}^{\beta} (a_k + b_k \beta') \cdot \frac{\beta' (1-k) h}{\beta} \cdot \frac{(1-k) h}{\beta} d\beta'$$

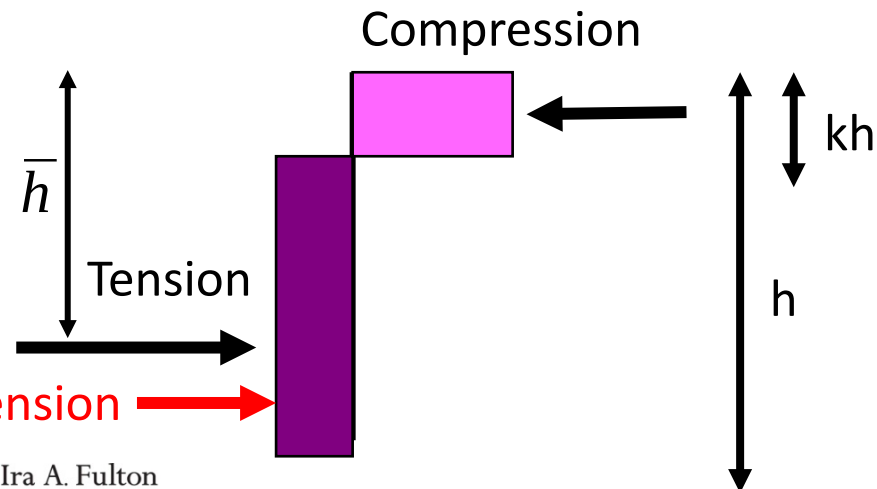
Moment-Curvature Envelop Curve from Modal Interaction



Mean Stress distribution in an intermediate stage



$$\sum Force = b \int_0^{\varepsilon_i} \sigma(\varepsilon) \frac{(1-k)h_0}{\beta \varepsilon_{cr}} d\varepsilon = F_{cc} + F_{ct} + F_{st} + F_{sc} = 0$$

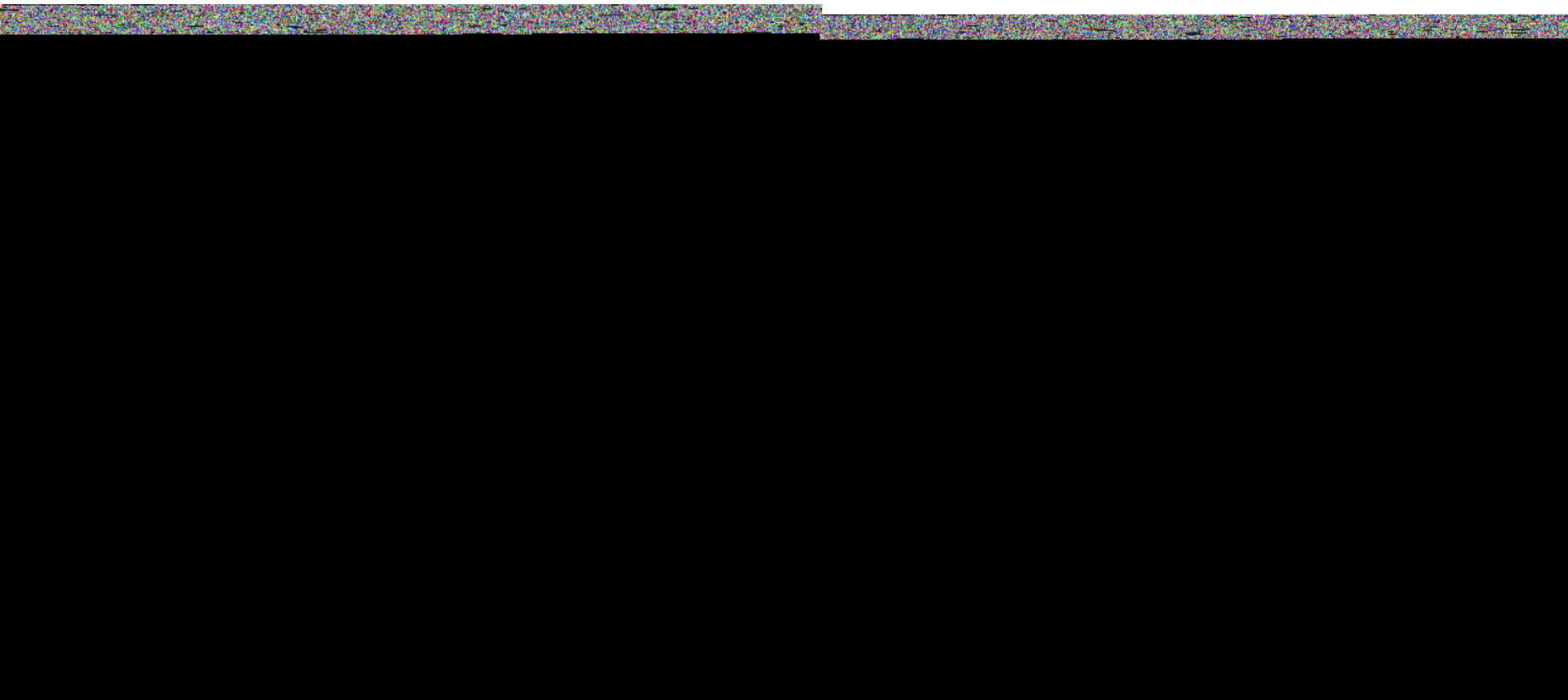


$$\bar{h} = \frac{b \int_0^{\varepsilon_i} h(\varepsilon) \sigma(x) dh}{\sum F_{ct}} = \frac{\int_0^{\varepsilon_i} h(\varepsilon) \sigma(\varepsilon) \frac{(1-k)h_0}{\beta \varepsilon_{cr}} d\varepsilon}{F_{ct}}$$

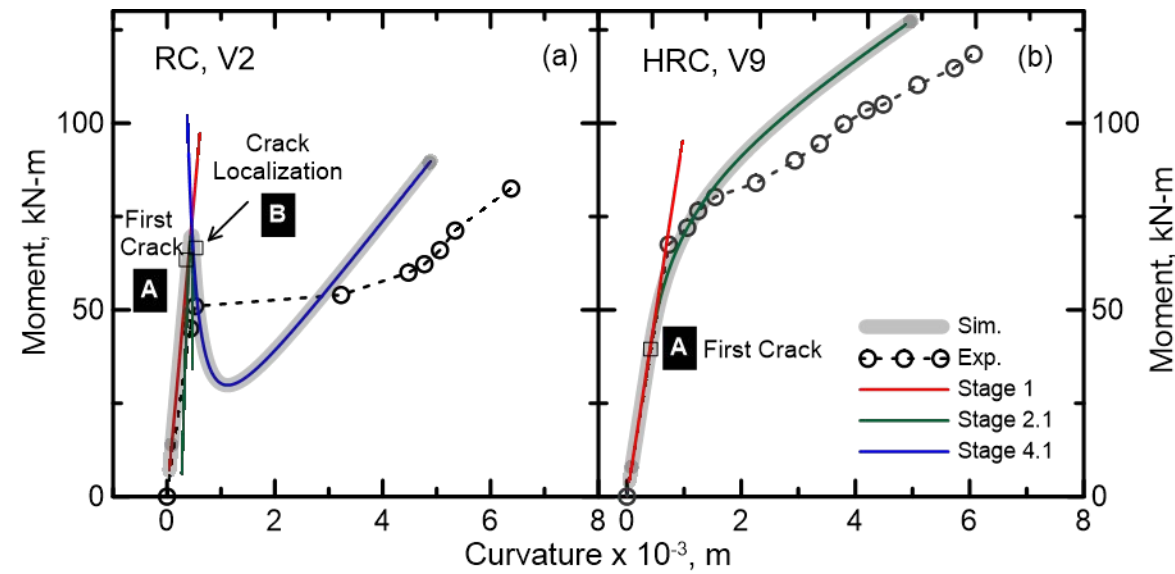
$$\sum M = b \int_0^{\varepsilon_i} \sigma(\varepsilon) \frac{y(1-k)h_0}{\beta \varepsilon_{cr}} d\varepsilon = M_{cc} + M_{ct} + M_{st} + M_{sc} = \bar{M}$$

Efficiency of the reinforcement is in not delaying the localization- yielding of steel main Synergy of fiber and rebar combination

- As soon as post cracking starts, concrete contribution diminishes to less than 10% in plain RC concrete
- Even after yielding of steel, Concrete in the hybrid system is carrying more than twice its steel
- With only half the amount of steel in the hybrid system, the rebars in hybrid extend the serviceability range to a higher overall stiffness and extend to twice the deflection range

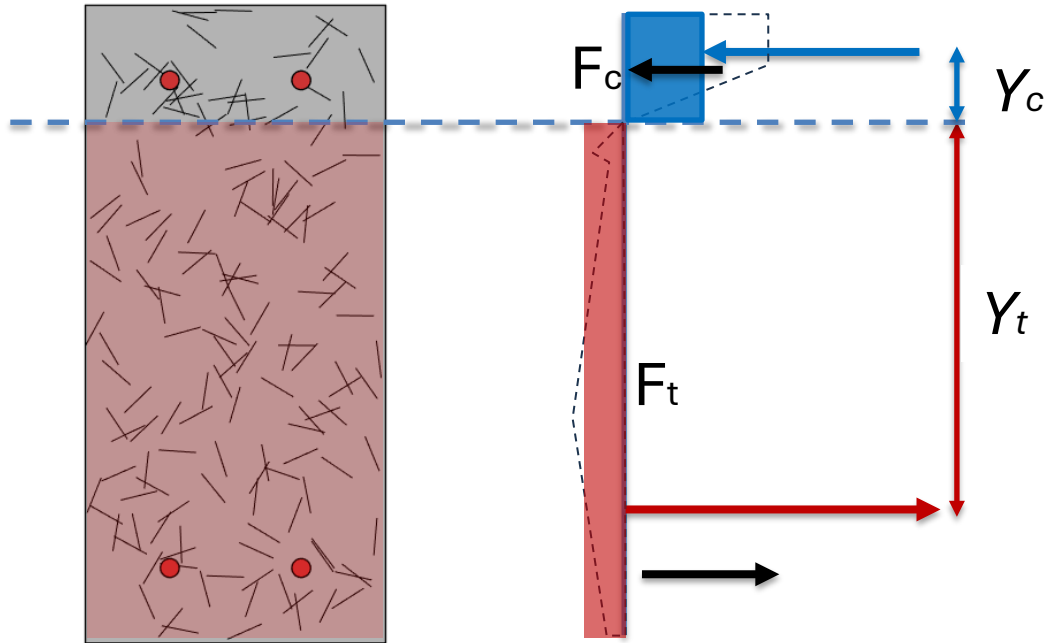


Instability in Reinforced Concrete



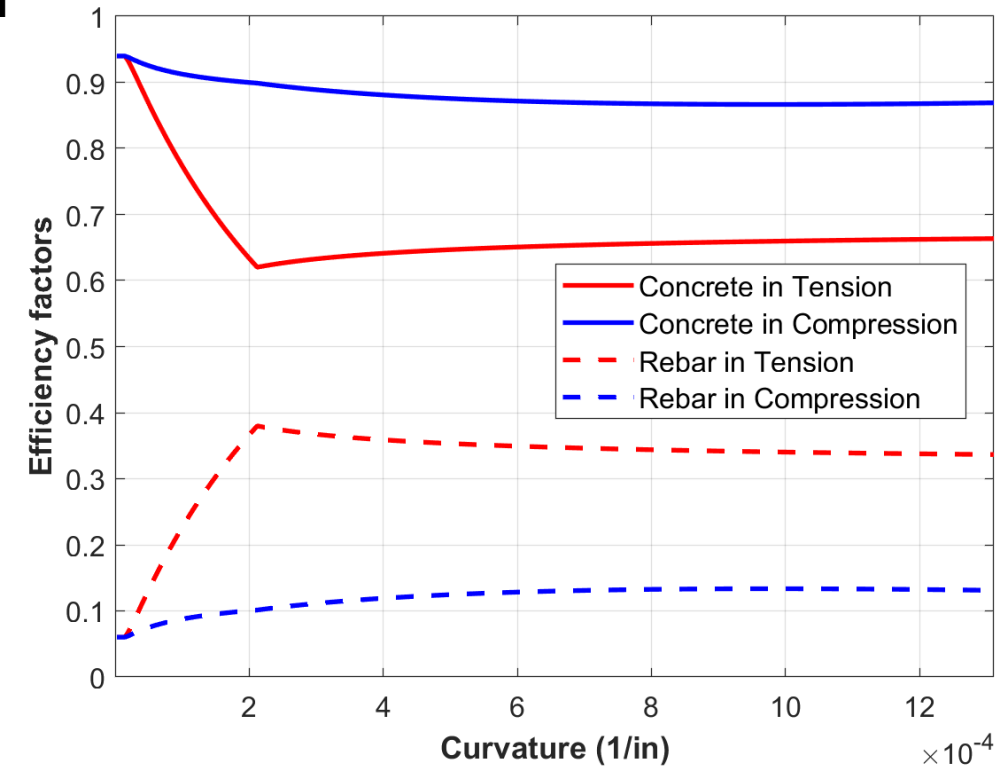
Efficiency of Reinforcements in the section

Efficiency is defined as the fraction of the component to the sum of tension and compression forces at that strain level



$$\eta_{CT} = \frac{(n(\rho_c - \rho_t)\kappa + \omega)\mu}{n(\mu\rho_c + \omega\rho_t)\kappa + \mu\omega}$$

$$\eta_{ST} = \frac{\kappa n \rho_t}{\kappa n \rho_t - k \mu + \mu}$$

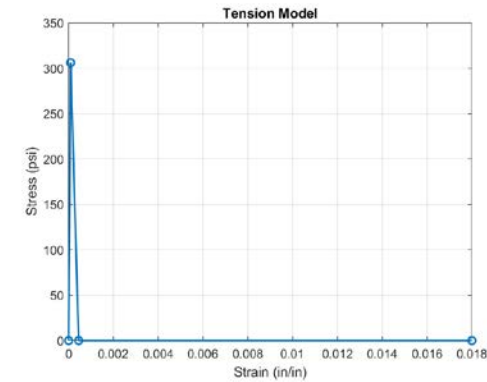
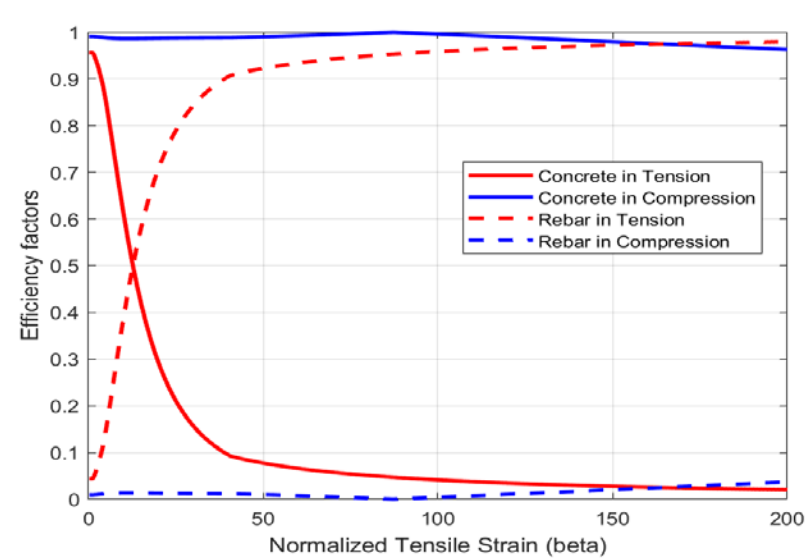
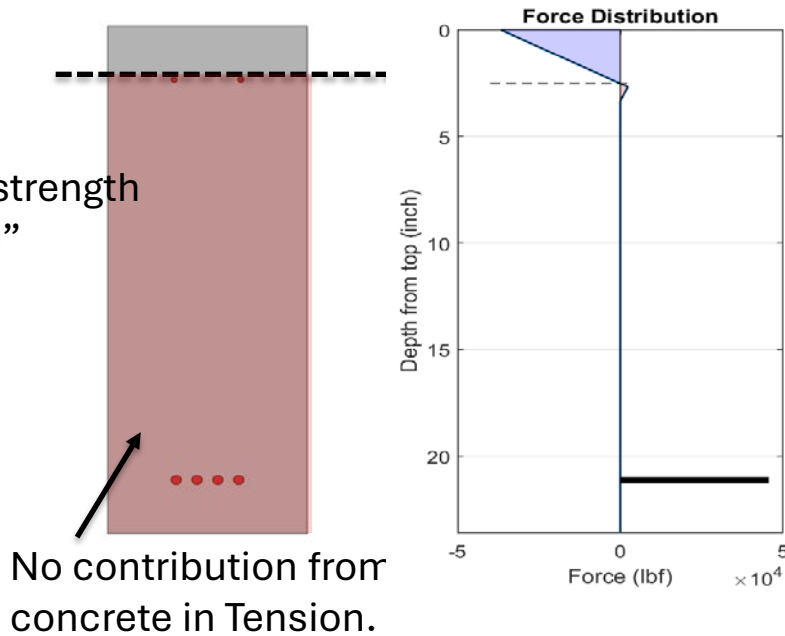


$$\eta_{SC} = \frac{\omega(n(\rho_c - \rho_t)\kappa - \mu)}{n(\mu\rho_c + \omega\rho_t)\kappa + \mu\omega}$$

$$\eta_{SC} = \frac{\rho_c n \kappa}{\omega k + \rho_c n \kappa}$$

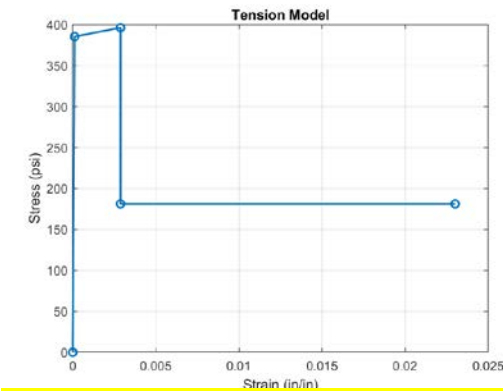
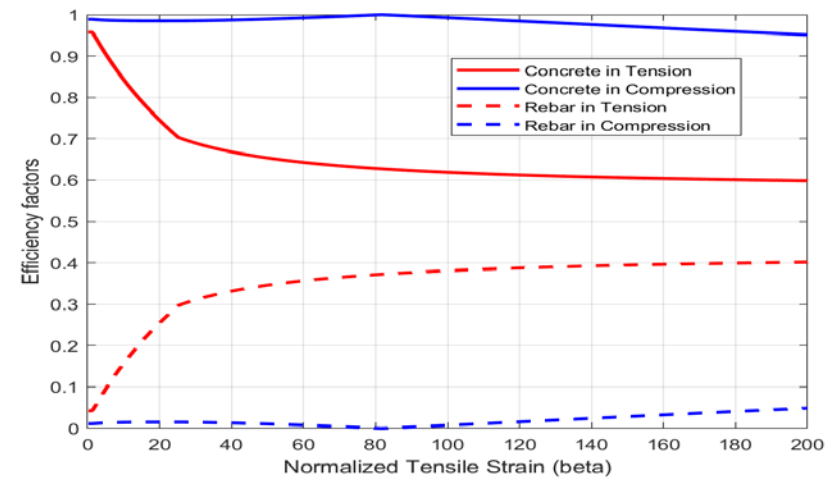
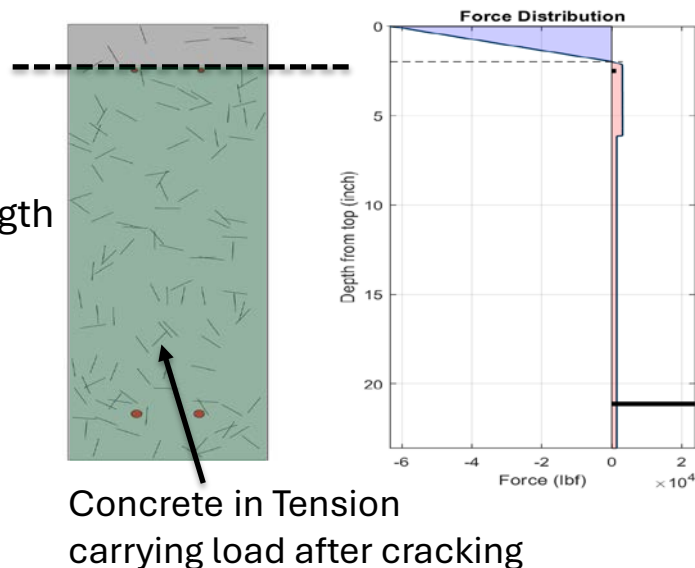
Contribution of Reinforcement in Conventional and Hybrid Design

$\rho = 0.28\%$
 $\mu = 0$ residual strength
 Size = 7.5" x 23.5"



Reinforcement is the primary tension mechanism

$\rho = 0.14\%$
 $\mu = 0.5$ res. strength
 Size = 7.5" x 23.5"



Hybrid Reinforced Section

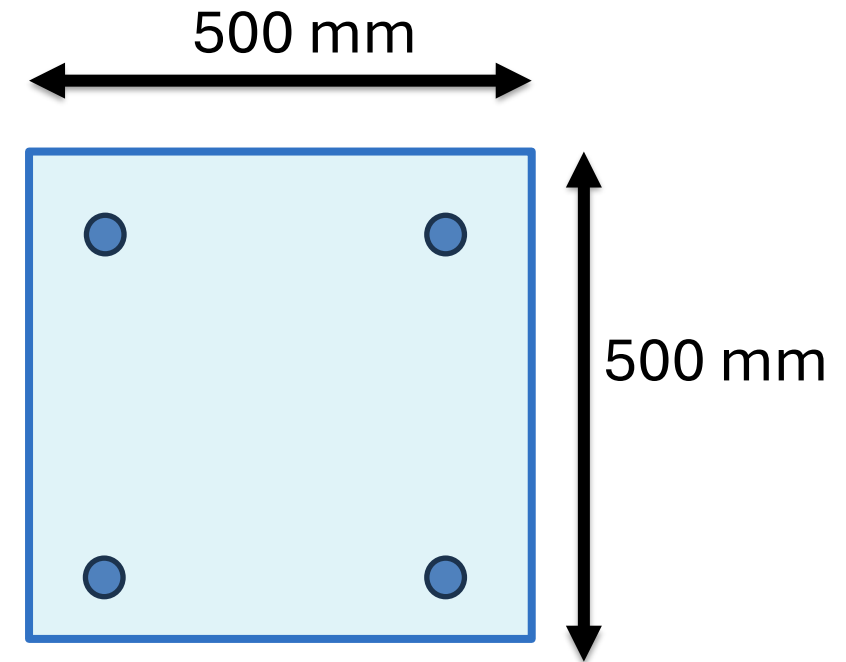
Even at the ultimate level, Concrete carries more than 60:40 wrt to steel at half the reinforcement

Synergy of fiber and rebar combination as Hybrid Reinforcement

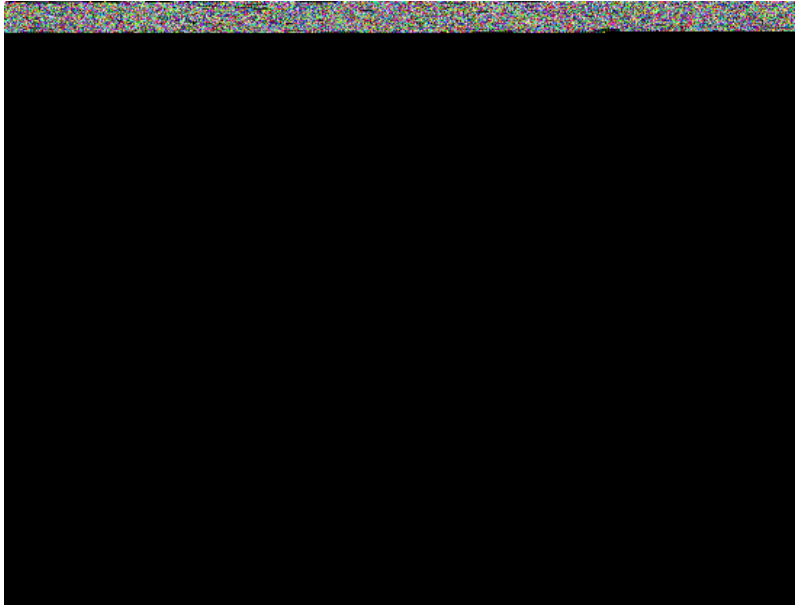
- As soon as post cracking starts, concrete contribution diminishes to less than 10% in plain RC concrete
- With only half the amount of steel in the hybrid system, the rebars in hybrid extend the serviceability range to a higher overall stiffness and extend to twice the deflection range
- Improving the efficiency of the reinforcement in delaying the yielding of steel
- Even after yielding of steel, Concrete in the hybrid system is carrying more than twice the steel



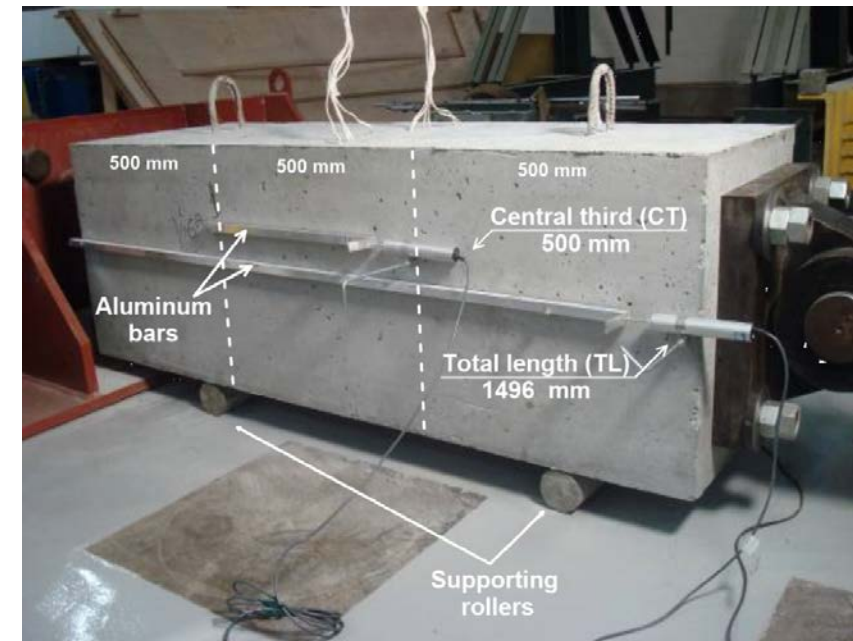
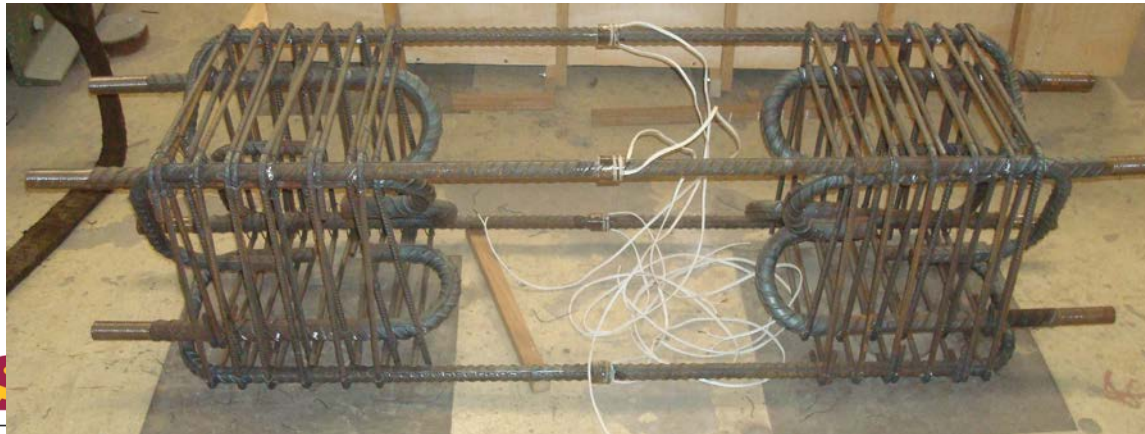
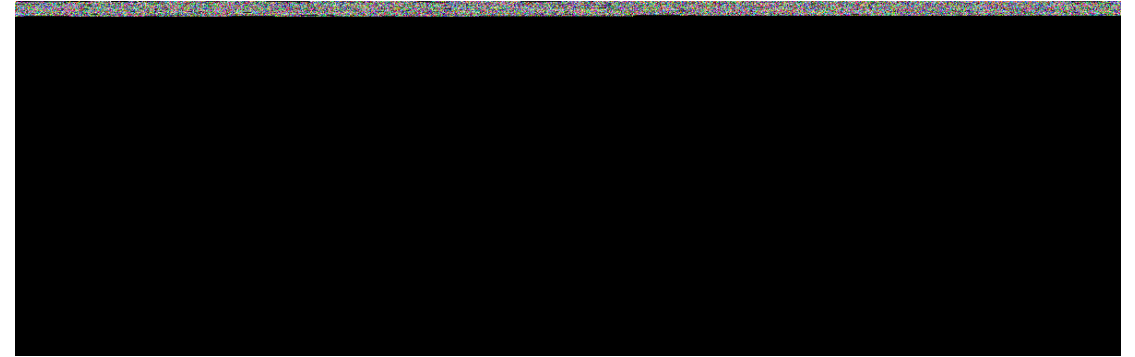
Tension Stiffening Experiments set up



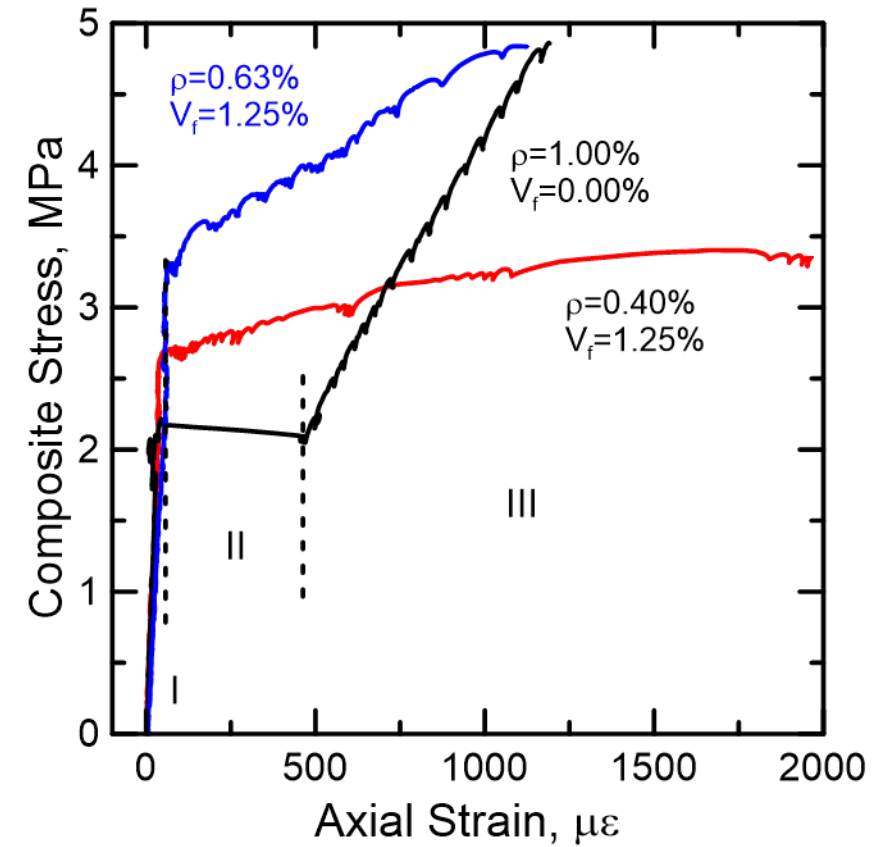
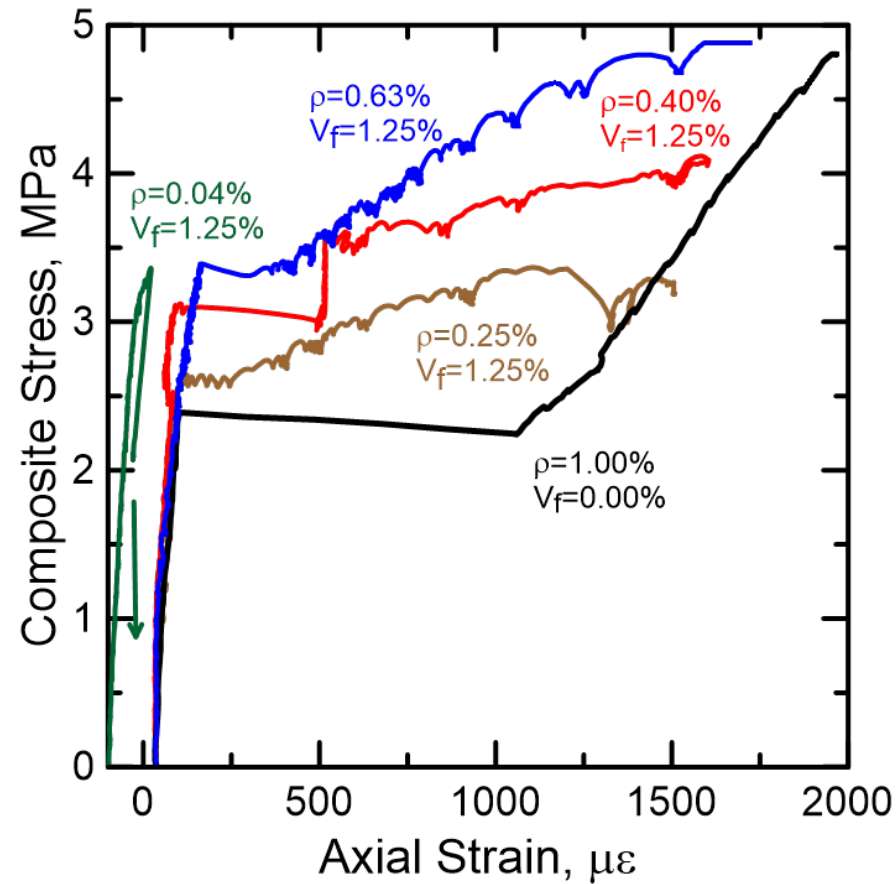
Sample manufacturing, strain gauges, and rebar reinforcements



T1, T2, C1, B1 and B2)



Uniaxial tension tests



Effect of steel fibers in the tensile behavior of concrete ties with different ratios of rebars, (a) conventional concrete, and (b) SCC

Nature of Microcracking

Sample: SCC-100-0



Sample: SCC 040 125



Sample: SCC-063 125



Sample: CON 100 00



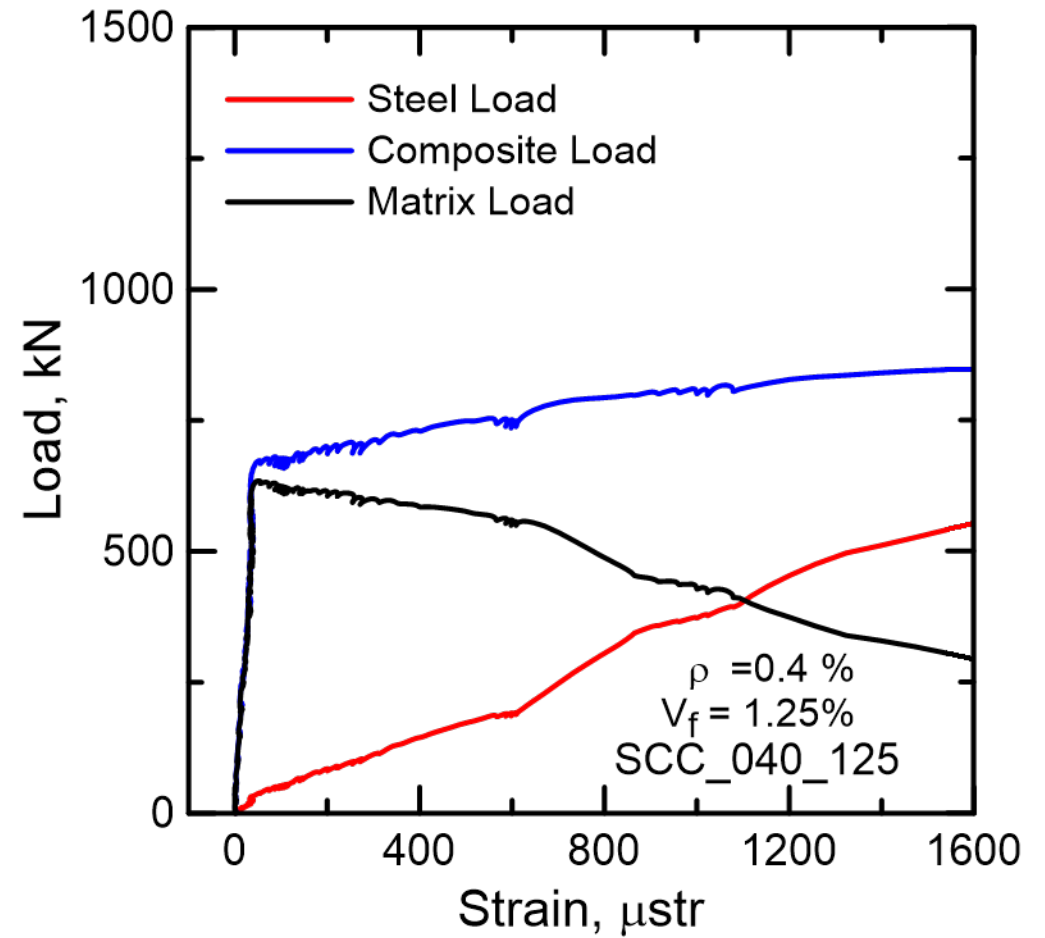
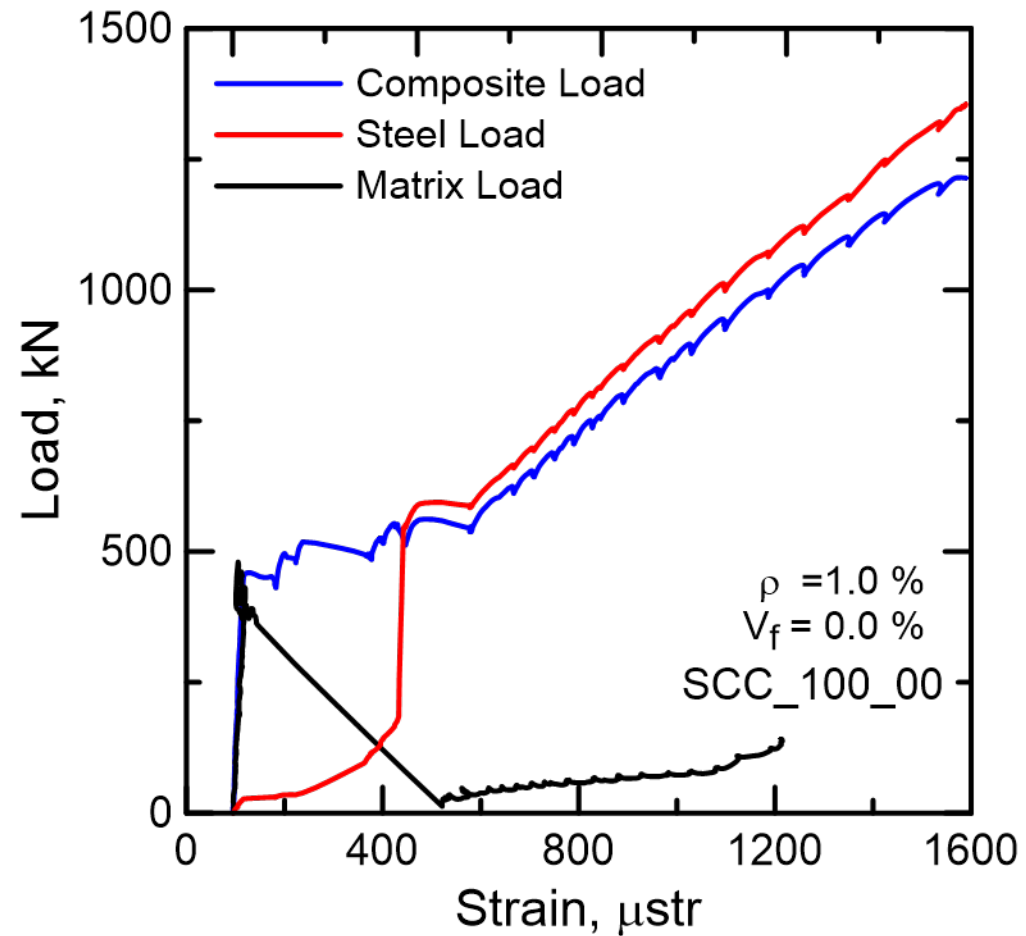
Sample: CON 025 125



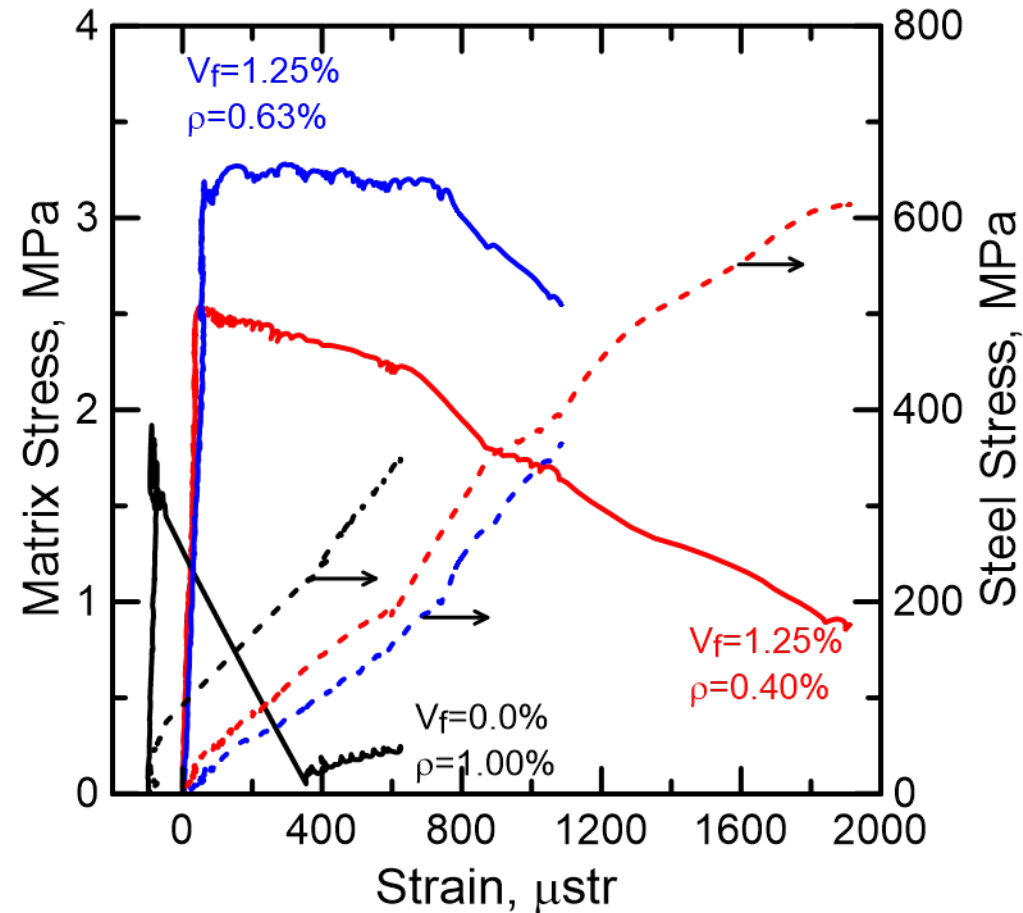
Sample: CON 040 125



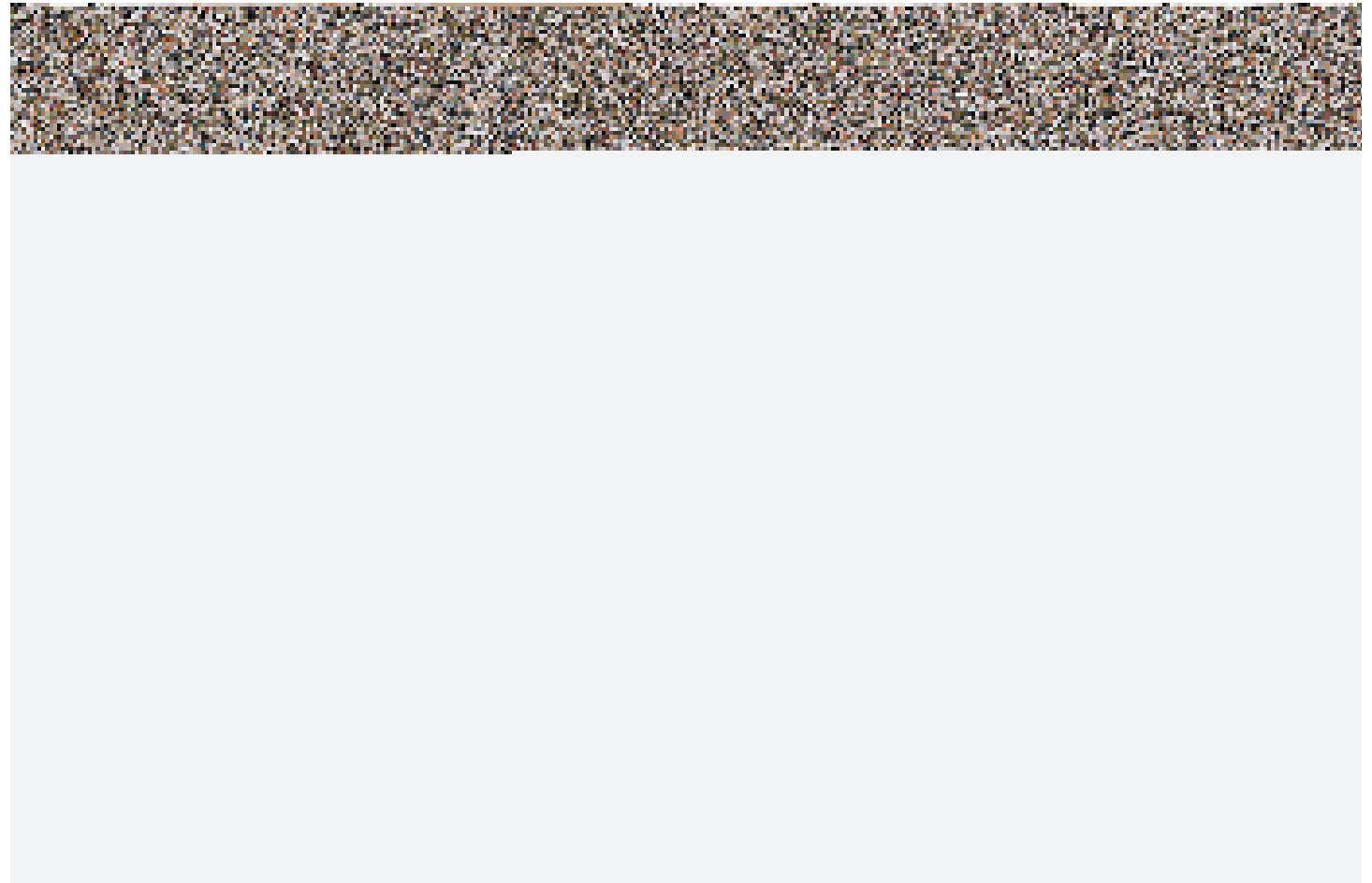
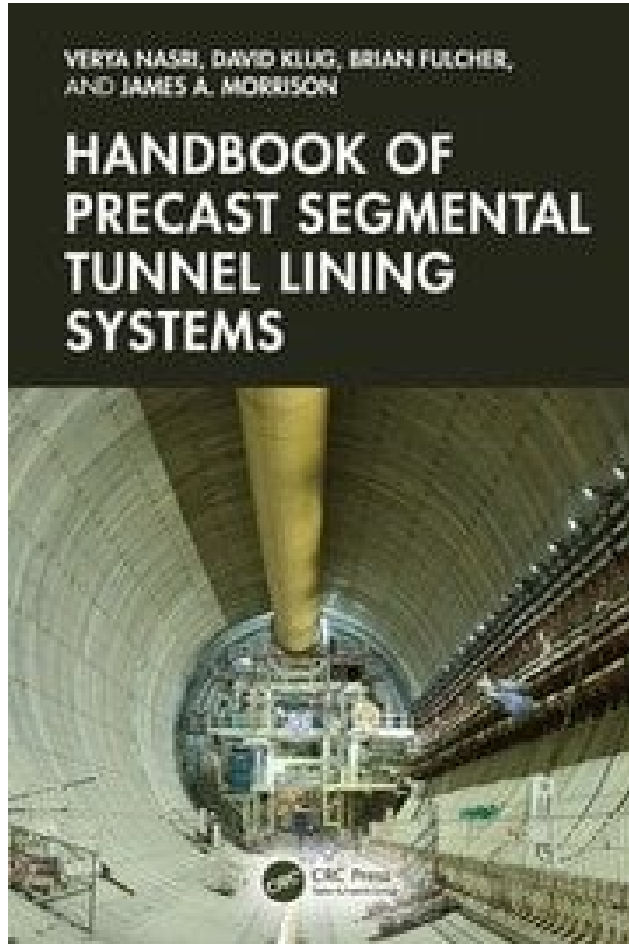
Load Distribution



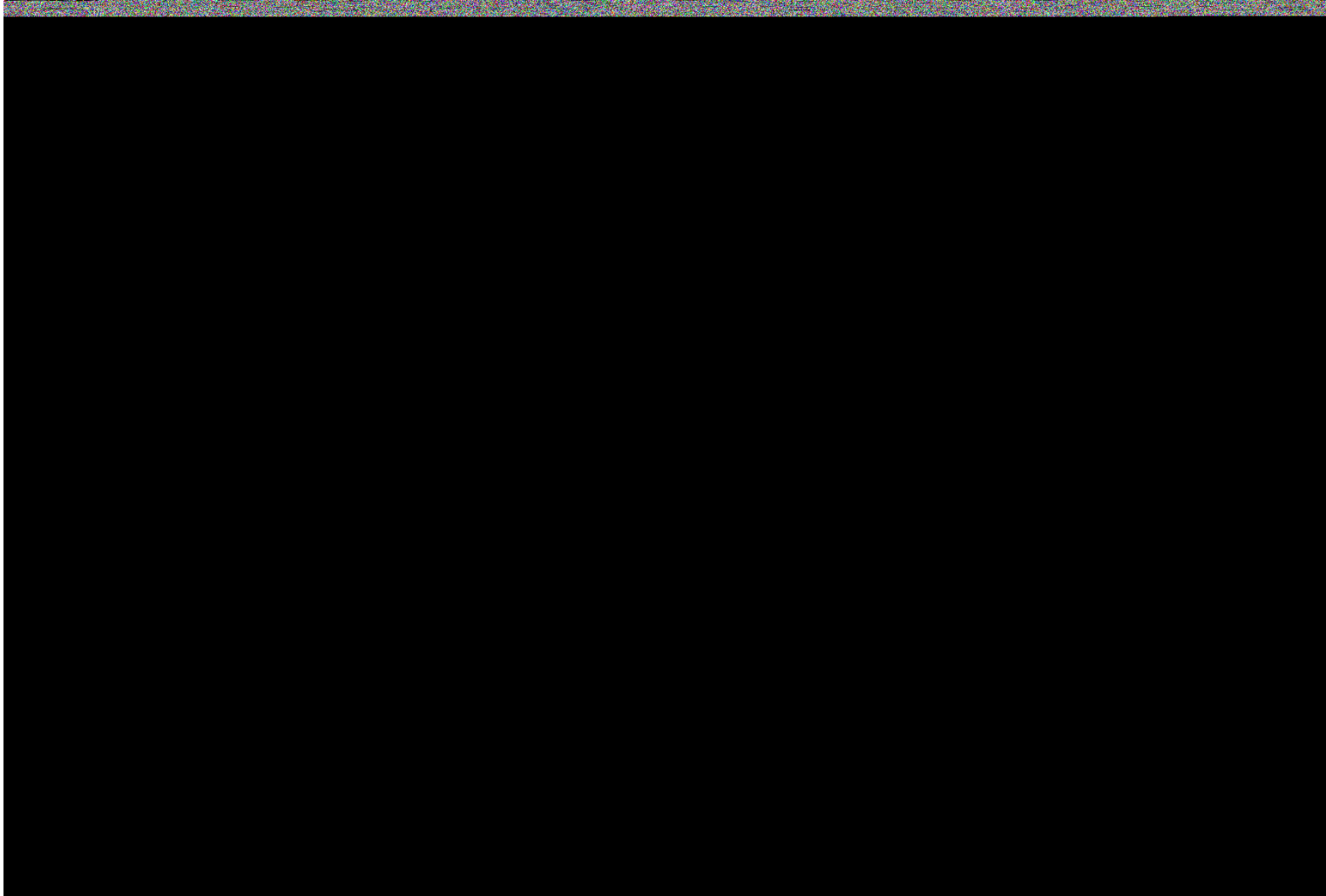
Fibers as Localization Inhibitor Carry Majority of the Load in the presence of Rebar



Precast Tunnel Sections



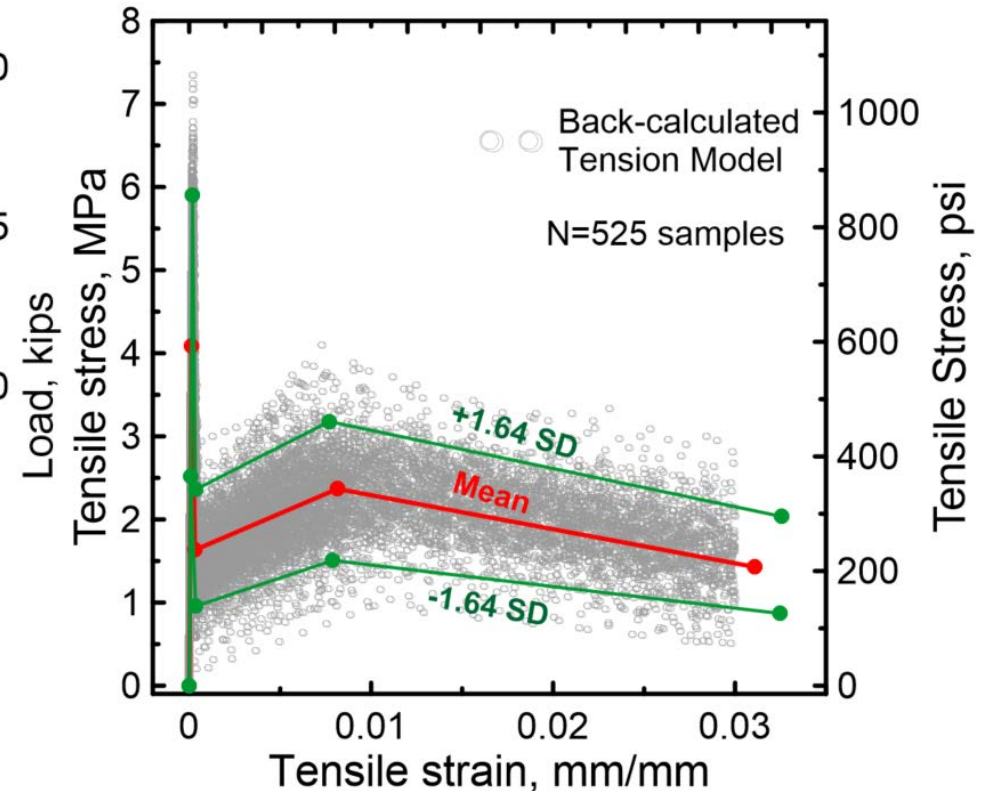
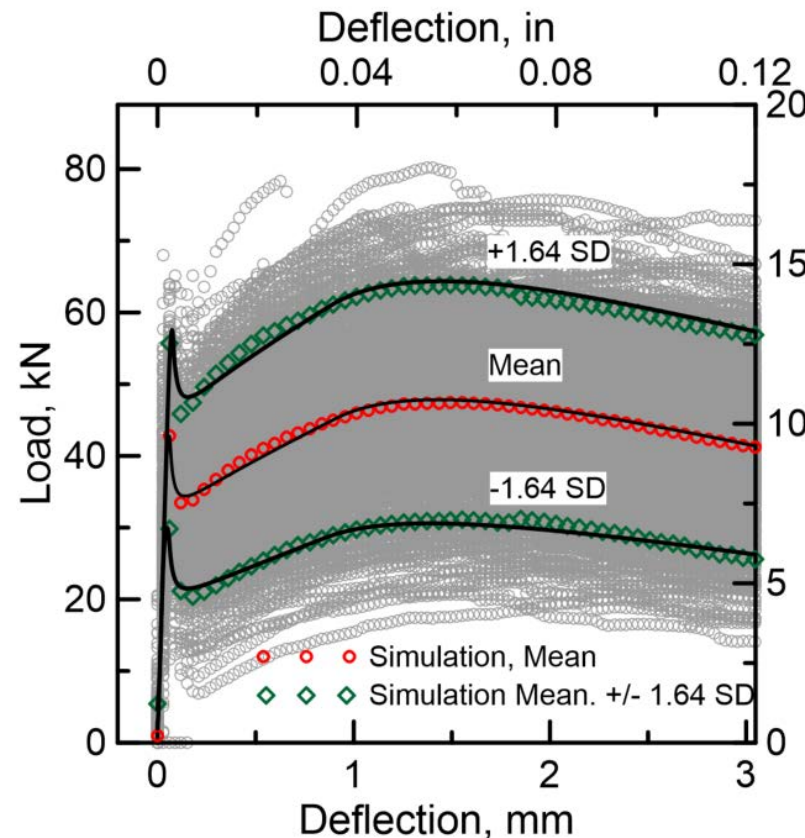
Precast Tunnel Segment Testing Facilities at ASU



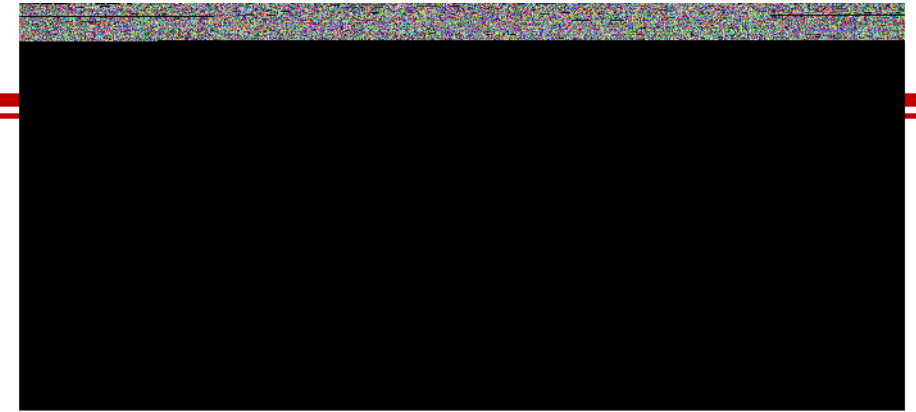
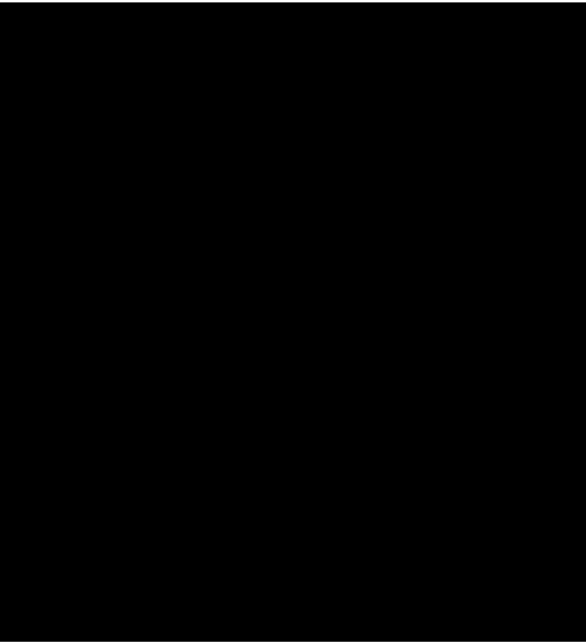
Material parameters of FRC for structural design

- Mechanical properties such as the compressive and tensile stress-strain behavior of fiber-reinforced concrete (FRC) can be obtained through back-calculation using load–deflection data from ASTM C1609 tests.
- What is the appropriate characteristic value for design?

Is the lower bound at -1.64 standard deviations (a 90% confidence interval) an appropriate choice for conservative design assumptions?

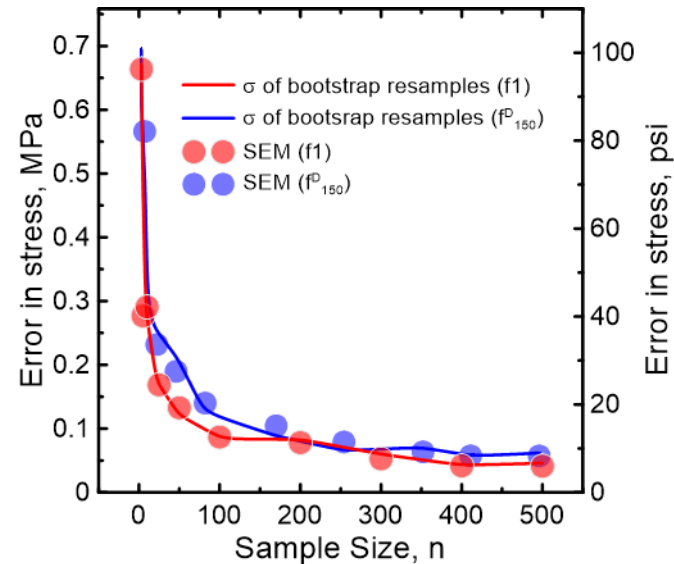


ASTM C1609 tests N=525

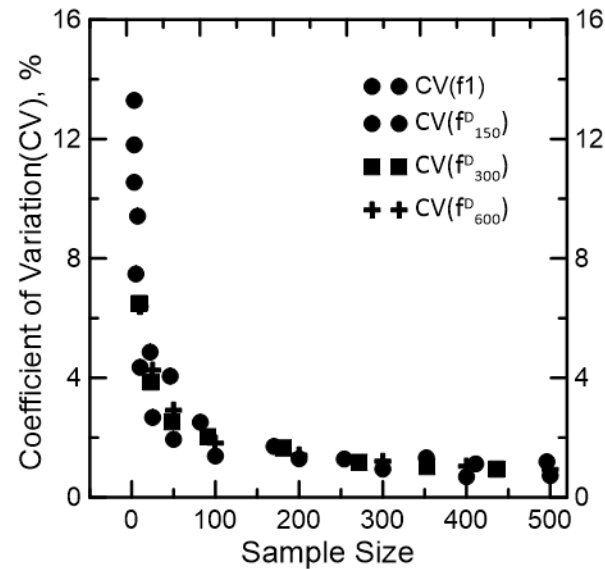


No way to stop the rapid crack propagation at the MOR level

Comparison of SD from Bootstrap analysis versus analytical standard error of mean



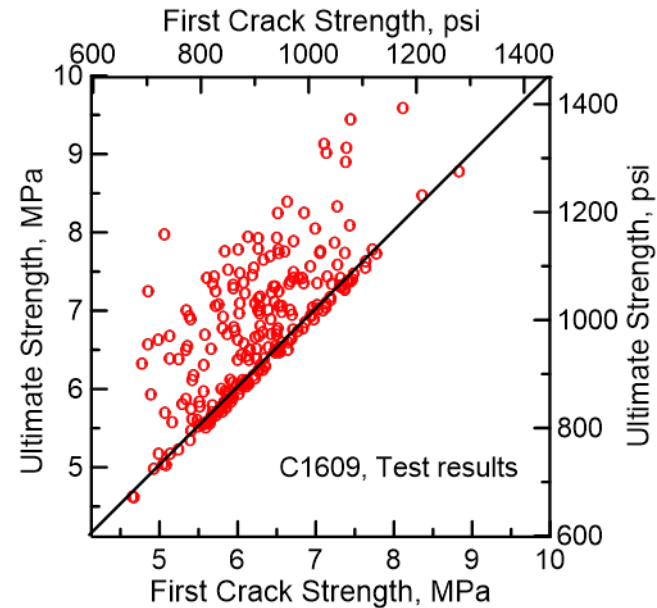
Comparing coefficient of variation in bootstrap samples.



- Appropriate variability factors can be derived from the CV for specific sample size.
- Around $n=25$ samples are required for CV to become 5% or lower

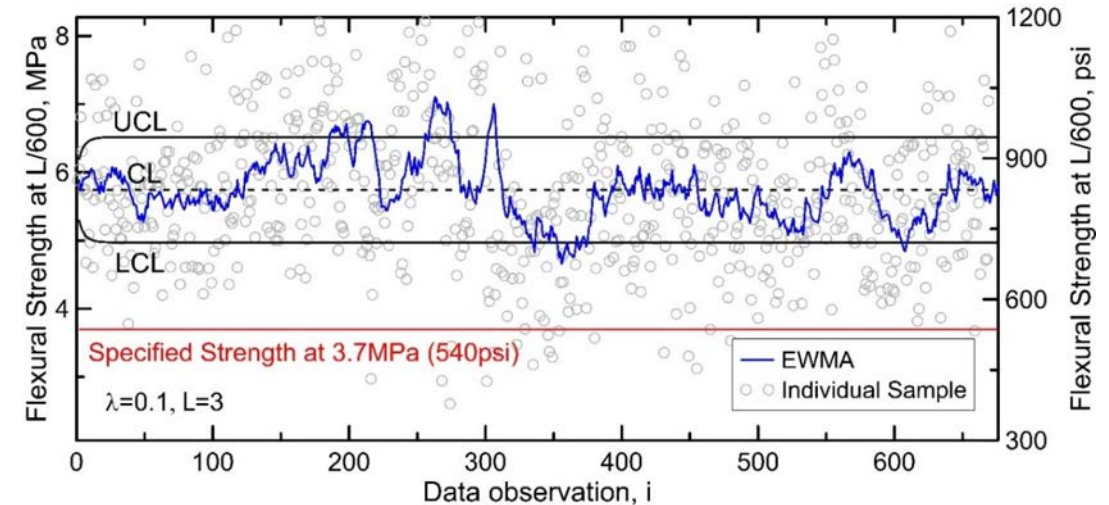
ASTM-C1609 results of 1000+ samples from precast segments 6/2018 -10/2019

Representative samples of different categories from same mix design but different date



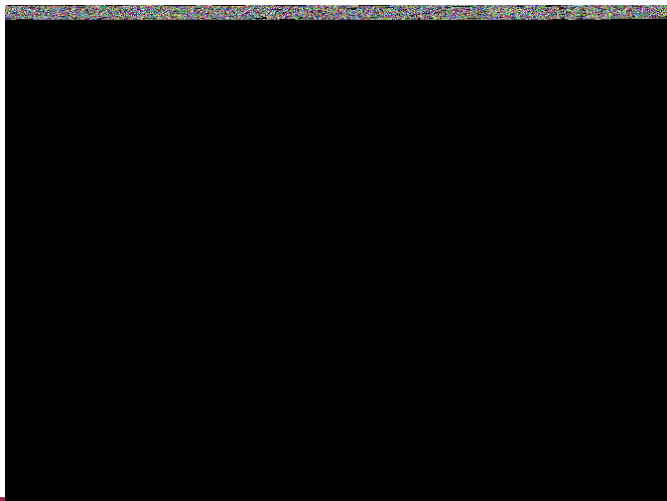
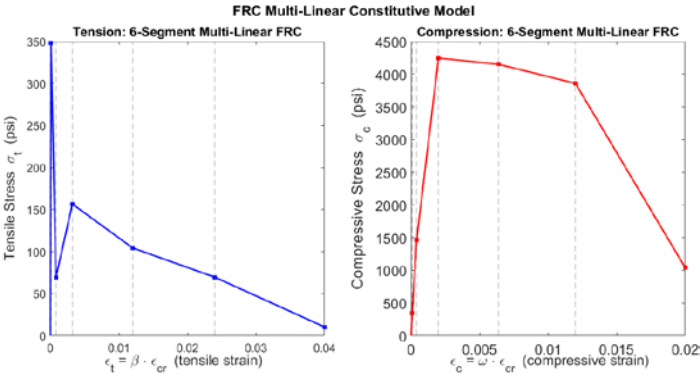
Correlation of first crack

The SPC model uses CUSUM and EWMA charts to indicate stability or instability of the manufacturing process

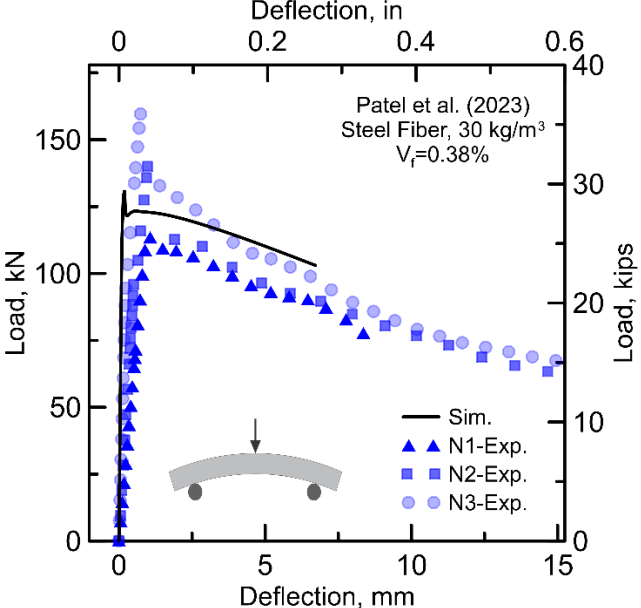


The control charts of residual strength at L/600 (f_{600}^D) at 28-day collected from June 2018 to October 2019; (a) CUSUM, (b) EWMA

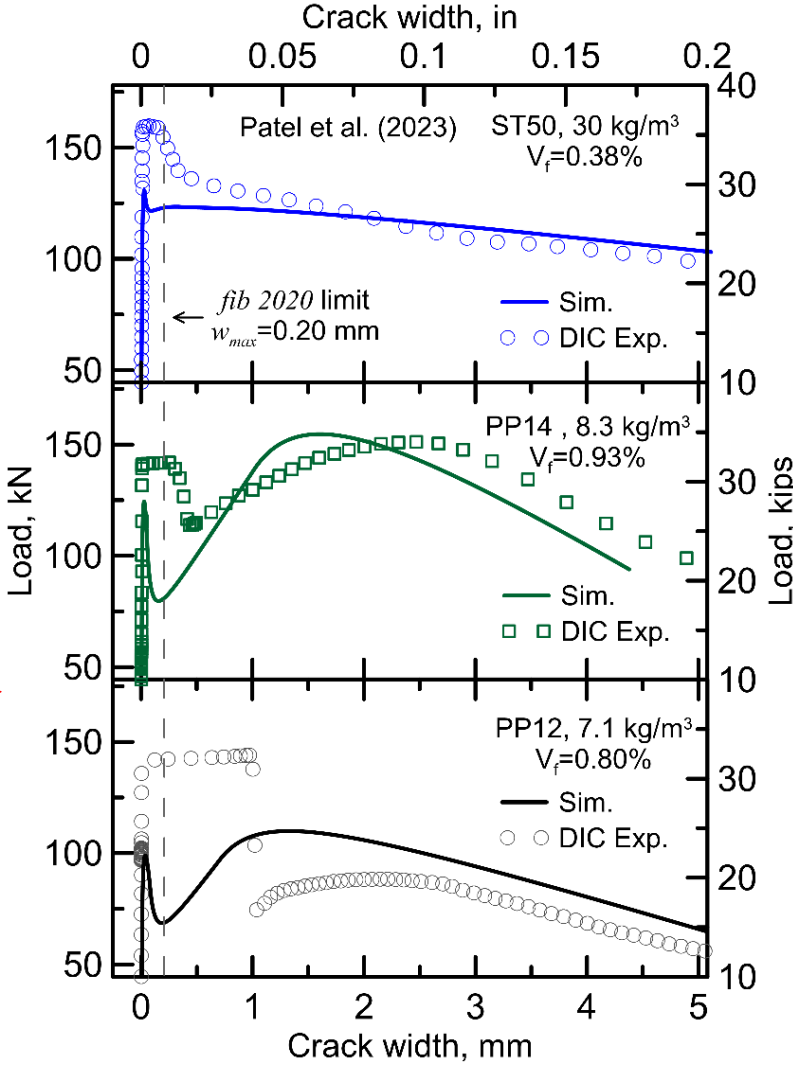
Standard Beam, ASTM C1609



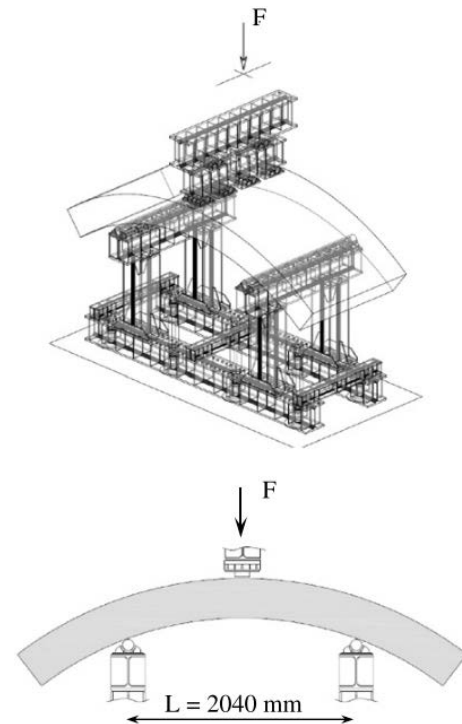
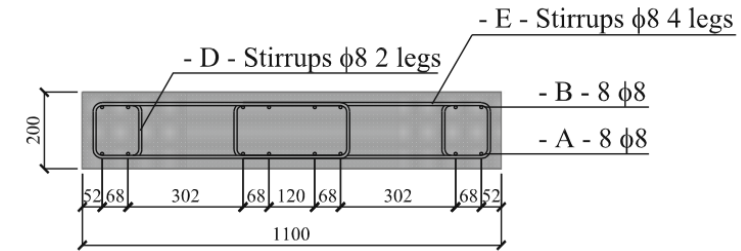
Full scale load-deflection response
By using the closed-form analytical solution



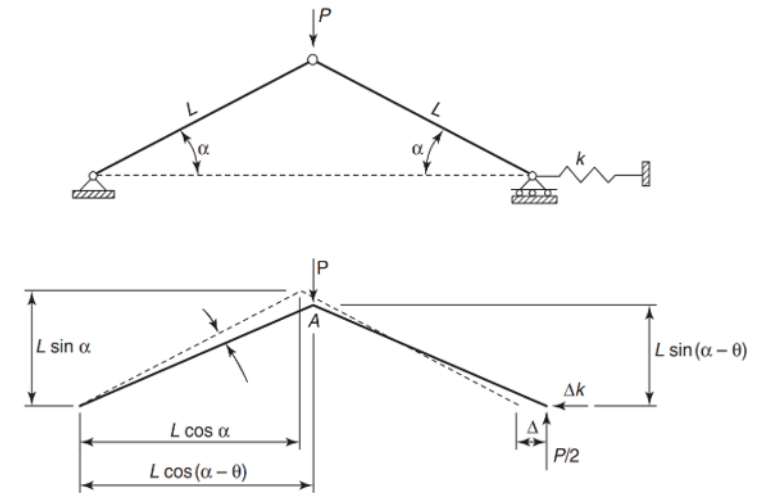
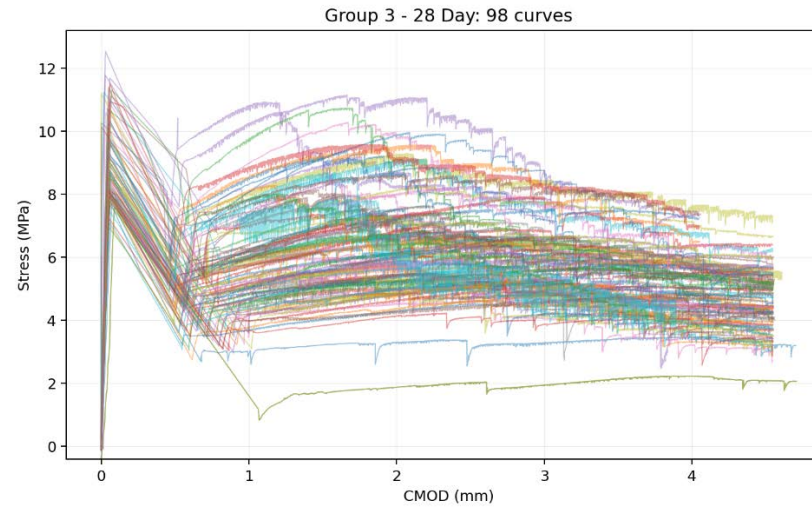
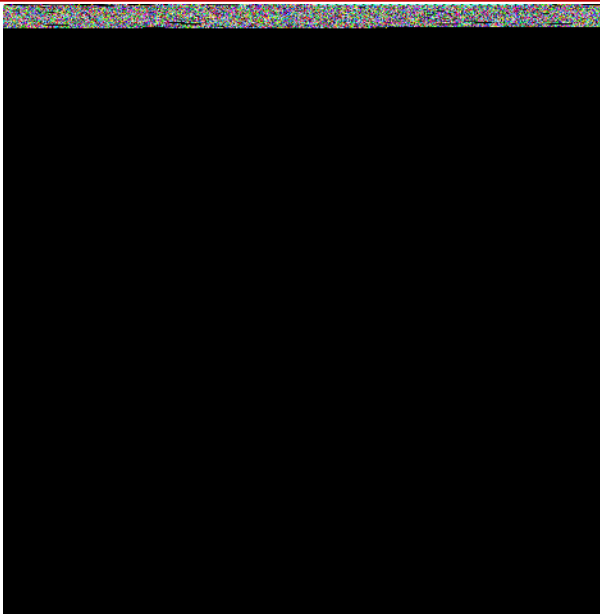
Full scale load-crack width response
This can be used in the SLS design



Simulation of load-deflection of tunnel segments using closed-form analytical solution

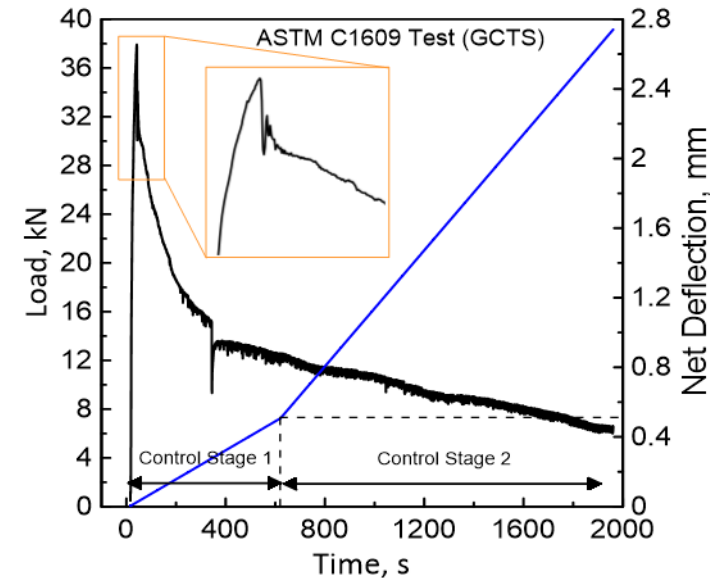
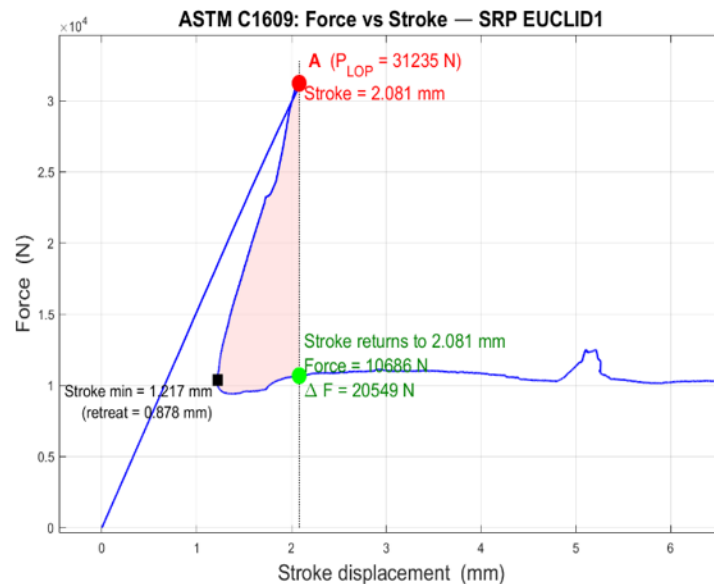


Tension Snap-Back or Snap-Through Instability Penalty Factor



Snap Through Simulation,

Snap back
Event

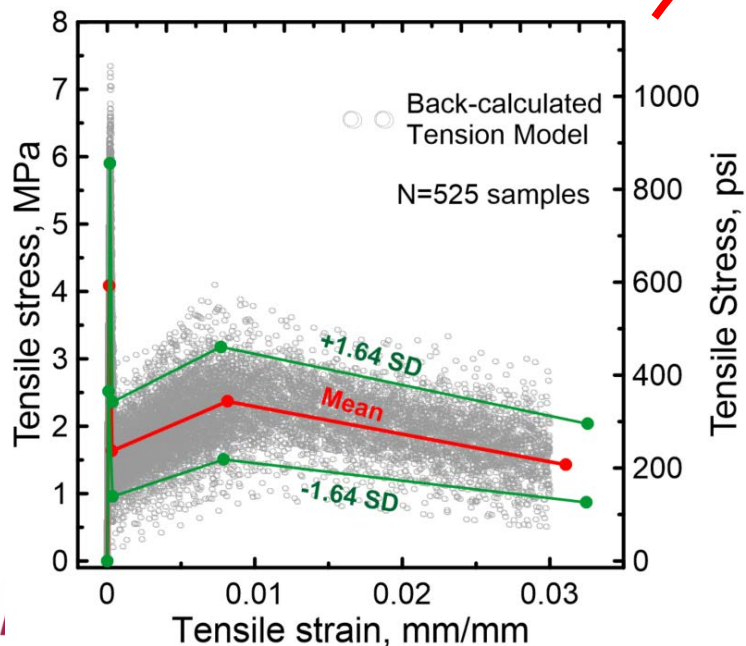


Controlled
test

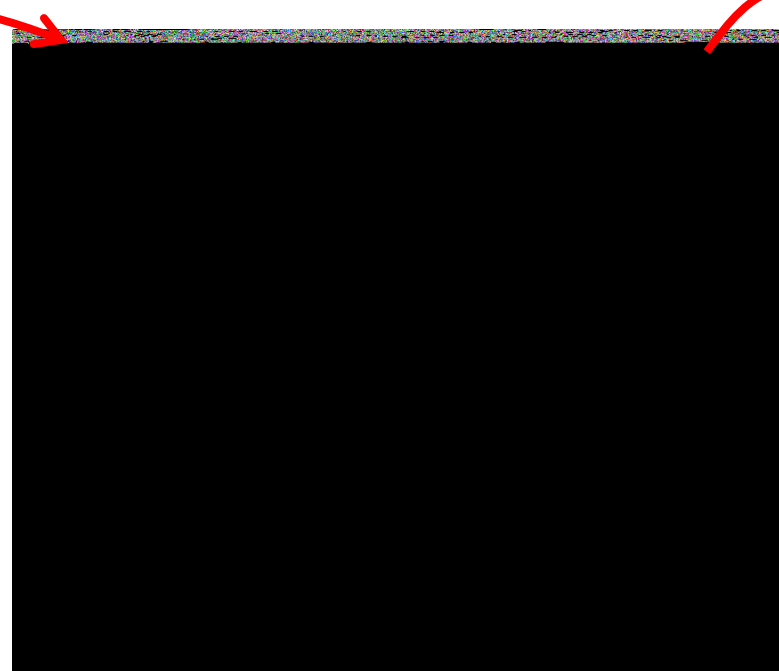
Simulation of Synthetic Section

- In this study, a computational model simulates the section response of ASTM C1609 beams by randomly assigning residual tensile strength values to a 100×1000 mesh grid. The location of the neutral axis (NA) is determined using a numerical method based on the Newton–Raphson algorithm. The corresponding moment–curvature and load–deflection responses are then calculated accordingly using numerical method.

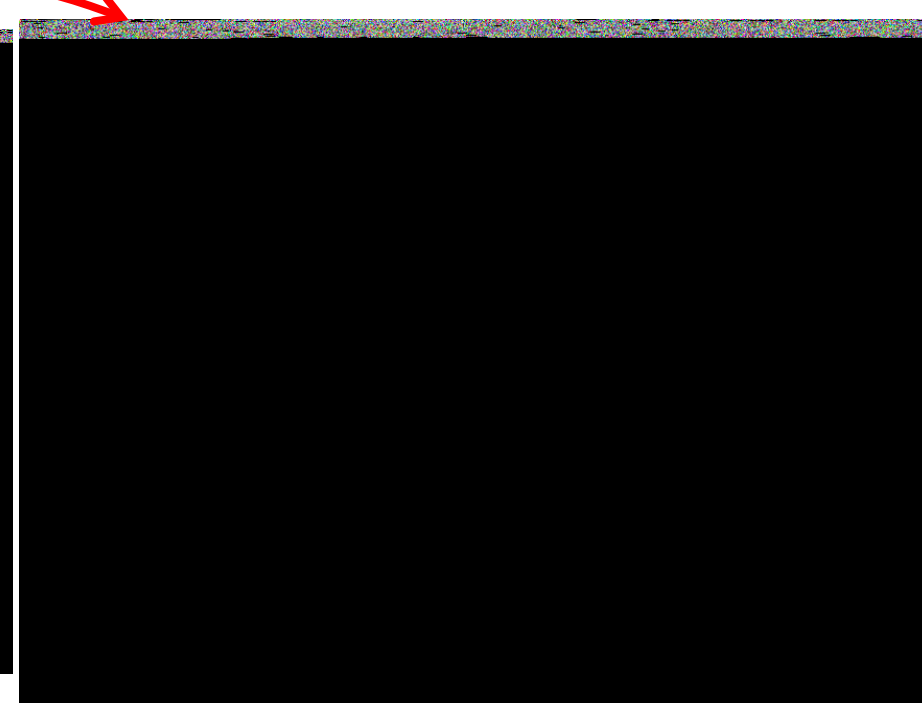
Back-Calculated Tensile Strength
Data from 525 Samples

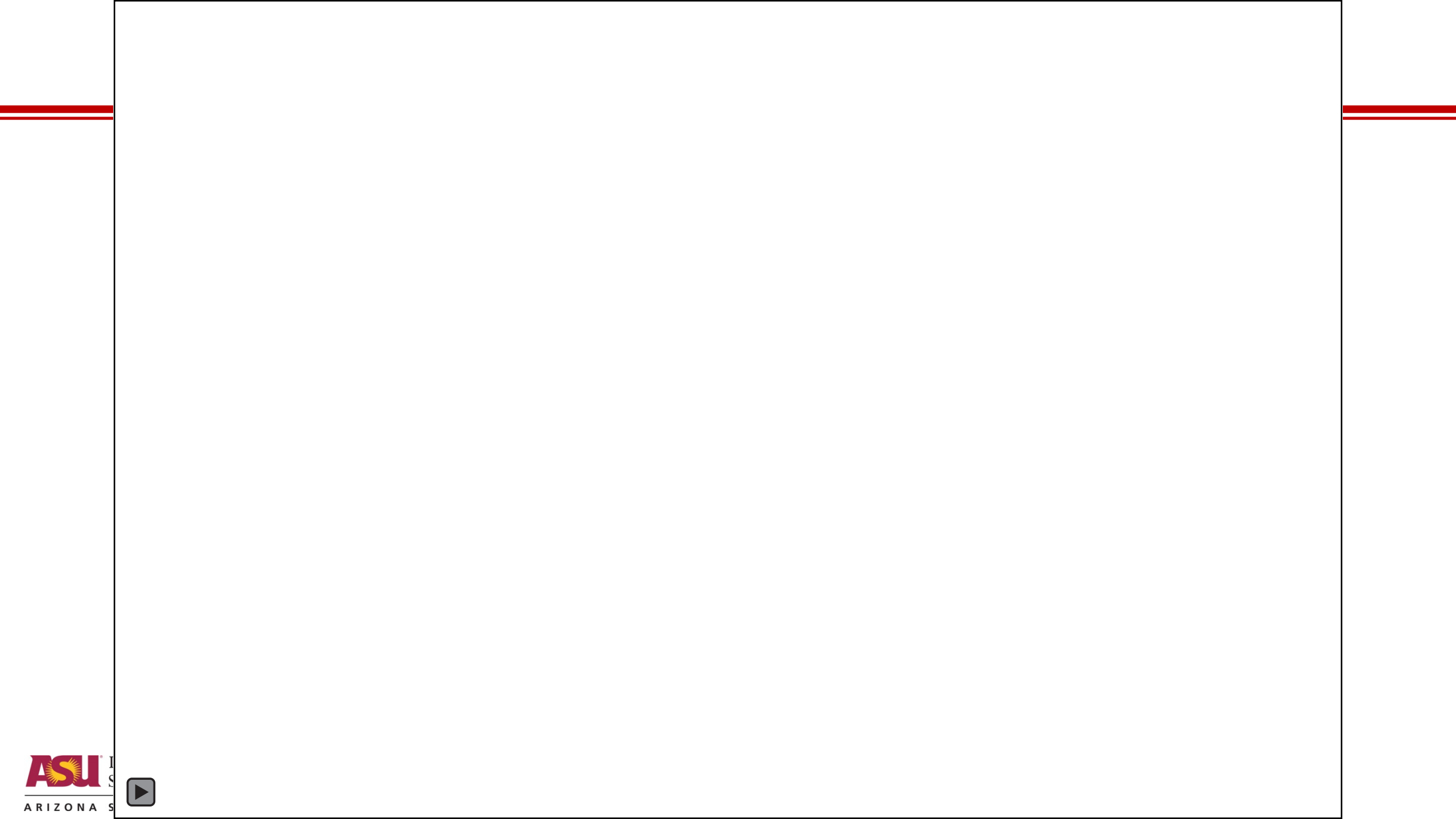


Stress distribution



Mesh X-Y=100×1000





Key Points

- When 5–10 ASTM C1609 tests are available, the simulated section response is approximately 22% higher than the mean response obtained from the full 525-sample dataset.
- With 20 to 525 samples available, the simulated section response moves closely to the population mean of 817 samples, improving the reliability of the analysis.
- Using the lower bound at -1.64 standard deviations (90% confidence interval) may be overly conservative, as the actual section response reflects an averaged behavior across the beam width, not just extreme values.

Thank you For your Attention

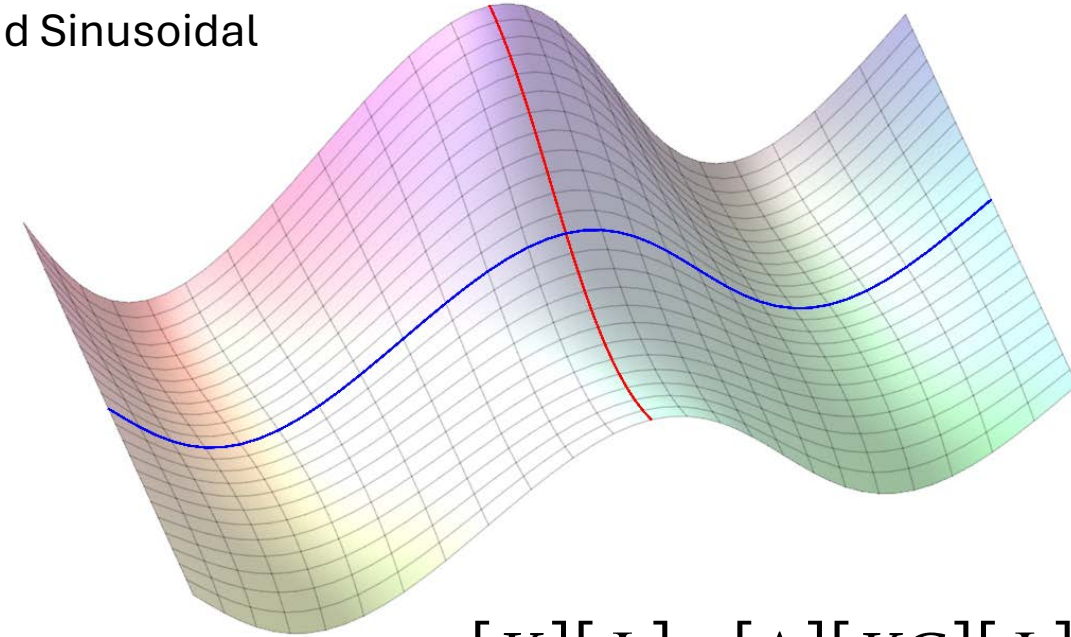
TRC L Shapes in compression



Finite Strip Method (FSM) Displacement Functions

Displacement function $W(x,y)$ with the Polynomial Function (red) and Sinusoidal Function (blue) in a simply supported strip.

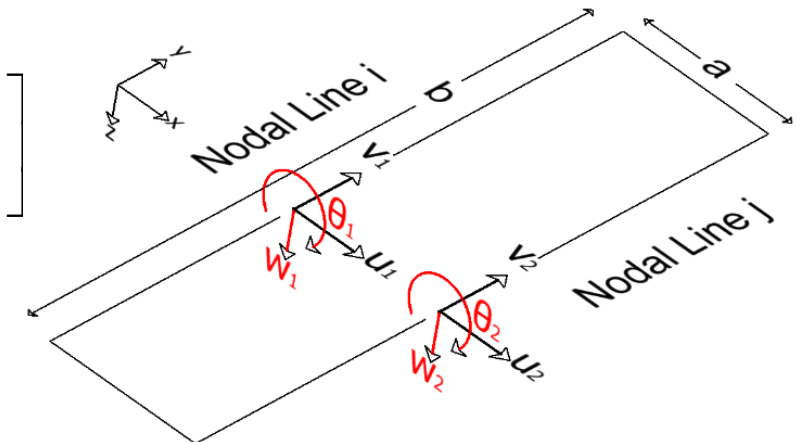
$$K = \begin{bmatrix} & & & & 0 & 0 & 0 & 0 \\ & & & & 0 & 0 & 0 & 0 \\ & & & & 0 & 0 & 0 & 0 \\ & & & & 0 & 0 & 0 & 0 \\ ke_{uv} & & & & & & & \\ & & & & & & & \\ 0 & 0 & 0 & 0 & & & & \\ 0 & 0 & 0 & 0 & & & & \\ 0 & 0 & 0 & 0 & & ke_{w\theta} & & \\ 0 & 0 & 0 & 0 & & & & \end{bmatrix} \quad KG = \begin{bmatrix} & & & & 0 & 0 & 0 & 0 \\ & & & & 0 & 0 & 0 & 0 \\ & & & & 0 & 0 & 0 & 0 \\ & & & & 0 & 0 & 0 & 0 \\ kg_{uv} & & & & & & & \\ & & & & & & & \\ 0 & 0 & 0 & 0 & & & & \\ 0 & 0 & 0 & 0 & & & & \\ 0 & 0 & 0 & 0 & & & & \\ 0 & 0 & 0 & 0 & & kg_{w\theta} & & \end{bmatrix}$$



$$[K][\Phi] = [\Lambda][KG][\Phi]$$

Red

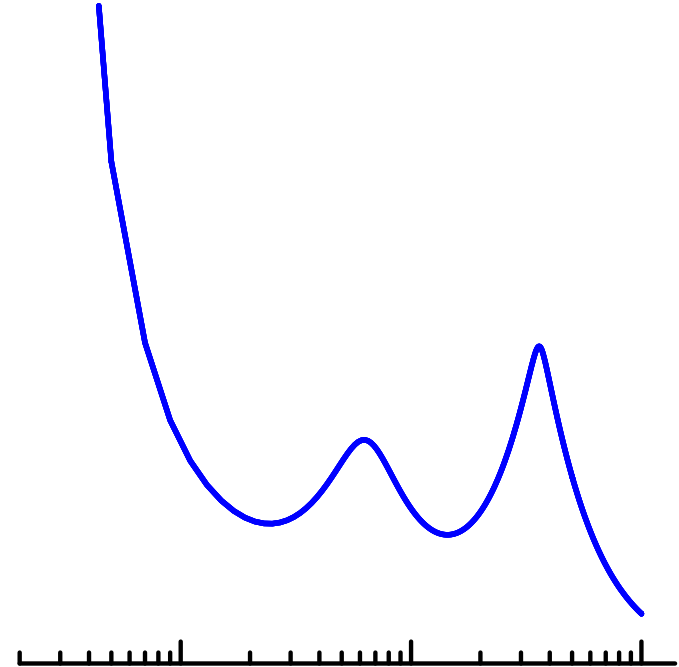
$$w(x) = \sum_{n=1}^r \left[\left(1 - \frac{3x^2}{a^2} + \frac{2x^3}{a^3} \right) w_{in} + \left(x - \frac{2x^2}{a} + \frac{x^3}{a^2} \right) \theta_{in} + \left(\frac{3x^2}{a^2} + \frac{2x^3}{a^3} \right) w_{jn} + \left(-\frac{x^2}{a} + \frac{x^3}{a^2} \right) \theta_{jn} \right]$$



Blue

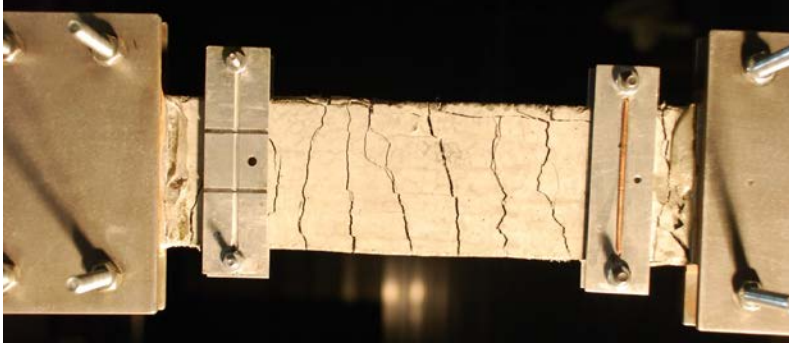
$$Y_n(y) = \sin \frac{\lambda_n y}{b} \quad \lambda_n = \pi, 2\pi, 3\pi \dots$$

Signature Curve

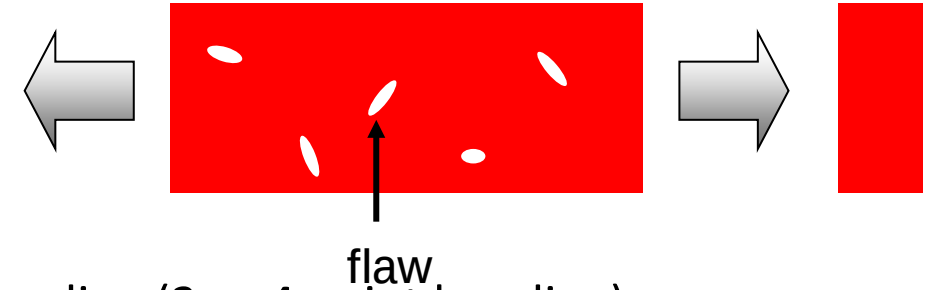


Multiple cracking in tension and flexure

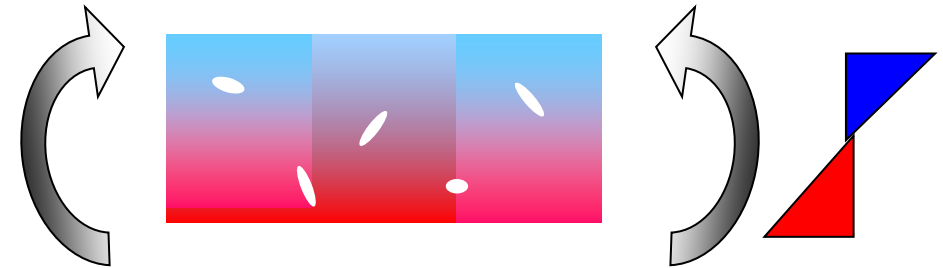
Tensile test



- Uniaxial Tension



- Flexural Bending (3 or 4 point bending)

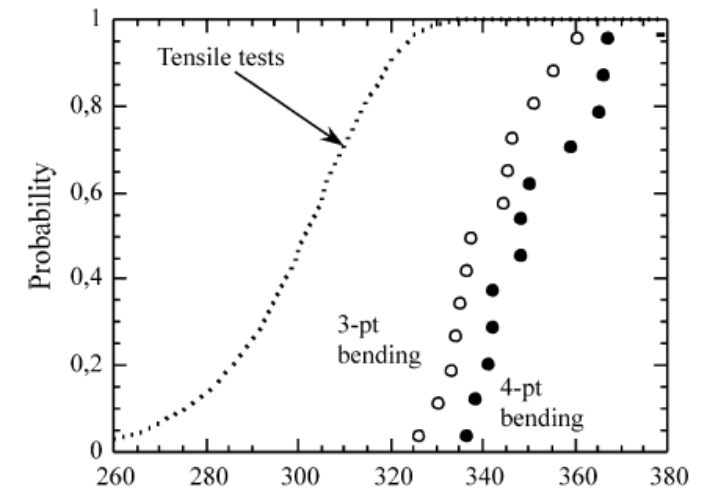


Moment-curvature



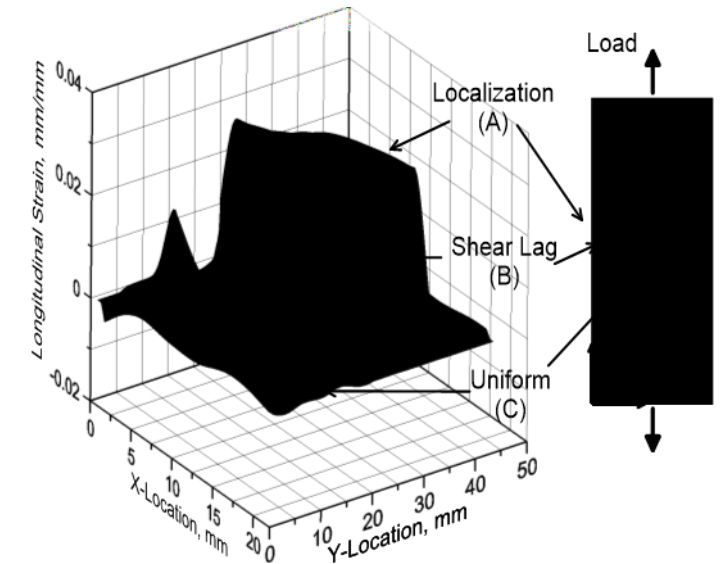
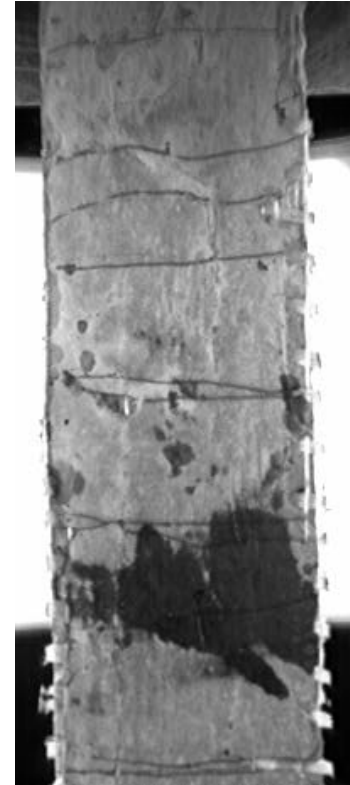
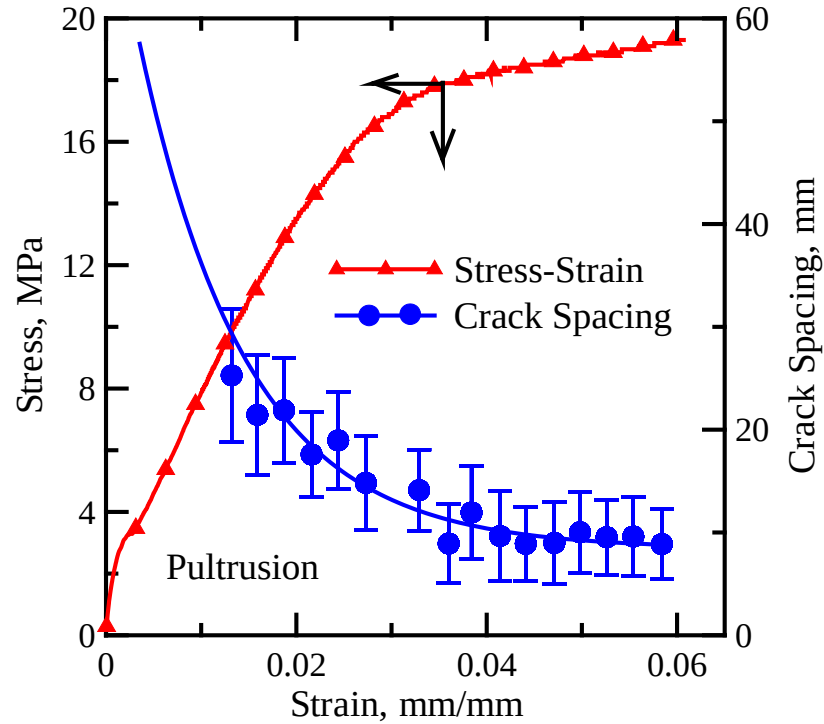
Flexural test

V.Calard, J.Lamon, "A probabilistic-statistical approach to the ultimate failure of ceramic-matrix composites—part I: experimental investigation of 2D woven SiC/SiC composites," , Composites Science and Technology 62 (2002) 385–393

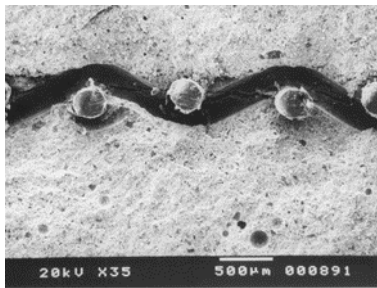


Textile Cement Composites

- Automated pultrusion system for manufacturing full scale TRC structural shapes from laminates
- large strain capacity and ductility



Peled, A. and Mobasher, B., (2005), "Pultruded Fabric-Cement Composites," ACI Materials Journal, Vol. 102 , No. 1, pp. 15-23.



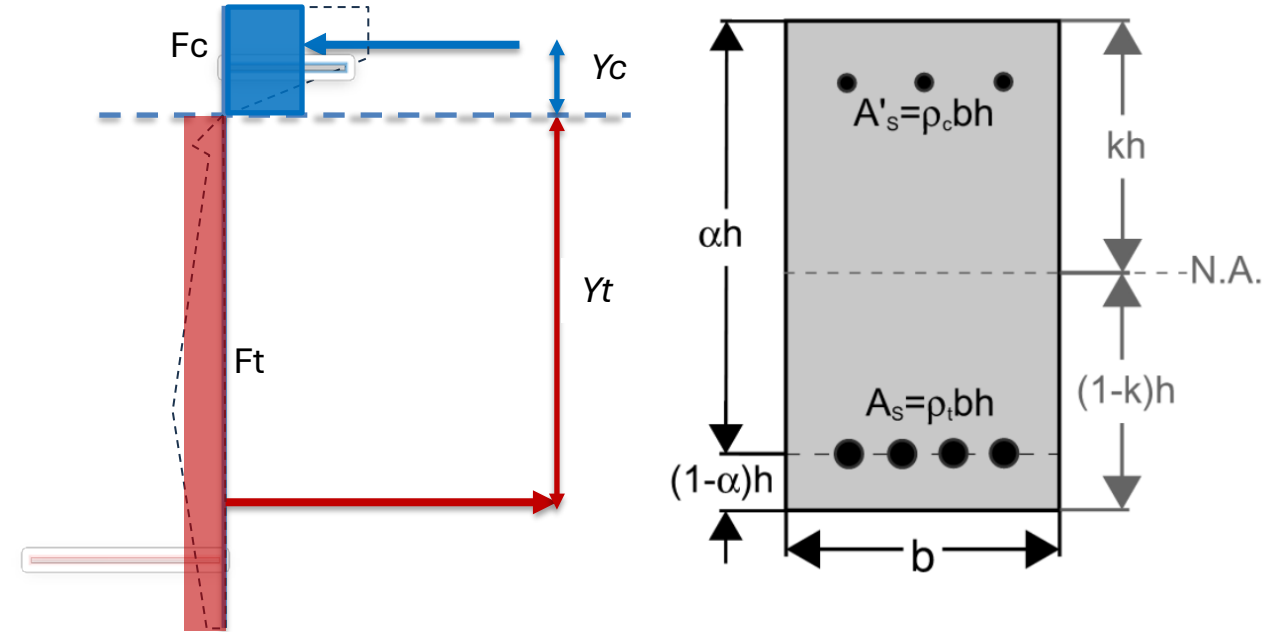
Simplified Equations for Doubly Reinforced Model (Normalized Form)

$$F_T = \kappa n \rho_t - k \mu + \mu$$

$$F_C = \omega k + \rho_c n \kappa$$

$$Y_T = \frac{(2\alpha\kappa n \rho_t - 2k\kappa n \rho_t + k^2\mu - 2k\mu + \mu)}{2(-\kappa n \rho_t + k\mu - \mu)}$$

$$Y_C = \frac{(2\rho_c n \kappa \alpha + \omega k^2 + 2\rho_c n \kappa k - 2\rho_c n \kappa)h}{2(\omega k + \rho_c n \kappa)}$$

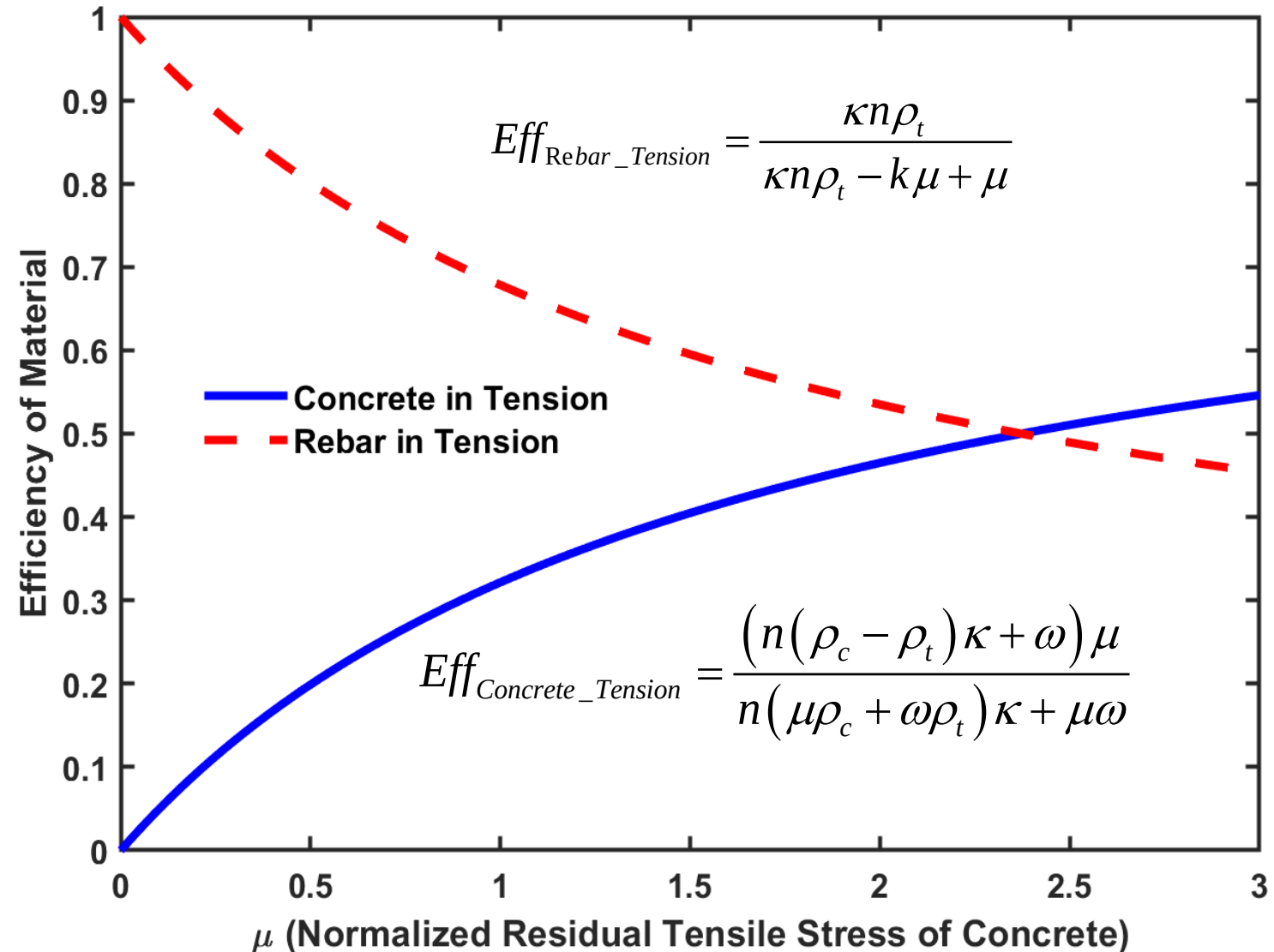


$$F_T - F_C = 0 \quad (\text{Solve for } k) \quad = \quad k = \frac{\rho_c n \kappa - \kappa n \rho_t - \mu}{\omega + \mu}$$

$$M_{ult} = F_C \times Y_C + F_T \times Y_T = M_{ult} = \frac{n^2 (\rho_c - \rho_t)^2 \kappa^2 - 2n \left(((\alpha - 1)\omega + \mu\alpha) \rho_c + \rho_t (\alpha\omega + \mu(\alpha - 1)) \right) \kappa - \mu\omega}{2\omega + 2\mu}$$

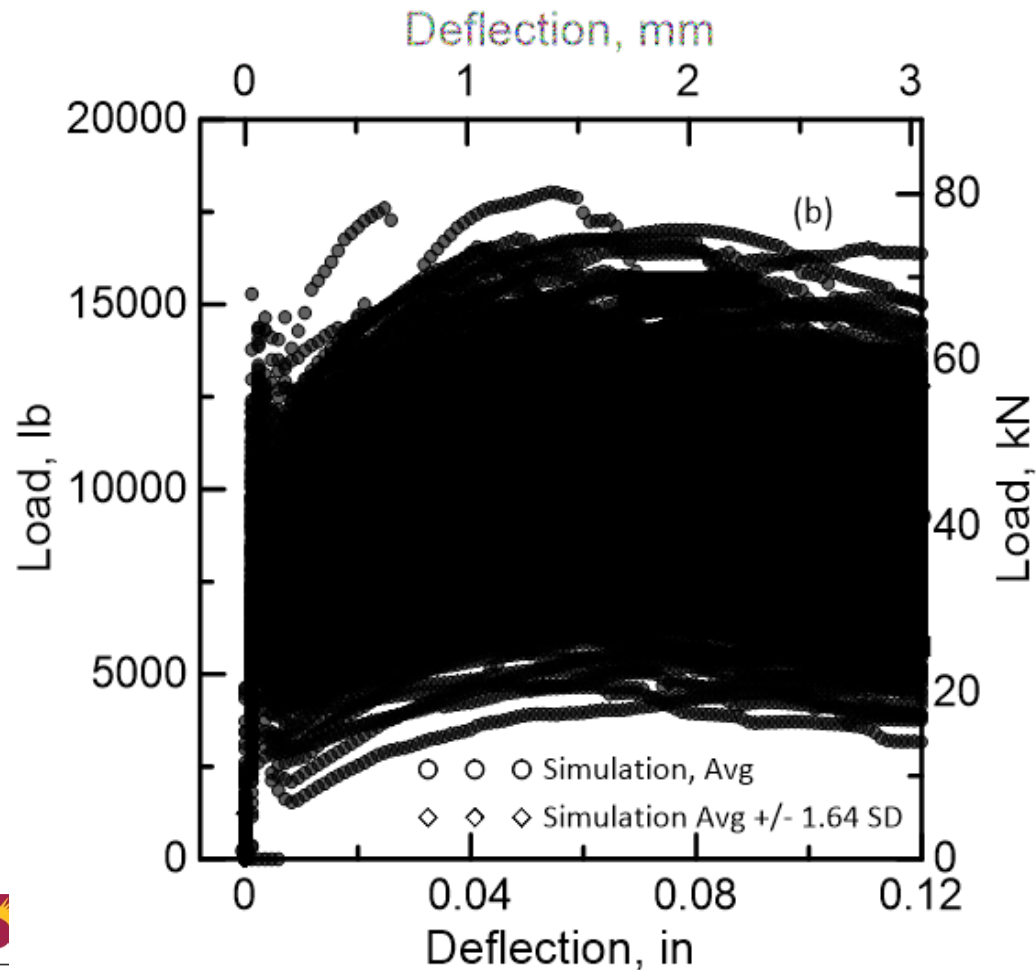
Increased efficiency and redundancy is design

- With addition of fibers, an increased average residual tensile strength, increases the efficiency of concrete section
- Values reported for the ultimate limit state, but can be calculated at any service level or curvature

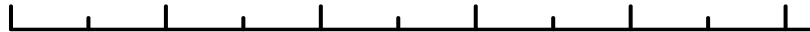


QC/QA Procedures for precast Tunnel Linings

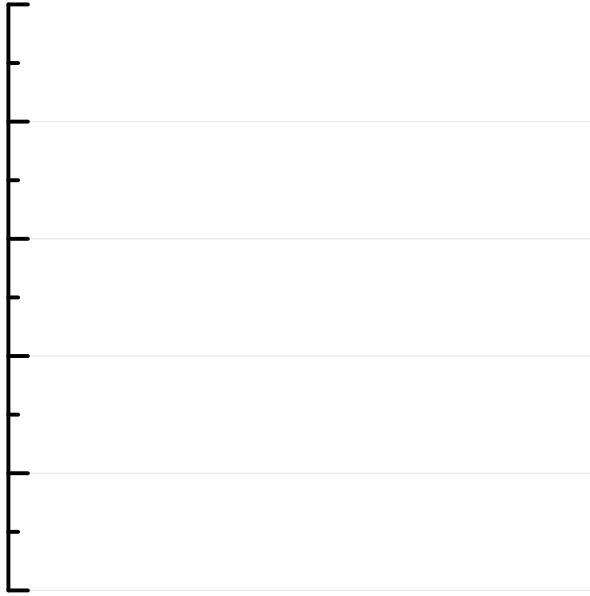
- Evaluation of more than 9 months of QC/QA beam samples, 1200+ flexural C1609 beams
- Address the variations using Statistical Process Control tools



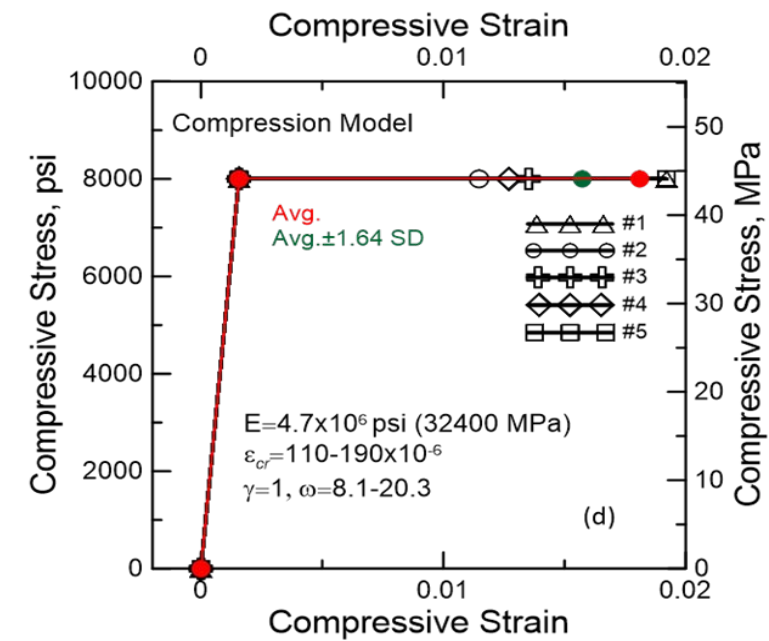
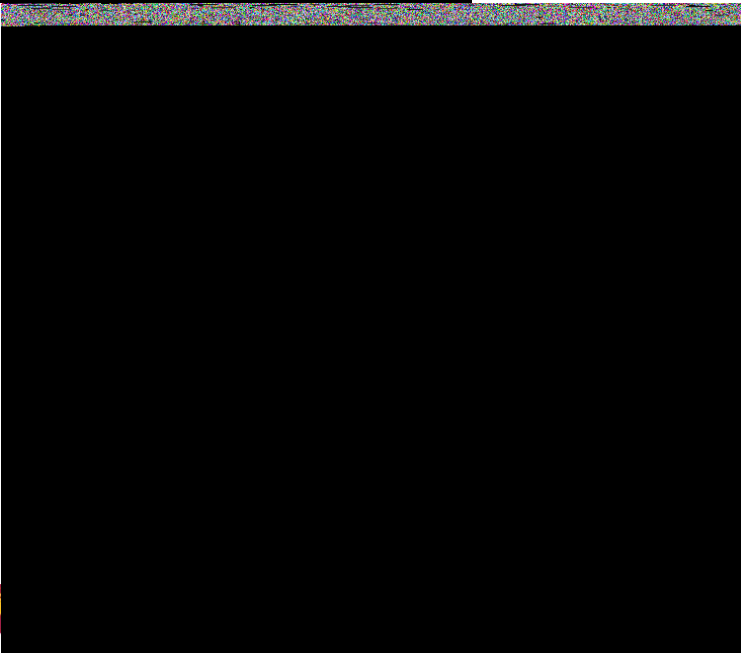
Statistical measures of distribution and correlation



Basis of Design- Tensile Properties obtained from Experimental testing, ASTM C1609

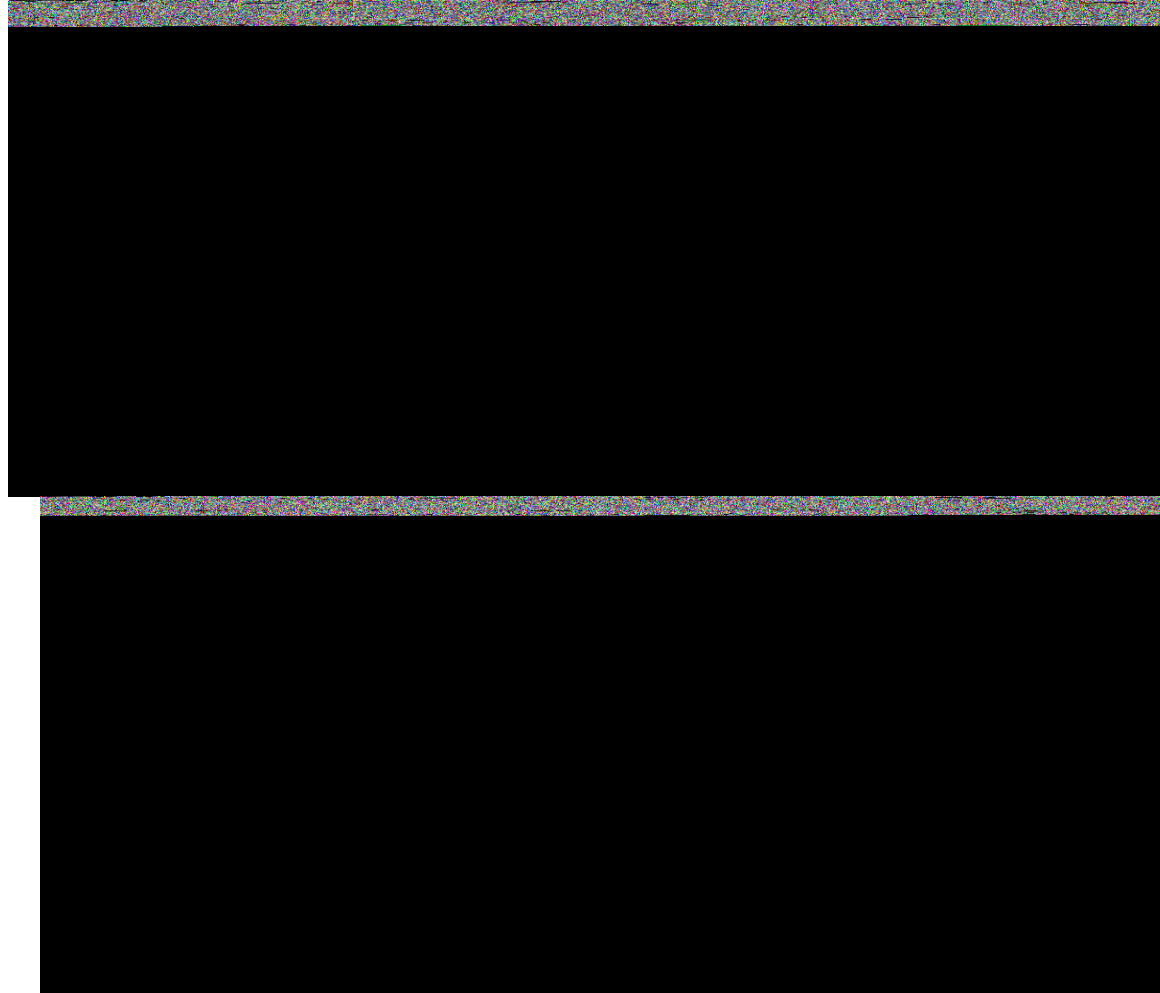


Simulation of 800+ QC data in chart

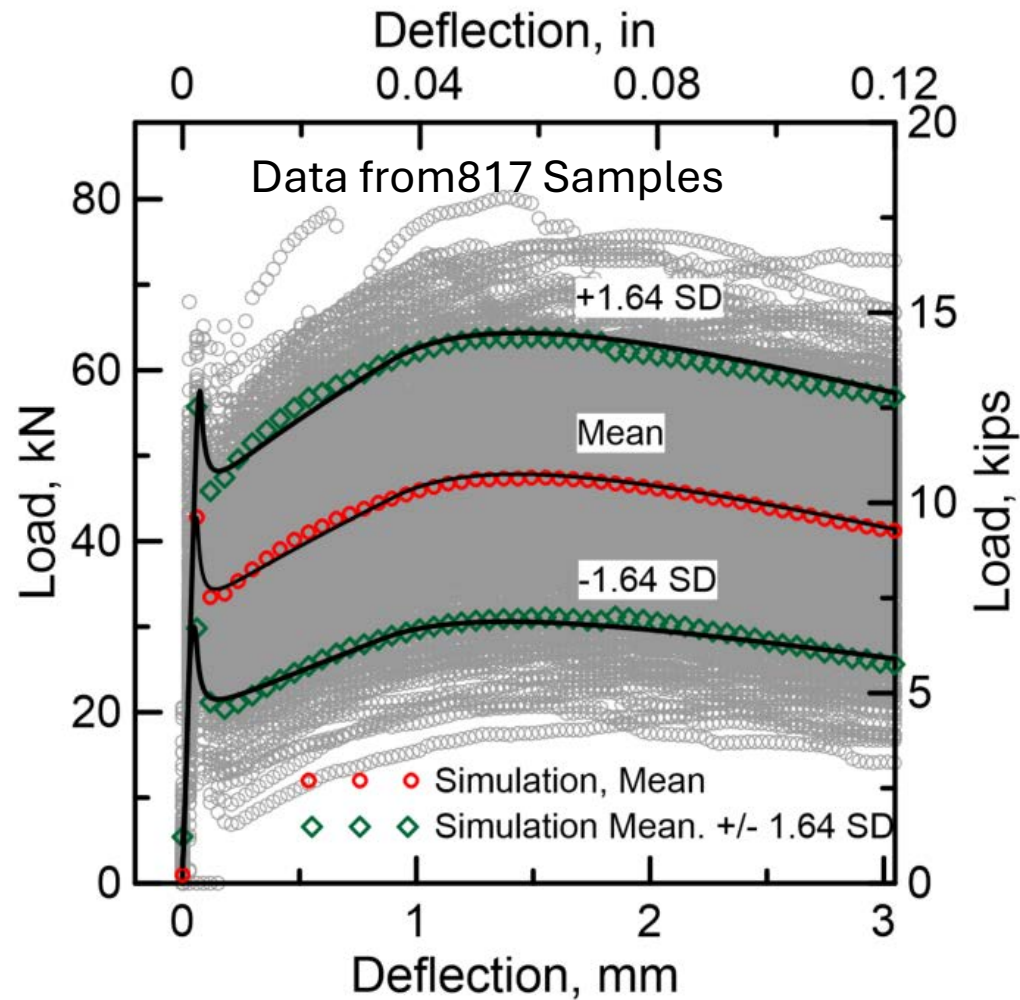


CUSUM and EWMA Run Charts to monitor and control Production problems and outlier analysis

Control Charts for 9 months of production



- Load-Deflection response from closed-form solution
- Using lower-bound tensile stress-strain from Back-Calculation



- Load-deflection response from numerical method
- Using different random sets of tensile stress-strain data from back-calculation
- Available sample sizes = [5,10,20,50,100,250,525]

